



# A Federated Learning Approach for Reliable Diabetes Prediction

Comparative Analysis of Machine Learning and Deep  
Learning Models with Probability Calibration

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This thesis was submitted to the Faculty of Computer Science and Engineering at Blekinge Institute of Technology to fulfill the requirements for a Master of Science degree in computer science. The study for this thesis extended over 20 weeks of full-time academic involvement.

The authors confirmed their authorship of this thesis and the use of only the sources specified in the bibliography and acknowledged them as references. Also, they state that this thesis has not been presented to any other academic institution in search of a degree.

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# Abstract

**Background:** Diabetes prediction using machine learning can help with early risk identification. However, healthcare data is spread across institutions and cannot always be shared centrally due to privacy and legal concerns. Federated learning offers a data-locality approach in which raw data can remain at the participating clients. It allows participating clients to train models collaboratively without exchanging raw patient data. However, predictive performance alone is insufficient for clinical use. Reliable probability estimates, interpretable explanations, fairness, and communication efficiency are all required for reliable deployment.

**Objectives:** TrustFed-Diabetes, a federated learning framework for accurate diabetes prediction, is evaluated in this thesis. The study examines whether federated learning can perform similarly to centralized learning under IID and non-IID data distributions. It also examines whether probability calibration, SHAP-based explainability, and fairness evaluation improve the trustworthiness of federated predictions.

**Method:** Experiments were carried out using the CDC Diabetes Health Indicators dataset. It contains 253,680 records and 21 characteristics. A simulated federated system with 3 and 5 clients was evaluated under IID and non-IID conditions. Dirichlet partitioning was used with  $\alpha = 0.1, 0.5$ , and  $1.0$ . Logistic Regression, Decision Tree, Random Forest, Gradient Boosting, XGBoost, and a shallow Multilayer Perceptron were tested. Six models were evaluated in centralized settings, while logistic regression and a shallow MLP were additionally evaluated under federated training. FedAvg and FedProx were compared for federated aggregation. The probability calibration was evaluated using sigmoid and isotonic regression, including federated variations. Expected Calibration Error (ECE) and Brier Score were used to measure calibration performance. SHAP was used to assess feature relevance and explanation stability across clients and communication rounds. The demographic parity difference and equalized odds difference across demographic groups were used to measure fairness. Communication cost was calculated based on the total number of bytes transmitted during federated training.

**Results:** Federated learning performed similarly to centralized performance under IID settings. The ROC-AUC difference was only 0.0007–0.0016 for the two federatable models. Under non-IID conditions, federated performance remained far more stable than local-only training. Local-only training reduced accuracy by more than 15 percentage points and F1-score by up to 20%. Isotonic calibration reduced ECE by around 74% while maintaining accuracy. However, sigmoid calibration did not provide the same improvement. Federated isotonic calibration retained the calibration benefit without requiring centralized calibration data. SHAP explanations were highly consistent across clients. The top-five feature overlap ranged from 98.4% to 100%. In the reported fairness comparison between federated logistic regression and centralized XGBoost, no additional fairness disadvantage was observed across the analyzed age, BMI, and sex subgroups. Logistic regression used around 42 KB of

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stantially greater communication cost.

**Conclusions:** The results show that federated learning may deliver competitive diabetes prediction while keeping patient data local. This was observed under both IID and non-IID conditions. Combining federated learning with probability calibration, SHAP-based explanation stability, and subgroup fairness evaluation provides a more complete view of model trustworthiness. The results also highlight the potential of federated learning as a privacy-preserving approach for clinical diabetes prediction. However, further validation is needed using real-world multi-institutional healthcare data. Additional privacy-enhancing techniques should be considered.

**Keywords:** Federated Learning, Diabetes Prediction, Machine Learning, Probability Calibration, SHAP, Explainable AI, Fairness, Non-IID Data, FedAvg, FedProx.

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# Contents

|                                                                        |           |
|------------------------------------------------------------------------|-----------|
| <b>Abstract</b>                                                        | <b>3</b>  |
| <b>Acknowledgments</b>                                                 | <b>6</b>  |
| <b>List of Abbreviations and Terms</b>                                 | <b>13</b> |
| <b>1 Introduction</b>                                                  | <b>1</b>  |
| 1.1 Ethical, societal, and sustainability aspects . . . . .            | 5         |
| 1.2 Problem Definition . . . . .                                       | 6         |
| 1.3 Aim and objectives . . . . .                                       | 7         |
| 1.3.1 Aim . . . . .                                                    | 7         |
| 1.3.2 Objectives . . . . .                                             | 7         |
| 1.4 Research questions . . . . .                                       | 7         |
| 1.5 Scope . . . . .                                                    | 8         |
| 1.6 Thesis Outline . . . . .                                           | 9         |
| <b>2 Background</b>                                                    | <b>12</b> |
| 2.1 Diabetes Mellitus . . . . .                                        | 12        |
| 2.2 Binary Classification . . . . .                                    | 12        |
| 2.2.1 Machine Learning Models . . . . .                                | 13        |
| 2.3 Data Preprocessing . . . . .                                       | 17        |
| 2.4 Centralized Learning . . . . .                                     | 18        |
| 2.5 Local-Only Learning . . . . .                                      | 18        |
| 2.6 Federated Learning . . . . .                                       | 19        |
| 2.7 Federated Averaging (FedAvg) . . . . .                             | 19        |
| 2.8 FedProx . . . . .                                                  | 20        |
| 2.9 Non-IID Data in Federated Learning . . . . .                       | 21        |
| 2.10 Communication Cost . . . . .                                      | 22        |
| 2.11 Probability Calibration . . . . .                                 | 23        |
| 2.12 Decision Threshold Selection . . . . .                            | 26        |
| 2.13 Evaluation Metrics . . . . .                                      | 26        |
| 2.14 Explainable AI-SHAP . . . . .                                     | 28        |
| 2.15 Algorithmic Fairness . . . . .                                    | 30        |
| <b>3 Related Work and Literature Review</b>                            | <b>32</b> |
| 3.1 Federated Learning in Healthcare and Diabetes Prediction . . . . . | 32        |
| 3.2 Non-IID Data Heterogeneity in Federated Learning . . . . .         | 35        |
| 3.3 Probability Calibration in Clinical Machine Learning . . . . .     | 36        |

|          |                                                                                                                 |           |
|----------|-----------------------------------------------------------------------------------------------------------------|-----------|
| 3.4      | Explainable AI for Federated Diabetes Prediction . . . . .                                                      | 38        |
| 3.5      | Fairness in Clinical Machine Learning and Federated Learning . . . .                                            | 39        |
| 3.6      | Summary of Research Gaps . . . . .                                                                              | 40        |
| <b>4</b> | <b>Methodology</b>                                                                                              | <b>42</b> |
| 4.1      | Methodological Approach . . . . .                                                                               | 42        |
| 4.2      | Problem Definition and Study Design . . . . .                                                                   | 43        |
| 4.3      | Dataset Description . . . . .                                                                                   | 44        |
| 4.3.1    | Data Availability . . . . .                                                                                     | 45        |
| 4.4      | Data Preprocessing . . . . .                                                                                    | 45        |
| 4.5      | Data Splitting Strategy . . . . .                                                                               | 46        |
| 4.6      | Federated Learning Environment Setup . . . . .                                                                  | 46        |
| 4.7      | Non-IID Simulation Design . . . . .                                                                             | 47        |
| 4.8      | Model Selection and Implementation . . . . .                                                                    | 47        |
| 4.9      | Hyperparameter Configuration . . . . .                                                                          | 48        |
| 4.10     | Federated Aggregation Procedure . . . . .                                                                       | 49        |
| 4.11     | Probability Calibration Procedure . . . . .                                                                     | 49        |
| 4.12     | Decision Threshold Selection . . . . .                                                                          | 50        |
| 4.13     | Explainability Implementation . . . . .                                                                         | 50        |
| 4.14     | Fairness Evaluation Procedure . . . . .                                                                         | 51        |
| 4.15     | Communication Cost Measurement . . . . .                                                                        | 51        |
| 4.16     | Evaluation Metrics Summary . . . . .                                                                            | 52        |
| 4.17     | Experimental Configuration and Reproducibility . . . . .                                                        | 52        |
| <b>5</b> | <b>Results and Analysis</b>                                                                                     | <b>54</b> |
| 5.1      | Introduction and Organization of Results . . . . .                                                              | 54        |
| 5.2      | Comparison 1: Local-Only vs. Federated Learning . . . . .                                                       | 54        |
| 5.2.1    | Local-Only vs. Federated Learning, 5 Clients . . . . .                                                          | 57        |
| 5.3      | Comparison 2: Centralized vs. Federated Learning . . . . .                                                      | 58        |
| 5.4      | Comparison 3: IID vs. Non-IID Data Distribution . . . . .                                                       | 60        |
| 5.4.1    | FedAvg vs. FedProx, 5 Clients . . . . .                                                                         | 61        |
| 5.5      | Comparison 4: Calibrated vs. Uncalibrated Models . . . . .                                                      | 62        |
| 5.6      | Decision Threshold and the Clinical Cost of the Default 0.5 Cutoff .                                            | 64        |
| 5.7      | Comparison 5: Machine Learning vs. Deep Learning Under Federation                                               | 67        |
| 5.8      | Explanation Stability Across Communication Rounds and Clients . .                                               | 69        |
| 5.9      | Fairness Analysis Across Age, BMI, and Sex Subgroups . . . . .                                                  | 71        |
| 5.10     | Communication Cost and Convergence . . . . .                                                                    | 73        |
| <b>6</b> | <b>Discussion</b>                                                                                               | <b>75</b> |
| 6.1      | Overview of This Chapter . . . . .                                                                              | 75        |
| 6.2      | Answering RQ1: Does Federated Learning Match Centralized Performance, and at What Communication Cost? . . . . . | 75        |
| 6.2.1    | What Counts as an “Acceptable” Performance Drop? . . . . .                                                      | 76        |
| 6.2.2    | Robustness to Heterogeneity, in Light of the Literature . . . . .                                               | 77        |
| 6.2.3    | Communication Cost in Practical Terms . . . . .                                                                 | 78        |
| 6.3      | Answering RQ2: Do Calibration and SHAP-Based Explainability Improve Trustworthiness? . . . . .                  | 79        |

---

|          |                                                                                             |           |
|----------|---------------------------------------------------------------------------------------------|-----------|
| 6.3.1    | Why the Calibration Improvement Matters Clinically, Not Just Numerically . . . . .          | 79        |
| 6.3.2    | SHAP Explanation Stability and What It Means for Cross-Institution Clinical Trust . . . . . | 81        |
| 6.3.3    | Fairness: Equalized Odds as the More Effective Approach . .                                 | 81        |
| 6.4      | Findings in Relation to the Broader Literature . . . . .                                    | 82        |
| 6.5      | Contextualizing the Fairness Findings Against Real Diabetes Prevalence                      | 83        |
| <b>7</b> | <b>Conclusions and Future Work</b>                                                          | <b>84</b> |
| 7.1      | Summary of Findings . . . . .                                                               | 84        |
| 7.2      | Contributions . . . . .                                                                     | 85        |
| 7.3      | Validity and Limitations . . . . .                                                          | 85        |
| 7.4      | Future Work . . . . .                                                                       | 86        |
|          | <b>References</b>                                                                           | <b>88</b> |
| <b>A</b> | <b>Generative AI-Assisted Theoretical and Research Discussion</b>                           | <b>96</b> |
| A.1      | Acceptable Performance Drop in Federated Learning . . . . .                                 | 96        |
| <b>B</b> | <b>Generative AI-Assisted Thesis Writing and Structure</b>                                  | <b>97</b> |
| B.1      | Planning Chapter 7 . . . . .                                                                | 97        |
| <b>C</b> | <b>Technical and Implementation Support</b>                                                 | <b>98</b> |
| C.1      | Federated Learning Implementation . . . . .                                                 | 98        |
| C.2      | Coding Doubt-Debugging Federated Aggregation . . . . .                                      | 98        |
| C.3      | Paraphrasing Example . . . . .                                                              | 99        |
| C.4      | Coding Doubt-Non-IID Partitioning . . . . .                                                 | 99        |
| C.5      | Generative AI Disclosure . . . . .                                                          | 99        |

---

## List of Figures

|      |                                                                                                                                                                          |    |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 4.1  | Federated learning in comparison to local learning and centralized learning [62] . . . . .                                                                               | 43 |
| 5.1  | Local-only vs. federated logistic regression under IID data . . . . .                                                                                                    | 56 |
| 5.2  | Local-only vs. federated logistic regression as heterogeneity increases (3 clients) . . . . .                                                                            | 57 |
| 5.3  | Local-only vs. federated logistic regression, 5 clients . . . . .                                                                                                        | 58 |
| 5.4  | Centralized model comparison of ROC-AUC and F1 . . . . .                                                                                                                 | 59 |
| 5.5  | Centralized vs. Federated ROC-AUC(IID) . . . . .                                                                                                                         | 60 |
| 5.6  | FedAvg vs. FedProx across heterogeneity levels for 3 clients. . . . .                                                                                                    | 61 |
| 5.7  | : FedAvg vs. FedProx across heterogeneity, 5 clients . . . . .                                                                                                           | 62 |
| 5.8  | Calibration method comparison using Expected Calibration Error for federated FedAvg Logistic Regression with 3 clients under IID data (lower is better). . . . .         | 64 |
| 5.9  | Fixed vs. F1-Optimal Threshold Performance Metrics . . . . .                                                                                                             | 65 |
| 5.10 | Confusion Matrix Composition by Threshold Strategy . . . . .                                                                                                             | 66 |
| 5.11 | Machine Learning vs. Deep Learning Predictive Performance (Federated) . . . . .                                                                                          | 67 |
| 5.12 | Top SHAP-ranked features for federated logistic regression and MLP under IID conditions. . . . .                                                                         | 69 |
| 5.13 | Cross-client SHAP feature-importance stability across heterogeneity levels. . . . .                                                                                      | 70 |
| 5.14 | Selection Rate by Subgroup-Federated vs. centralized . . . . .                                                                                                           | 71 |
| 5.15 | Demographic parity difference and equalized odds difference by subgroup for federated logistic regression and centralized XGBoost with 3 clients under IID data. . . . . | 72 |
| 5.16 | Convergence and cumulative communication cost by round for FedAvg Logistic Regression with 3 clients under IID data. . . . .                                             | 73 |

---

## List of Tables

|      |                                                                                                                                 |    |
|------|---------------------------------------------------------------------------------------------------------------------------------|----|
| 1    | Machine Learning Models . . . . .                                                                                               | 13 |
| 2    | Training Paradigms . . . . .                                                                                                    | 14 |
| 3    | Data Heterogeneity . . . . .                                                                                                    | 14 |
| 4    | Communication Cost . . . . .                                                                                                    | 14 |
| 5    | Probability Calibration . . . . .                                                                                               | 15 |
| 6    | Decision Threshold . . . . .                                                                                                    | 15 |
| 7    | Explainable AI . . . . .                                                                                                        | 15 |
| 8    | Algorithmic Fairness . . . . .                                                                                                  | 16 |
| 4.1  | Main experimental configuration for the federated experiments. . . .                                                            | 53 |
| 4.2  | Federated hyperparameter search configuration. . . . .                                                                          | 53 |
| 5.1  | Summary of the Experimental Grid . . . . .                                                                                      | 55 |
| 5.2  | Local-only vs. federated logistic regression (IID partition, 3 clients,<br>no calibration, F1-optimal threshold) . . . . .      | 55 |
| 5.3  | Local-only vs. federated logistic regression as heterogeneity increases<br>(3 clients) . . . . .                                | 56 |
| 5.4  | Local-only vs. federated logistic regression, 5 clients . . . . .                                                               | 57 |
| 5.5  | Centralized model comparison (calibration = none, F1-optimal thresh-<br>old) . . . . .                                          | 58 |
| 5.6  | Centralized vs. federated performance for the two federatable models<br>(IID partition) . . . . .                               | 59 |
| 5.7  | FedAvg vs. FedProx across increasing non-IID severity . . . . .                                                                 | 60 |
| 5.8  | : FedAvg vs. FedProx across heterogeneity, 5 clients . . . . .                                                                  | 62 |
| 5.9  | Calibrated vs. uncalibrated comparison (federated FedAvg, logistic<br>regression, 3 clients, IID) . . . . .                     | 63 |
| 5.10 | Fixed 0.5 threshold vs. F1-optimal threshold . . . . .                                                                          | 64 |
| 5.11 | Confusion matrix counts at each threshold (held-out test fold) . . . .                                                          | 65 |
| 5.12 | Logistic regression (ML) vs. shallow MLP (DL) under federated train-<br>ing (FedAvg, 3 clients, IID, no calibration) . . . . .  | 67 |
| 5.13 | Top SHAP-ranked features, federated logistic regression vs. federated<br>MLP (FedAvg, 3 clients, IID) . . . . .                 | 68 |
| 5.14 | Cross-client feature-importance stability (top-5 overlap and Spearman<br>correlation, logistic regression, FedAvg . . . . .     | 70 |
| 5.15 | Fairness gaps by subgroup, federated logistic regression vs. centralized<br>XGBoost . . . . .                                   | 71 |
| 5.16 | Convergence and cumulative communication cost by round for FedAvg<br>Logistic Regression with 3 clients under IID data. . . . . | 73 |

---

|     |                                                                                    |    |
|-----|------------------------------------------------------------------------------------|----|
| 6.1 | How the reviewed literature implicitly treats federated performance gaps . . . . . | 76 |
| 7.1 | This thesis's contributions against the five gaps identified in Chapter 3          | 85 |

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## List of Abbreviations and Terms

| Term                                       | Definition (as used in this thesis)                                                                                                                                                                                                                                        |
|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Logistic Regression</b>                 | A linear model that assigns each clinical feature a weight and passes their weighted sum through a sigmoid function to produce a probability; cheap to communicate, which is why it is one of only two models used in this thesis's federated experiments (Section 2.2.1). |
| <b>Decision Tree</b>                       | A model that repeatedly splits patients into groups based on feature thresholds chosen to best separate diabetic from non-diabetic cases, using Gini impurity to score each split (Section 2.2.1).                                                                         |
| <b>Random Forest</b>                       | An ensemble of many decision trees, each trained on a random bootstrap sample and a random subset of features, whose predictions are averaged to reduce the overfitting a single tree is prone to (Section 2.2.1).                                                         |
| <b>Gradient Boosting Classifier</b>        | An ensemble method that builds trees sequentially, with each new tree specifically trained to correct the errors of the trees built before it (Section 2.2.1).                                                                                                             |
| <b>XGBoost</b>                             | An optimized gradient boosting implementation that adds explicit regularisation and a second-order loss approximation; the strongest centralized model in this thesis by ROC-AUC (Section 2.2.1, Table 5.4).                                                               |
| <b>Shallow Multilayer Perceptron (MLP)</b> | A feedforward neural network with one hidden layer, able to capture non-linear feature interactions that logistic regression cannot; the only neural network used in this thesis, and one of the two models compatible with federated averaging (Section 2.2.1).           |

Table 1: Machine Learning Models

| Term                                | Definition (as used in this thesis)                                                                                                                                                                                                                    |
|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Centralised Learning</b>         | Training a single model on all institutions' patient data pooled together in one place; used in this thesis only as a theoretical performance ceiling, since real regulations (GDPR, HIPAA, India's DPDPA 2023) prohibit it in practice (Section 2.4). |
| <b>Local-Only Learning</b>          | Each simulated institution trains its own private model using only its own data, with nothing shared between institutions; the privacy-preserving baseline this thesis tests federated learning against (Section 2.5).                                 |
| <b>Federated Learning (FL)</b>      | A training paradigm in which institutions collaboratively train one shared model by exchanging only model parameters, never raw patient data, with a central server aggregating each round's updates (Section 2.6).                                    |
| <b>FedAvg (Federated Averaging)</b> | The foundational federated algorithm: each client trains locally for several epochs, then the server combines every client's updated parameters via a weighted average (weighted by each client's data volume) (Section 2.7).                          |
| <b>FedProx</b>                      | A variant of FedAvg that adds a proximal penalty term pulling each client's local model back toward the current global model during training, intended to reduce client drift under heterogeneous data (Section 2.8).                                  |
| <b>Client Drift</b>                 | The tendency of clients' local models to diverge from one another during federated training when their local data distributions differ, which can degrade the quality of the aggregated global model (Section 2.9).                                    |

Table 2: Training Paradigms

| Term                                                        | Definition (as used in this thesis)                                                                                                                                                                                                      |
|-------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>IID (Independently and Identically Distributed)</b>      | The assumption that every client's local data is a representative random sample of the same overall population, the easiest and least realistic condition for federated learning (Section 2.9).                                          |
| <b>Non-IID Data</b>                                         | The realistic condition in which different institutions see systematically different patient populations (for example, different age or prevalence mixes), which this thesis simulates using Dirichlet partitioning (Section 2.9).       |
| <b>Dirichlet Distribution / <math>\alpha</math> (alpha)</b> | The statistical distribution used to simulate non-IID client data splits; its concentration parameter $\alpha$ controls heterogeneity severity, where a lower $\alpha$ (e.g. 0.1) means more skewed, less similar clients (Section 2.9). |

Table 3: Data Heterogeneity

| Term                      | Definition (as used in this thesis)                                                                                                                                                                                                         |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Communication Cost</b> | The total network traffic (in bytes) generated by exchanging model parameters between clients and the server across all federated training rounds; this thesis is the first federated diabetes study to measure it directly (Section 2.10). |
| <b>Convergence Round</b>  | The training round beyond which further rounds no longer meaningfully improve model performance, marking the point past which continued training only adds communication cost without benefit (Section 2.10).                               |

Table 4: Communication Cost

| Term                                       | Definition (as used in this thesis)                                                                                                                                                                                                      |
|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Probability Calibration</b>             | The property that a model's predicted risk percentages match real-world outcome frequencies: for example, among all patients given a 70% predicted risk, roughly 70% should actually have diabetes (Section 2.11).                       |
| <b>Reliability Diagram</b>                 | A plot comparing a model's average predicted probability against the observed proportion of positive outcomes within probability bins, used to visualise calibration quality (Section 2.11).                                             |
| <b>Brier Score</b>                         | The mean squared difference between predicted probabilities and true binary outcomes; lower is better, and it penalises both poor discrimination and poor calibration (Section 2.11).                                                    |
| <b>Expected Calibration Error (ECE)</b>    | The primary calibration metric in this thesis: the weighted average gap between a model's stated confidence and its actual accuracy across probability bins; 0 means perfectly calibrated (Section 2.11).                                |
| <b>Log Loss</b>                            | A metric that heavily penalises confident but wrong predictions, rewarding models that express appropriate uncertainty (Section 2.11).                                                                                                   |
| <b>Sigmoid Calibration (Platt Scaling)</b> | A calibration method that fits a simple two-parameter logistic correction to a model's raw scores; works well for S-shaped miscalibration patterns but underperformed isotonic calibration throughout this thesis (Section 2.11).        |
| <b>Isotonic Regression Calibration</b>     | A more flexible, non-parametric calibration method that fits any non-decreasing correction function to a model's raw scores; the calibration method recommended by this thesis's results (Section 2.11).                                 |
| <b>Global Calibration</b>                  | Calibration fitted on one shared, centrally pooled calibration dataset; the best-case calibration ceiling, but not privacy-preserving (Section 2.11).                                                                                    |
| <b>Federated Calibration</b>               | A privacy-preserving calibration procedure in which each client fits its own local calibrator, and predictions are calibrated by averaging across all clients' calibrators; patient data never needs to leave any client (Section 2.11). |

Table 5: Probability Calibration

| Term                                          | Definition (as used in this thesis)                                                                                                                                                                                                 |
|-----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Fixed Threshold (<math>t = 0.5</math>)</b> | The conventional default decision rule, classifying any patient with predicted risk $\geq 50\%$ as positive; shown in this thesis to badly under-detect diabetic patients on this imbalanced dataset (Section 2.12, Section 5.5.1). |
| <b>F1-Optimal Threshold</b>                   | A threshold chosen to maximise the F1 score on a held-out calibration split, then applied unchanged to the test set; the threshold strategy this thesis recommends for this dataset's class imbalance (Section 2.12).               |

Table 6: Decision Threshold

| Term                                                  | Definition (as used in this thesis)                                                                                                                                                                                        |
|-------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>SHAP (SHapley Additive exPlanations)</b>           | A game-theoretic method that decomposes any individual prediction into the contribution each input feature made toward it, used in this thesis to explain both federated and centralized model predictions (Section 2.14). |
| <b>Global Feature Importance</b>                      | The average magnitude of a feature's SHAP contribution across an entire patient population, used to identify which clinical indicators matter most overall (Section 2.14).                                                 |
| <b>Explanation Stability</b>                          | How consistently a model's SHAP-based feature ranking stays the same across federated communication rounds, measured in this thesis using Spearman rank correlation (Section 2.14, Section 5.7).                           |
| <b>Cross-Client Feature Agreement (Top-k Overlap)</b> | The proportion of top-ranked features two federated clients agree on, used to check whether clients at different simulated institutions would be shown consistent explanations (Section 2.14, Section 5.7).                |

Table 7: Explainable AI

| Term                                       | Definition (as used in this thesis)                                                                                                                                                                                                                   |
|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Algorithmic Fairness</b>                | The study of whether a model's predictions or errors differ systematically across demographic subgroups such as age, BMI, or sex (Section 2.15).                                                                                                      |
| <b>Demographic Parity Difference (DPD)</b> | The gap between the highest and lowest positive-prediction rate across subgroups; a large DPD can reflect genuine prevalence differences rather than bias, which is why this thesis reports it alongside EOD rather than alone (Section 2.15).        |
| <b>Equalized Odds Difference (EOD)</b>     | The gap in true-positive and false-positive rates across subgroups; the fairness metric this thesis treats as more clinically meaningful, since it allows for real prevalence differences while still requiring equal error treatment (Section 2.15). |

Table 8: Algorithmic Fairness

Diabetes mellitus is one of the most important chronic health concerns of today's world. It affects about 537 million individuals worldwide and is expected to reach 783 million by 2045 if current trends continue. The disease occurs primarily in two forms: Type 1 diabetes, in which the immune system destroys insulin-producing cells. Type 2 diabetes, in which the body becomes increasingly resistant to insulin. Type 2 diabetes accounts for 90-95 percent of all cases and is closely associated with changeable lifestyle factors such as obesity, physical inactivity, and poor eating habits. If left untreated, it gradually destroys organs over time. It causes serious problems such as cardiovascular disease, renal failure, nerve damage, and eyesight loss. Because many cases go unnoticed for long periods of time. Therefore, early and accurate diabetes risk identification is important for preventing complications and improving health outcomes. [5].

Diabetes diagnosis requires the analysis of multiple clinical and demographic variables. These include fasting glucose concentration, body mass index, blood pressure, insulin levels, and age. As patient data volumes increase, analyzing these variables manually becomes more difficult for healthcare professionals. Machine learning has emerged as an effective method for assisting with early diagnosis. It can automatically understand complex correlations between these variables and diabetes outcomes. Classical models such as logistic regression, decision trees, random forests, and gradient boosting have shown strong and interpretable performance on structured diabetes datasets [49] [5]. Deep learning models such as multilayer perceptrons can also capture non-linear relationships with competitive predictive accuracy [58] [40]. Despite these advancements, the great majority of published diabetes prediction models rely on centralized data collection. Healthcare organizations operate under strict data protection and privacy requirements that can restrict the sharing and centralization of patient data. The applicable requirements depend on the jurisdiction, legal basis, purpose of processing, and safeguards used.

### • *The Privacy Barrier in Healthcare Data*

Healthcare organizations are subject to strict data protection laws around the world that prevent sharing of patient data beyond organizational boundaries. In Europe, the General Data Protection Regulation establishes rigorous criteria for the authorized processing of personal health data. It also places restrictions on cross-institutional data transfers. In the United States, the Health Insurance Portability

and Accountability Act provides standards for safeguarding individually identifiable health information. India's Digital Personal Data Protection Act 2023 provides safeguards for personal data [42] [64]. These requirements can make cross-institutional data sharing complex and may create practical and regulatory barriers to centralizing patient records. This makes it difficult to design accurate and generalizable prediction models based on data from multiple healthcare organizations.

This generates a basic conflict at the core of healthcare AI. The data required to build effective, generalizable diabetes prediction models is spread among hundreds of institutions. Each institution with a limited and potentially biased view of the patient population. Machine learning models trained on data from a single institution usually have low generalizability. They might include demographic biases specific to that institution, decreasing their effectiveness and fairness when used in larger clinical contexts [56]. The legislative system that safeguards patient privacy also restricts large-scale data collaboration, which would enable AI-powered diabetes prediction to be genuinely reliable on a population scale. Resolving this conflict is the primary motivation for the federated learning approach used in this research.

### • *Federated Learning as a Solution*

Federated learning was introduced by McMahan et al. (2017) [38] through the Federated Averaging method. It provides a principled solution to the privacy-performance issue. Rather than centralizing patient records, federated learning allows many institutions, known as clients, to train a shared global model. Each client trains a local model on its own private dataset. The resulting model weights are then sent to a central aggregation server, which receives an updated global model in exchange. This iterative procedure is repeated across numerous communication cycles until the global model converges. Throughout the training process, the patient records remain within the source institution. [54].

Over the last five years, the healthcare research community has become increasingly enthusiastic about this model. Nguyen et al. [43] conducted a comprehensive review of over 150 federated healthcare research studies and documented its successful application in health data management, remote patient monitoring, medical imaging, and disease prediction. Pati et al. [47], examined privacy protection in federated healthcare AI. Their study showed that federated learning considerably decreases privacy risks while enabling collaborative model development across institutions. For diabetes prediction, Islam et al. [23] demonstrated that federated models trained on real clinical electronic health record data from different healthcare organizations can reach performance comparable to centralized training. Tang et al. [60] applied this conclusion to real cross-provincial primary care data from Canada, with no patient data leaving the originating province. Su et al. [59] demonstrated that secure federated protocols for diabetes risk prediction may prevent gradient leakage attacks while retaining competitive accuracy. These works collectively demonstrate that federated learning is technically viable for diabetes prediction. However, they also have a major common drawback, which this thesis addresses directly.

- *The Trustworthiness Problem*

The existing federated diabetes prediction study assesses its models using classification accuracy metrics such as accuracy, F1-score, and AUC. These metrics assess whether a model predicts the correct diagnosis. However, they do not address whether the model’s estimated probabilities are reliable, if its reasoning is understandable to doctors, or whether it treats different patient groups fairly. Probability reliability, explainability, and fairness are exactly what help decide whether a model can be trusted and used in a clinical setting. However, they are consistently absent from the existing federated diabetes research.

Probability reliability is important because clinical choices for diabetes treatment are made using predicted risk scores rather than binary predictions. A physician’s decision to start preventive treatment is based on whether a predicted 75 percent risk actually means 75 percent, not 55 or 90 percent. This is known as calibration. Silva Filho et al. [57] demonstrated that modern machine learning classifiers such as Random Forest and Gradient Boosting are systematically overconfident in their probability estimates. These models may produce probabilities that are close to extremes instead of reflecting true outcome frequencies. In healthcare applications prediction confidence is as important as prediction accuracy. Overconfidence or incorrect probability estimations can result in hazardous clinical interventions or missed early diagnosis [25]. Despite this, every federated diabetic study analyzed in this thesis [23], [60], [59], [17], [12], [70], and [19] completely ignores probability calibration.

Explainability is also important in healthcare. Medical practitioners must first comprehend and trust AI-generated predictions before acting on them. Black-box models that provide accurate predictions, but their lack of interpretability can reduce clinician confidence and create challenges for real-world implementation [8]. Regulatory frameworks such as GDPR also highlight the importance of explainability. Article 22 provides a legal basis for explanation in automated decision-making systems, making explainability an important consideration in European healthcare settings. SHAP, or Shapley Additive Explanations, has emerged as the main post hoc explainability method. It provides feature contribution scores that can be interpreted at both the local instance and global population levels. In a federated context, SHAP must be computed locally at each client. This raises concerns about explanation consistency when data distributions differ across clients. Ducange et al. [15] highlighted that non-IID conditions potentially produce contradictory clinical guidance from the same global model across different institutions. Among existing federated diabetes studies, only Gawande et al. [17] uses an explainability technique. However, their study used LIME rather than SHAP and does not measure explanation consistency between clients or rounds.

Fairness is another important consideration in diabetes prediction. A model that performs well on aggregate measures may behave differently across demographic subgroups defined by age, BMI, or gender. Equalized odds were defined as the recommended fairness metric for clinical prediction tasks by Xu et al. [69]. Chen et

al. [13] observed that federated learning can serve as a bias reduction method by allowing training across demographically varied institutional datasets. Despite this, no existing federated diabetes study assesses demographic parity or equalized odds across any subgroups. This creates a gap in clinical readiness assessments for all published approaches.

### • *Identified Research Gap*

The lack of probability calibration, explainability, and fairness evaluations in federated diabetes research creates a gap between technical performance and clinical availability. Some studies look into explainable AI, but only in centralized settings. Alam et al. [6] used SHAP to predict diabetes without a federated component. Tasin et al. [61] demonstrated strong SHAP and LIME explanations on the Pima Indians dataset without considering privacy preservation. Others focus solely on federated learning, discarding explainability and calibration completely. Kumar et al. [29] provides the closest existing integration, combining federated learning, SHAP, and isotonic calibration for liver disease prediction. However, their work addresses a different disease domain. It also did not compare classical ML models to deep learning under federated calibration. It does not assess explanation stability across communication rounds and does not include a fairness analysis. In diabetes prediction, no study has coupled federated learning with probability calibration, SHAP-based explainability, and fairness evaluation in one framework. This thesis addresses this gap by evaluating these aspects under both IID and non-IID data distributions.

Another weakness recognized by the existing literature is the constant lack of communication cost analysis in federated diabetic studies. Communication overhead was identified as one of the five unresolved challenges in healthcare federated learning by Nguyen et al. [43]. McMahan et al. [38] noted that total communication cost depends on factors such as model size, client count, and the number of rounds to convergence. There is no previous federated diabetes study that reports this cost, making it hard to determine the practical communication cost required for deployment.

### • *Contributions*

This thesis addresses all five identified gaps by proposing and evaluating TrustFed Diabetes, a federated learning framework for diabetes prediction. The framework evaluates model trustworthiness across five dimensions simultaneously. The framework trains five classical machine learning models: logistic regression, random forest, gradient boosting classifier, decision tree, and XGBoost. A shallow multilayer perceptron is also evaluated. FedAvg and FedProx are used for federated aggregation. Isotonic regression and sigmoid calibration applied locally at each client. SHAP was computed independently at each client to preserve data locality. The experiments are conducted under IID and non-IID conditions. Non-IID data is created using Dirichlet partitioning at three heterogeneity levels. The experiments use the CDC Diabetes Health Indicators dataset, which contains 253,680 records [3]. The five evaluation dimensions are predictive performance, probability calibration quality, SHAP

explanation stability across federated rounds, subgroup fairness across age, BMI, and sex categories, and total communication cost to convergence.

## 1.1 Ethical, societal, and sustainability aspects

- **Ethical Aspects:** This thesis uses sensitive medical data for the prediction of diabetes. Which poses significant ethical considerations, particularly about patient privacy, confidentiality, and data security. Medical data cannot be freely shared due to legal and ethical constraints. For this reason, this study employs a federated learning approach. In this approach, patient data is kept inside local healthcare institutions such as hospitals or clinics. The data is never moved to a central server. Instead, only model updates (learned patterns) are exchanged between institutions. This helps protect patient identity and ensure compliance with data protection standards such as GDPR and HIPAA.

Although data is not stored centrally, the system still enables collaborative learning across many healthcare facilities. This makes the approach suitable for privacy-sensitive medical applications. Explainable AI approaches are used in the model to increase transparency. This lets clinicians understand why a prediction (such as diabetes risk) is produced, rather than viewing the model as a "black box." This builds trust and promotes better clinical decision-making. Probability calibration is used to guarantee that the predicted probabilities are accurate and realistic. This reduces the risk of making overconfident or misleading predictions, which is especially significant in healthcare. Overall, the combination of federated learning, explainability, and calibration leads to a more transparent and reliable diabetes prediction system.

- **Societal considerations:** From a societal perspective, the recommended approach encourages early detection of diabetes. Early detection may help reduce major long-term health problems and improve overall patient outcomes. Federated learning allows healthcare institutions to collaborate without revealing sensitive patient data. This enables the development of more robust and generalizable models. Such models benefit both major hospitals with advanced resources and smaller or resource-limited healthcare organizations. Furthermore, the study takes into account real-world variances in patient data between hospitals (known as non-IID data). Evaluating the model under these settings helps assess its ability to handle population diversity while reducing potential bias. This ensures that projections are fair and reliable across patient groups, resulting in more equitable healthcare decision-making.
- **Sustainability Aspects:** The thesis emphasizes sustainability by using efficient classical machine learning models and a shallow neural network. These

models require fewer computational resources than big deep learning architectures. Federated learning promotes sustainability by minimizing the requirement for centralized data storage. It avoids the continual transmission of raw sensitive information to a central server. However, it adds communication cost due to iterative model updates between clients and servers. Despite this, the decentralized nature of federated learning enables scalable and privacy-preserving deployment of AI systems in healthcare. This makes it a potential approach for long-term use with less reliance on centralized infrastructure and data storage systems.

## 1.2 Problem Definition

The rising global prevalence of diabetes has made early and accurate diagnosis a major concern in modern healthcare. However, developing successful AI-driven prediction systems involves major hurdles that extend beyond obtaining high accuracy. One of the most serious concerns is data privacy. Patient health records are extremely sensitive, and healthcare organizations are legally and ethically required to keep them secure. Traditional machine learning algorithms require centralizing data from various sources to train a model. This can create privacy hazards and frequently violates rules such as GDPR and HIPAA. As a result, many institutions are hesitant to share their data. This limits the amount and variety of data accessible for training reliable prediction models.

Even when models are created, they frequently behave as black boxes. These models make predictions but provide no explanation for why a certain decision was made. This is a significant issue in the clinical setting. A doctor cannot act properly on a forecast that they do not understand. Patients need to know how decisions concerning their health are made. The lack of interpretability reduces trust in AI systems, limiting their use in real-world healthcare situations.

Beyond interpretability, probability reliability is another crucial issue. Many machine learning models generate predicted probabilities that do not accurately reflect a patient's actual risk of diabetes. For example, a model may assign a 90% risk score while the true probability is much lower. Such miscalibration can lead to inappropriate decisions, especially when risk estimates influence treatment planning. Furthermore, real-world healthcare data is rarely consistent. Different hospitals serve distinct patient populations, resulting in non-uniform or non-IID data distributions among institutions. Most existing research evaluates models under ideal, balanced data conditions, which do not reflect the reality of clinical settings.

This thesis addresses all of these challenges by introducing a federated learning architecture. It enables various healthcare institutions to train diabetes prediction models together without ever exchanging raw patient data. The framework also integrates explainable AI approaches and probability calibration methods. The aim is to create a system that is not only accurate but also provides more reliable probabilities, in-

interpretable explanations, and fairness information that could support future clinical evaluation.

## 1.3 Aim and objectives

### 1.3.1 Aim

The purpose of this thesis is to build and construct a federated learning framework for diabetes prediction. The framework simulates collaborative training across multiple clients while keeping raw data local. The study compares the performance of classical machine learning models and deep learning approaches in both IID and non-IID data sets. It also incorporates explainable AI techniques to improve model transparency and probability calibration methods to ensure that predicted risk scores are more reliable and clinically meaningful.

### 1.3.2 Objectives

- To design and implement a federated learning framework that keeps raw data local so that simulated healthcare clients can collaboratively develop diabetes prediction models without exchanging raw patient data.
- To create and evaluate a shallow multilayer perceptron and traditional machine learning models for diabetes risk classification in both IID and non-IID data distribution scenarios.
- To incorporate explainable AI methods based on SHAP to improve model transparency and guarantee prediction stability throughout multiple training rounds.
- To assess model fairness across demographic subgroups using demographic parity and equalized odds metrics, and to increase the accuracy of predicted risk probabilities using isotonic regression and sigmoid-based probability calibration.
- To evaluate the federated framework's communication efficiency and provide a unified framework that evaluates predictive performance, calibration, explainability, fairness, and communication efficiency within a controlled simulated setting.

## 1.4 Research questions

**RQ1.** *How closely can federated learning match centralized performance while determining the clinically acceptable performance drop and the communication cost required for convergence?*

**Motivation:** Early and accurate diabetes prediction requires huge amounts of patient data from numerous healthcare institutions. However, data security rules such as GDPR, HIPAA, and India’s DPDP 2023 explicitly prohibit centralized data exchange across organizational borders. Federated learning addresses this by allowing collaborative model training via model parameter sharing rather than raw patient data. This protects privacy while allowing institutions to benefit from each other’s data. However, the potential future use of federated learning for clinical deployment depends on several unanswered questions. How closely does its predictive performance match that of a centralized model trained on combined data? What level of performance drop in terms of AUROC and F1 is acceptable without compromising clinical utility? How much communication cost in cumulative bytes is required to achieve stable model convergence? Real-world hospital data is often consistently distributed across institutions. Different sites serve diverse patient demographics, resulting in non-IID data situations that can reduce federated model performance when compared to optimal IID settings. Before considering federated learning as a potential alternative for future clinical deployment, it is important to understand whether it maintains performance under both IID and non-IID settings. The communication cost required to achieve stable model convergence should also be considered.

**RQ2.** *How does isotonic calibration and SHAP-based explainability improve the trustworthiness of federated diabetes prediction in terms of calibration error, explanation stability, and fairness?*

**Motivation:** Accurate diabetes prediction alone is insufficient for potential clinical use. Healthcare professionals must trust the model’s results in order to act on them. A model that produces systematically overconfident risk probabilities may result in unnecessary healthcare interventions. whereas a model whose feature explanations shift unexpectedly between several training rounds cannot be reliably evaluated by clinicians. Fairness is another important concern. A model that performs well overall but acts differently across age, BMI, and sex subgroups. This can raise fairness concerns in a clinical context where equal treatment for all patient groups is essential. Despite these issues, no existing federated diabetes study has examined probability reliability, explanation consistency, and subgroup fairness within a single framework. This research addresses this gap by examining how isotonic regression calibration improves probability trustworthiness via ECE and Brier Score. It also evaluates the stability of SHAP-based explanations across federated rounds and examines fairness across age, BMI, and sex subgroups as measured by equalized odds difference.

## 1.5 Scope

This thesis focuses on developing and testing a federated learning framework for diabetes prediction using the CDC Diabetes Health Indicators dataset [3]. The dataset is a publicly available health-survey dataset containing 253,680 records with 21 features. It covers health indicators such as blood pressure, BMI, age, physical activity, and general health status. The experiments are conducted in a simulated federated

environment with 3 and 5 clients representing healthcare institutions. Data is partitioned under both IID and non-IID conditions using a Dirichlet distribution at concentration levels  $\alpha = 0.1, 0.5, \text{ and } 1.0$  to simulate realistic inter-institutional data heterogeneity.

The study compares five conventional machine learning models: logistic regression, random forest, gradient boosting classifier, decision tree, and XGBoost with a shallow multilayer perceptron as the deep learning baseline. Two federated aggregating methods, FedAvg and FedProx, are compared across all setups. To improve the dependability of estimated risk probabilities, probability calibration is used at the client level. This includes isotonic regression and Platt scaling. SHAP is employed for post-hoc explainability. Each client computes both local instance-level and global feature-level attributions independently in order to preserve data locality.

The evaluation framework simultaneously covers five dimensions. Predictive performance measured through AUROC, F1, precision, recall, and accuracy. Probability reliability is measured through Expected Calibration Error and Brier Score. Explanation stability is measured through SHAP feature importance rank correlation across federated communication rounds. Subgroup fairness was measured through demographic parity difference and equalized odds difference across age, BMI, and sex.

This thesis does not include actual hospital facilities, live patient data, or clinician surveys. All results are based on controlled simulation tests using the CDC dataset. The scope is purposely confined to binary diabetes classification on structured tabular clinical data. It does not include multi-modal data, longitudinal patient records, continuous glucose monitoring, or other chronic disease prediction tasks. Advanced privacy-enhancing approaches such as differential privacy, secure multi-party computation, and homomorphic encryption are recognized as essential directions, but they fall outside the scope of this work. Similarly, big deep learning architectures beyond shallow MLPs, personalized federated learning techniques, and real-time federated deployment across physical healthcare networks are proposed as future research directions rather than being addressed in this thesis.

## 1.6 Thesis Outline

- **Chapter 1 Introduction**

The first chapter provides an introduction to the research topic by discussing the clinical importance of early diabetes prediction. It also discusses the regulatory challenges that prevent centralized data sharing among healthcare institutions under GDPR, HIPAA, and India's DPDPA 2023. It defines the central issue that existing federated diabetes prediction frameworks focus solely on accuracy while ignoring probability reliability, explainability, and fairness. This chapter also presents the two research questions, their motivations, the study's goals and objectives, and their motivations. The chapter finishes by explaining the study's significance in the context of privacy-preserving clinical AI and

reliable medical decision assistance.

- **Chapter 2 Background**

The theoretical framework and mathematical foundations required to comprehend the procedures and outcomes of this thesis are discussed in Chapter 2. The clinical background of diabetes mellitus is covered. This chapter also discusses machine learning models for binary diabetes classification, such as logistic regression, random forest, gradient boosting, decision tree, XGBoost, and the shallow multilayer perceptron. The mathematical foundations of federated learning, such as the FedAvg and FedProx algorithms, are also discussed. Non-IID data simulation using Dirichlet partitioning is covered. Probability calibration theory is discussed, such as isotonic regression and sigmoid calibration. SHAP-based explainability and algorithmic fairness metrics, such as equalized odds and demographic parity, are also covered. Finally, communication cost measurement in federated systems is also covered.

- **Chapter 3 Related Work**

Chapter 3 presents the related work and literature review, covering federated learning in healthcare and diabetes prediction, non-IID data, probability calibration, explainability, and fairness. The review is organized into four topical clusters: federated learning in healthcare and diabetes prediction, non-IID data heterogeneity and aggregation algorithms, probability calibration in clinical machine learning, and explainable AI in federated healthcare systems. Each cluster is closely analyzed to see what existing research contributes and what remains unclear. The chapter concludes with a comprehensive comparison table of all reviewed studies. It also provides a synthesis of the five specific knowledge gaps addressed by this thesis. These include the absence of calibration in federated diabetes studies, the lack of integrated FL and XAI and calibration frameworks, unmeasured SHAP explanation stability across federated rounds, no ML versus DL comparison under federated calibration, and the lack of fairness evaluation in federated diabetes prediction.

- **Chapter 4 Methodology**

The research method is fully described in technical detail in Chapter 4. The CDC Diabetes Health Indicators dataset is presented along with its preprocessing. This includes class imbalance handling and feature scaling. The chapter also describes federated learning environment setup with 3 and 5 simulated clients. IID and non-IID data partitioning using Dirichlet distribution at  $\alpha = 0.1, 0.5, \text{ and } 1.0$ . Local model training and hyperparameter optimization at each client. Probability calibration using isotonic regression and sigmoid calibration. SHAP explainability is integrated at the client level. Federated aggregation using FedAvg and FedProx. Finally, the chapter presents a five-dimensional evaluation framework covering predictive performance, calibration quality, explanation stability, subgroup fairness, and communication cost.

- **Chapter 5 Results and Analysis**

Chapter 5 provides and examines the research findings. The results are arranged around five evaluation dimensions and five comparison analyses. It

compares local and federated learning to determine the advantages of collaborative learning over isolated local training in terms of accuracy, F1-score, and ROC-AUC. It then compares centralized and federated training to assess the privacy-performance trade-off. This comparison examines how closely federated training matches a pooled-data centralized baseline in terms of accuracy, F1-score, and ROC-AUC while maintaining total data localization. The calibrated versus uncalibrated comparison assesses the improvement in prediction reliability brought about by isotonic regression and sigmoid calibration. Brier score and expected calibration error are used for comparison. The IID versus non-IID comparison assesses the federated framework's robustness in the face of realistic hospital data heterogeneity at all three Dirichlet concentration levels. It also examines performance and calibration consistency. Finally, the machine learning versus deep learning comparison looks at the differences in predictive behavior, calibration quality, and SHAP-based interpretability between the five conventional ML models and the shallow MLP in federated training.

- **Chapter 6 Discussion**

In Chapter 6, the results are discussed in relation to the expectations set out in the literature study and how they address the two research questions. It discusses the implications of SHAP explanation stability across federated rounds for clinical trust in federated systems. It also contextualizes the fairness findings across age, BMI, and sex subgroups in relation to actual demographic differences in diabetes prevalence. The centralized versus federated performance gap against the acceptable threshold for clinical deployment established in RQ1. The chapter also explains the significance of the calibration improvement across all model types in terms of clinical probability reliability. Finally, the chapter considers the limitations of the simulated environment and the generalizability of outcomes outside of the CDC dataset. It links each result to the five knowledge gaps noted in Chapter 3.

- **Chapter 7 Conclusions and Future Work**

Chapter 7 concludes the thesis by summarizing the answers to both research questions and presenting the main contributions of the work. It presents the framework as a unified federated approach for diabetes prediction that addresses privacy, calibration, explainability, fairness, and communication cost measurement. It recognizes the limitations of using a single public dataset in a simulated rather than real federated deployment. Finally, it proposes concrete directions for future research. These include real multi-institutional deployment across physical hospitals, integration of differential privacy and secure multi-party computation, personalized federated learning strategies for strongly non-IID settings, extension to other chronic disease datasets, and longitudinal patient risk.

### 2.1 Diabetes Mellitus

Diabetes mellitus is a chronic condition that changes how the body processes blood sugar. When you eat, your blood glucose levels rise, and the pancreas responds by releasing insulin. Insulin functions like a key, allowing cells to absorb glucose and use it for energy. In diabetes, this mechanism fails. Either the pancreas is not producing enough insulin, or the body's cells have developed resistance to its effects, or both. As a result, glucose accumulates in the bloodstream rather than being absorbed. This can cause long-term damage to organs throughout the body. Diabetes is especially dangerous for public health because it develops quietly. A person can have dangerously high blood glucose levels for years without experiencing any symptoms. When symptoms arise, increased thirst, frequent urination, exhaustion, and hazy vision and significant problems such as cardiovascular disease, kidney failure, peripheral nerve damage, and vision loss may have already occurred [5].

#### *Why Prediction Matters*

Early identification is not just medically beneficial. It may change a patient's path. Pre-diabetic interventions, such as lifestyle changes or early medication, might delay or even prevent the onset of complete diabetes. They can also lower the risk of serious consequences [49]. The problem is to identify which people are at high risk before they exhibit symptoms. This is where machine learning comes in. A well-trained model can identify high-risk individuals by learning complex, non-linear relationships between clinical measurements such as blood glucose levels, BMI, blood pressure, age, and family history and known diabetes outcomes in large patient datasets [58]. The issue, however, is that the patient data required to train robust models is kept within hospitals and clinics that are prohibited by law from sharing it. That structural challenge serves as the primary reason for everything in this thesis.

### 2.2 Binary Classification

Diabetes prediction is modeled as a binary classification task. Each patient is defined by a collection of clinical measures (referred to as a vector  $x$ ). The model's purpose is to assess the patient's likelihood of having diabetes. Formally, the model learns a function  $f$  that relates clinical measurements to probabilities.

$$p = f(x) \in [0, 1] \quad (2.1)$$

A result of 0.9 indicates that the model believes there is a 90% probability this patient has diabetes. A number of 0.15 indicates that it believes there is just a 15% possibility.

### ***Turning Probabilities Into Decisions***

To make an actionable clinical decision whether to refer for more testing or not, the likelihood must be transformed into a binary label. This is done using a decision threshold ( $t$ ):

$$\hat{y} = \begin{cases} 1 & \text{if } p \geq t \\ 0 & \text{if } p < t \end{cases} \quad (2.2)$$

where  $\hat{y} = 1$  means the model predicts diabetes positive and  $\hat{y} = 0$  means diabetes negative.

Choosing  $t$  is not a machine learning challenge but rather a clinical one. A lower threshold identifies more cases (greater recall), but it also flags more healthy patients (lower precision). A higher threshold decreases false alerts but risks overlooking people who truly require care. These trade-offs will be handled differently in different medical situations, such as mass screening programs versus specialist referral pathways. The methods chapter goes into much information about this threshold selection problem.

## **2.2.1 Machine Learning Models**

This thesis compares six model families: logistic regression, decision tree, random forest, gradient boosting, XGBoost, and a shallow multilayer perceptron (MLP). The six models were chosen to provide a realistic comparison of various machine-learning approaches to diabetes prediction. Logistic regression was chosen as an interpretable baseline because it gives a straightforward reference for evaluating more sophisticated models. Decision trees were used to capture non-linear correlations while staying relatively simple to comprehend, whereas random forests were used to see if mixing several decision trees may result in more robust predictions. Gradient boosting and XGBoost are more advanced boosting-based techniques, allowing the study to assess the performance of progressively powerful tree-based models, with XGBoost serving as a robust centralized benchmark. Finally, the shallow MLP was included to represent a neural-network approach and to determine whether a more adaptable model may deliver competitive performance while staying suited for federated parameter aggregation. Overall, the selection was designed to include a variety of model types, levels of complexity, and suitability for federated learning, providing a balanced foundation for comparing centralized versus federated diabetes prediction.

### ***Logistic Regression***

Logistic regression is one of the oldest and most commonly used classification algorithms in medicine. The idea is straightforward. Each clinical indicator (blood pressure, glucose level, BMI, and so on) is assigned a weight that represents how strongly it influences the prediction towards or away from diabetes. The model calculates a weighted total of all feature values. The sigmoid function converts this sum into a probability between 0 and 1:

$$p = \frac{1}{1 + e^{-(w^T x + b)}} \quad (2.3)$$

During training, the model adjusts its weights to minimize how inaccurate its probability estimations are throughout the training set. This is measured using the binary cross-entropy loss. The loss increases dramatically when the model is confidently incorrect.

$$L = -\frac{1}{n} \sum_{i=1}^n [y_i \log p_i + (1 - y_i) \log(1 - p_i)] \quad (2.4)$$

Logistic regression is particularly well-suited to federated learning for practical reasons. The entire model consists of only one number per feature plus a bias term. Sending these numbers over a network is essentially cost-free [38]. Its fundamental disadvantage is that it assumes the distinction between diabetic and non-diabetic patients can be shown as a straight line through the feature space. This is an approximation that ignores complicated feature interactions. The sklearn implementation is utilized in both centralized and local experiments. A unique NumPy implementation is created for federated experiments. This allows direct gradient computation in accordance with the federated averaging protocol.

### *Decision Tree*

A decision tree learns by continuously breaking down the patient dataset into smaller groups based on feature thresholds. At each split, it asks which attribute and which threshold for that feature results in the clearest differentiation of diabetic and non-diabetic patients. The quality of a split is measured by Gini impurity. Gini impurity measures how mixed diabetic and non-diabetic individuals are in a group. During training, the decision tree evaluates several features and threshold values before using Gini impurity to determine the appropriate split. A lower Gini impurity indicates that the resulting groups are more clearly separated as diabetic and non-diabetic patients. The tree continues to split to create the final Decision Tree, which is subsequently tested on the test data.

$$\text{Gini}(S) = 1 - \sum_{c \in \{0,1\}} p_c^2 \quad (2.5)$$

$$\Delta \text{Gini} = \text{Gini}(S) - \frac{|S_L|}{|S|} \text{Gini}(S_L) - \frac{|S_R|}{|S|} \text{Gini}(S_R) \quad (2.6)$$

The tree expands until each leaf has a nearly similar population of patients. The estimated chance of a new patient reaching a leaf is determined by the proportion of

diabetes patients in that leaf during training.

Decision trees are quite interpretable. You can pinpoint exactly which feature values resulted in a prediction. However, they are known for overfitting. When a leaf has a small number of training examples, the resulting probability estimate can be quite incorrect. This makes them a particularly interesting scenario for probability calibration, which is a major focus in this thesis.

### *Random Forest*

Random Forest solves the overfitting problem of single decision trees by training a huge ensemble of trees. Each is based on a randomly picked bootstrap sample of the training data and only evaluates a random subset of features at each split. This creates deliberate variation among the trees. The final predicted probability is the average over all  $T$  trees:

$$p_{RF}(x) = \frac{1}{T} \sum_{t=1}^T p_t(x) \quad (2.7)$$

Averaging across multiple different trees greatly reduces variation when compared to a single tree. This results in more stable and typically stronger predictions. Random Forest consistently performs well in medical datasets. However, it has a well-documented calibration problem. Its probability estimates can be pushed towards the extremes, causing the model to be overconfident. Post-hoc calibration is thus very important for this model [57] [44].

### *Gradient Boosting Classifier*

Gradient boosting constructs decision trees sequentially rather than training them independently and averaging their predictions. The first tree provides an initial approximation. Each subsequent tree is trained to correct the errors made by the existing ensemble. Consequently, each new tree focuses its learning capacity on instances where the current model performs poorly. The learning rate,  $\eta$ , controls the contribution of each new tree to the overall ensemble and therefore determines the step size of the boosting process.

$$F_m(x) = F_{m-1}(x) + \eta \cdot h_m(x) \quad (2.8)$$

The final probability is generated by passing the collected score through the sigmoid function.

$$p(x) = \frac{1}{1 + e^{-F_M(x)}} \quad (2.9)$$

$$p(x) = \frac{1}{1 + e^{-F_M(x)}}$$

Gradient boosting is among the most powerful models for structured tabular data. Like Random Forest, its probability estimates are overconfident and benefit greatly from calibration [45].

### ***XGBoost***

XGBoost (Extreme Gradient Boosting) improves standard gradient boosting in two significant ways. First, it adds explicit regularization terms to the objective function, penalizing model complexity to prevent overfitting.

$$\text{Obj} = \sum_{i=1}^n l(y_i, \hat{y}_i) + \sum_k \Omega(f_k) \quad (2.10)$$

$$\Omega(f) = \gamma T + \frac{1}{2} \lambda \sum_{j=1}^T w_j^2 \quad (2.11)$$

Second, XGBoost builds each new tree using a second-order Taylor approximation of the loss function. It makes use of both the gradient and the second-order derivative, or Hessian, rather than only the first-order gradient. The Hessian gives information about how the loss varies around that point, whereas the gradient indicates the direction in which the predictions should be modified. Using both pieces of information helps XGBoost select appropriate tree splits and leaf values during training and improves its estimation of the impact of an update. Compared to employing the first-order gradient alone, this improves the optimization process's knowledge.

XGBoost is only utilized in centralized training, where it consistently produces the greatest AUROC of all models. This makes it the major performance upper bound throughout the thesis. It serves as the benchmark against which federated techniques are measured.

### ***Shallow Multilayer Perceptron***

The multilayer perceptron is a feedforward neural network that may describe non-linear interactions between inputs and outputs by routing data via one or more hidden layers of neurons. The shallow MLP utilized here has only one hidden layer. In the federated NumPy implementation, the forward pass operates as follows:

$$\text{hidden} = \tanh(XW_1 + b_1) \quad (2.12)$$

$$p = \sigma(\text{hidden} \cdot W_2 + b_2) \quad (2.13)$$

The centralized scikit-learn version employs a ReLU activation in the hidden layer:

$$\text{hidden} = \text{ReLU}(XW_1 + b_1) = \max(0, XW_1 + b_1) \quad (2.14)$$

The model's total number of trainable parameters is

$$P = d \times h + 2h + 1 \quad (2.15)$$

where  $h$  represents the number of hidden neurons, and  $d$  represents the number of input characteristics. The communication overhead every federated cycle is directly determined by the overall model size, which is  $P \times 8$  bytes, since each parameter is

stored as a 64-bit floating point number that takes up 8 bytes.

The MLP is the thesis’s only neural network, and it serves as the deep learning baseline in the federated architecture. It is uniquely compatible with federated averaging because its parameters are continuous, real-valued matrices that can be meaningfully averaged across clients. In contrast, tree-based models have branching architectures, and threshold rules cannot be integrated by averaging [38]. Neural networks are also often overconfident in their probability outputs, making them an important calibration test case [18].

## 2.3 Data Preprocessing

Before any model is trained, raw input data is preprocessed in three standard processes. This ensures that the data can be reliably used for model training.

**Missing value imputation:** Real clinical datasets frequently contain gaps; a patient may not have had a specific test. Missing numeric values are substituted with the feature’s median derived from training data:

$$x_{\text{imputed}} = \text{median}(X_{\text{train column}}) \quad (2.16)$$

The median is preferred over the mean because it is less affected by extreme values and outliers, which are common in clinical measures.

**Feature Standardization:** Different clinical aspects exist on vastly different numerical scales. Blood glucose levels can range from 50 to 500 mg/dL, while ages range from 20 to 90. Models that rely on distance or gradient calculations may behave strangely when features have vastly varying magnitudes. Standardizing each feature to zero mean and unit variance puts them all on an equal scale:

$$z = \frac{x - \mu_{\text{train}}}{\sigma_{\text{train}}} \quad (2.17)$$

**One-hot encoding:** Categorical data, such as a feature identifying gender or ethnicity, cannot be supplied into a model as raw category labels. One-hot encoding transforms each category into a binary indicator column.

**A fundamental rule:** All preprocessing statistics, such as the median for imputation and the mean and standard deviation for standardization, are computed exclusively on the training data. They are then applied unchanged to the calibration and test sets. Fitting the preprocessor to the entire dataset, including test data, might result in information leakage and inflate the measured performance [26].

## 2.4 Centralized Learning

Centralized learning is the most straightforward training paradigm to understand. All patient data from all institutions is pooled in a single location, and one model is trained on the entire combined dataset. Mathematically, it minimizes one global loss:

$$w^* = \arg \min_w \frac{1}{n} \sum_{k=1}^K \sum_{i \in D_k} L(f_w(x_i), y_i) \quad (2.18)$$

In this thesis, centralized training is used as a pooled-data performance benchmark because the model has access to all available training examples simultaneously. It provides a reference against which the federated results can be compared, but it is not assumed to be a universal upper bound on performance. [23]. Throughout this thesis, federated results are measured against the results obtained from centralized training.

Centralized training serves as a theoretical benchmark. It represents the best performance that could be achieved if data could legally and practically be pooled under the applicable regulatory and organizational requirements [21] [42] [64]. The legal barrier is real and cannot be ignored. Centralized training therefore acts as a theoretical benchmark. It represents the best performance that could be achieved if data could flow freely rather than anything usable in practice. Federated learning is specifically designed to address the gap between what is possible in theory and what is legally acceptable.

## 2.5 Local-Only Learning

Local-only learning represents the polar opposite. Each institution develops its own separate model using only its own data. No data is shared with other institutions.

$$w_k^* = \arg \min_{w_k} \frac{1}{n_k} \sum_{i \in D_k} L(f_{w_k}(x_i), y_i) \quad (2.19)$$

From a privacy perspective, this is ideal. No data or model parameters ever leave the institution. However, from a predictive performance perspective, the consequences can be severe. A small rural clinic with 200 patients and minimal population variety generates a model that may operate pretty well for the specific population it has seen but performs badly in general. A hospital that primarily serves one age group or ethnic group will create a model with systematic blind spots.

This thesis's limit for privacy-preserving performance is local-only training. It addresses the most fundamental argument for federated learning: Does federating actually help? The average performance of all K local-only models is calculated as follows:

$$\text{Avg Local-Only} = \frac{1}{K} \sum_{k=1}^K \text{score}_k \quad (2.20)$$

If federated learning cannot consistently exceed this average, the overall communication overhead of federation is difficult to justify [30].

## 2.6 Federated Learning

McMahan et al. (2017) [38] presented federated learning as an approach to the problem of training on big multi-institutional datasets without centralizing the data. The fundamental discovery is straightforward. Patient records cannot be shared, but model parameters learned from the data can be freely transmitted between institutions.

Here’s how it goes in reality. Each institution trains a local model on its own private data, which remains behind the institution’s firewall. The institution then transmits only the model’s learned parameters (weights and biases) to a central server. The server aggregates the parameters from all participating institutions to create an updated global model. This model is then sent back to all institutions. This technique is repeated in multiple cycles. No patient record leaves its original institution [47].

The global federated training objective is to minimize a weighted combination of each client’s loss on its own data [33]:

$$\min_w F(w) = \sum_{k=1}^K \frac{n_k}{N} F_k(w) \quad (2.21)$$

Where  $F_k(w)$  is the local loss of client  $k$ ,  $n_k$  is the number of training examples at client  $k$ , and  $N = \sum_k n_k$  is the total training data across all clients. The issue is to reduce this global target without revealing any client’s local data [43].

## 2.7 Federated Averaging (FedAvg)

FedAvg is the foundational federated optimization algorithm and the benchmark against which all federated approaches are measured. Each communication round has four steps:

1. The server sends the current global model to all participating clients.
2. Each client trains the received model on its own local data across numerous epochs.
3. Each client communicates its modified model parameters to the server.
4. The server uses weighted averaging to incorporate all received parameters into a new global model.

$$\theta_{\text{global}} \leftarrow \sum_{k=1}^K \frac{n_k}{N} \theta_k \quad (2.22)$$

The weight matrix and bias vector of each layer are aggregated independently. Larger clients, those with more training data, contribute proportionally more to the overall model.

The primary practical advantage of allowing several local training phases prior to each aggregation is improved communication efficiency. McMahan et al. (2017) [38] found that this strategy required 10 to 100 times fewer communication rounds than simply averaging gradients after each local step. In reality, fewer rounds lead to less network overhead and faster convergence.

**An important limitation:** In this thesis, FedAvg is applied to logistic regression and MLP because their numerical parameters can be directly averaged across clients. Tree-based models such as decision trees, random forests, gradient boosting, and XGBoost cannot be aggregated using this same direct parameter-averaging approach because their branching structures, feature splits, and thresholds cannot be meaningfully averaged. Specialized federated tree-training methods are outside the scope of this thesis. [38].

## 2.8 FedProx

FedProx was presented by Li et al. (2020) [31] to solve a weakness of FedAvg that becomes evident when data distributions differ between clients. This is often the case in real healthcare settings.

The problem is known as client drift. FedAvg allows each client to optimize its local target separately for several steps before providing its parameters to the server. When clients have various data distributions, they are effectively optimizing for multiple objectives. After multiple local steps, their models may drift far from one another. When the server attempts to average these disparate models, the outcome may not be favorable to any client’s real data distribution. Under sufficiently high heterogeneity, this drift can substantially compromise global model quality [30].

FedProx addresses this by incorporating a proximal term into each client’s local training objective. As training progresses, this term functions as a flexible constraint. It pulls each client’s model back towards the current global model.

$$\min_{w_k} \left[ F_k(w_k) + \frac{\mu}{2} \|w_k - \theta_{\text{global}}\|^2 \right] \quad (2.23)$$

In gradient terms, each local update step becomes [32]:

$$\text{gradient} = \text{local gradient} + \mu(w_k - \theta_{\text{global}}) \quad (2.24)$$

The parameter  $\mu$  controls how tight the leash is pulled. When  $\mu = 0$ , FedProx reduces to FedAvg. Clients’ ability to deviate from the global model becomes more limited as  $\mu$  increases. The proximal term is computed entirely locally, resulting in no additional communication cost. The parameters exchanged between clients and the server are the same size as regular FedAvg.

Asad et al. (2026) and Mathew and Asha (2024) [7] and [37] revealed that FedProx outperforms FedAvg on non-IID clinical tabular data. This provides empirical support for its inclusion in this thesis.

## 2.9 Non-IID Data in Federated Learning

### *The Heterogeneity Problem*

The acronym IID stands for independently and identically distributed. It is a statistical assumption that each client’s data is a representative random sample from the same global distribution. In reality, this is rarely the case in healthcare. Different hospitals serve different patient populations. A hospital in the city center may see a younger and more diversified population of patients. A rural clinic could serve an older and more similar community. A specialist diabetic center encounters a more different case mix than a normal practice. Because of these demographic and prevalence disparities, data distributions among clients are truly diverse [1].

This is important theoretically since FedAvg’s theoretical convergence guarantees the assumption of IID circumstances. Under non-IID conditions, the algorithm may act irrationally, converge slowly, or converge to a global model that does not serve any individual client especially well. On medical datasets, Babar et al. (2024) [9] found accuracy decreases ranging from 5 to 15 percentage points when non-IID circumstances were used instead of IID. Milasheuski et al. (2024) [39] showed that the impact of heterogeneity differs significantly across federated algorithms on tabular clinical data.

### *IID Setting*

In an IID scenario, each client receives data sampled evenly from the global distribution. This results in similar class proportions and feature distributions. In this idealized situation, all hospitals would have statistically identical patient populations. This is the most beneficial situation for federated learning and is used as the experiment’s baseline distribution.

### *Non-IID via Dirichlet Distribution*

Controlled experiments cannot be based on real-world institutional differences. Instead, non-IID situations are simulated using the Dirichlet distribution, which has become the community-standard approach in the federated learning literature vali-

dated across over 235 federated learning studies [67] [16].

The class distribution among clients is derived from [2]:

$$\pi \sim \text{Dirichlet}(\alpha, \alpha, \dots, \alpha) \quad (2.25)$$

The concentration value ( $\alpha$ ) is the important parameter. Consider it as managing how similar the customers are to each other:

Large  $\alpha$  (e.g., 100): All clients have approximately equal class proportions close to IID.

$\alpha = 1.0$  **Mild heterogeneity** - clients differ visibly but not much

$\alpha = 0.5$  **Moderate heterogeneity** indicates significant distributional differences amongst clients.

$\alpha = 0.1$  **Strong heterogeneity** - certain clients may have a high proportion of diabetic patients, while others may have relatively few. This closely resembles genuine disparities across hospitals serving quite diverse regions [71].

### *Client Drift*

Under non-IID conditions, each client's local training gradient represents its own specific data distribution [27]. After numerous local gradient steps, the models at various clients moved in directions shaped by their respective data. This pulls them apart and further from the global optimum. When the server averages these disparate models, it combines updates that were individually optimized for different objectives. The resulting global model may not be particularly effective in meeting any client's objectives. This is the client drift problem, which is directly related to FedProx's proximal penalty term [31] [30].

## 2.10 Communication Cost

Every round of federated training requires the server to communicate the current global model to all clients and each client to return its updated model. Over several rounds and multiple clients, this bidirectional parameter exchange builds into a large communication overhead. This is a genuine deployment cost that must be quantified honestly rather than ignored [43].

The total communication cost during training is computed as

$$\text{params\_bytes} = P \times B \quad (2.26)$$

$$\text{round\_bytes} = \text{params\_bytes} \times \text{clients} \times 2 \quad (2.27)$$

$$\text{cumulative\_bytes} = \sum_{r=1}^R \text{round\_bytes}_r \quad (2.28)$$

where  $P$  is the number of model parameters,  $B$  is the size of each parameter (8 bytes for 64-bit double precision), and  $R$  is the number of training rounds. The factor of two reflects the bidirectional nature of each round. The server sends the model down, and each client sends its updated model back to the server [53].

The model architecture has a direct impact on communication costs. Logistic regression with  $d$  features has only  $d + 1$  parameters, with one per feature plus a bias. This makes it incredibly cheap to transmit. The shallow MLP has  $d \times h + 2h + 1$  parameters. The number of parameters increases with input size and hidden layer width, making each round much more expensive.

An important concept is the convergence round. This occurs when model performance stabilizes and further training yields no meaningful progress. Beyond this point, the system spends communication resources with no return on model quality. Identifying the convergence point helps determine the optimal operating point for a federated deployment [55].

It's important to understand what federated learning does and does not achieve in terms of communication. It does not lower total data transfer compared to isolated local training. Instead, it introduces iterative overhead that would not exist otherwise. It replaces the legally restricted one-time communication of huge volumes of raw patient records with the iterative transmission of model parameters [38] [43].

## 2.11 Probability Calibration

### *The Calibration Problem*

A probability model makes a claim, such as "I believe there is a 70% chance this patient has diabetes." Calibration examines whether that assertion is valid in the real world. A correctly calibrated model satisfies the following condition: of all patients assigned a 70% risk, exactly 70% have diabetes. Formally:

$$P(Y = 1 \mid f(x) = p) = p \quad \forall p \in [0, 1] \quad (2.29)$$

The difficulty is that most current classifiers, particularly ensemble approaches such as Random Forest and Gradient Boosting, as well as neural networks, can be miscalibrated. Their estimated probabilities may cluster toward the extremes. Patients may be assigned risks near 0% or near 100% more frequently than the evidence suggests. The model is overconfident and appears to be more certain than it actually is [18].

Overconfidence is unhelpful in general prediction tasks. In a clinical setting, it is a true patient safety concern. Healthcare experts make treatment decisions based on risk levels, not merely whether the model says "positive" or "negative." A patient who is 85% likely to develop diabetes may be prioritized for more severe action, while a patient who is 40% likely to get diabetes may receive less frequent monitoring. If

the probabilities are not calibrated, the actual risks may differ dramatically from the predicted values. This may impact resource allocation, treatment intensity, and clinical priority [25].

Calibration has mostly been lacking from the federated diabetes prediction literature. No existing federated diabetes study employs post-hoc calibration of predicted probabilities [23] [60] [59] [17]. One of the thesis’s main contributions is to address this gap.

### *Brier Score*

The Brier score is the mean squared difference between expected probability and actual binary outcome.

$$\text{Brier} = \frac{1}{n} \sum_{i=1}^n (p_i - y_i)^2 \quad (2.30)$$

The lower the better. A model that assigns 0.9 to a non-diabetic patient suffers a significant penalty (0.81). A model that assigns 0.5 to the same patient pays a much lower penalty (0.25). The Brier Score penalizes both bad discrimination and poor calibration, making it a more comprehensive measure of probabilistic prediction quality [22].

### *Expected Calibration Error (ECE)*

The main calibration metric used in this thesis is ECE. It calculates the average difference between observed and predicted rates, weighted by the number of patients in each bin:

$$\text{ECE} = \sum_{m=1}^M \frac{|B_m|}{n} |\text{acc}(B_m) - \text{conf}(B_m)| \quad (2.31)$$

The model is correctly calibrated if its ECE is 0. Greater systematic differences between stated confidence and actual accuracy are indicated by higher values. ECE is susceptible to miscalibration not only at the extremes but throughout the entire probability range [57].

### *Uncalibrated Baseline*

When no calibration is performed, raw model probabilities are used directly. This serves as the standard against which calibrated versions are measured. For ensemble models and neural networks, uncalibrated probabilities are typically clustered near 0 and 1 [18] [44].

### *Sigmoid Calibration (Platt Scaling)*

Sigmoid calibration, also known as Platt scaling, learns a simple two-parameter transformation of the model’s raw output scores. It applies a logistic function to a held-out calibration dataset.

$$p_{\text{calibrated}} = \frac{1}{1 + e^{-(a \cdot p_{\text{raw}} + b)}} \quad (2.32)$$

The parameters  $a$  and  $b$  are determined by reducing cross-entropy on the calibration data. Initially, it expands or compresses the probability scale, shifting it up or down. Sigmoid calibration works well when the miscalibration pattern is roughly S-shaped, which is common for logistic regression. It may not be sufficient for ensemble methods and neural networks, which have more complex miscalibration patterns [45].

### *Isotonic Regression Calibration*

Isotonic regression calibration offers more flexibility than Platt scaling. Rather than assuming any specific functional form, it fits a completely general static non-decreasing function to the calibration data.

$$p_{\text{calibrated}} = g(p_{\text{raw}}) \quad (2.33)$$

$$\min_g \sum_{i=1}^m (y_i - g(s_i))^2 \quad \text{subject to} \quad g(s_i) \leq g(s_j) \text{ when } s_i \leq s_j \quad (2.34)$$

The asymmetry constraint is significant because it ensures that if the model ranks patient A as higher risk than patient B, the calibrated probabilities will maintain that ranking. Platt scaling cannot represent complex non-linear miscalibration patterns, although isotonic regression can. Ojeda et al. (2023) [45] demonstrated that it consistently outperforms sigmoid calibration for non-sigmoidal miscalibration. Odesola et al. (2025) [44] verified this for ensemble models using structured clinical data.

### *Global Calibration*

Global calibration fits a single calibration function to the entire shared calibration dataset, which includes all patients and institutions combined. It has the most data and achieves the best statistically stable calibration fit. Both sigmoid and isotonic algorithms can be used in global mode. Global calibration indicates the calibration performance limit, or what is possible when calibration data can be centralized [36].

### *Federated Calibration*

Calibration data cannot be centralized in a federated context since it is subject to the same legal constraints as training data. After the global model has converged, each client separately applies its own calibration function to its own local calibration partition. For federated sigmoid, each client  $c$  learns its own local parameters:

$$p_{\text{calibrated},c}(s) = \frac{1}{1 + e^{-(a_c s + b_c)}} \quad (2.35)$$

In federated isotonic, each client fits its own sequential function  $g_c$ . When a new patient requires a forecast, the calibrated probability is the mean of all client calibrators:

$$p_{\text{calibrated}}(s) = \frac{1}{K} \sum_{c=1}^K \text{calibrator}_c(s) \quad (2.36)$$

This approach ensures data locality throughout the pipeline. Patient data never leaves the originating institution during model training or calibration [36].

## 2.12 Decision Threshold Selection

After establishing calibrated probabilities, a decision threshold transforms these into binary clinical decisions. Two strategies are being evaluated.

**Fixed threshold ( $t = 0.5$ ):** A patient with a forecasted risk of 50% or above is considered positive. This is the most common default and is simple to understand. However, it is not always the appropriate threshold for a certain clinical situation or dataset imbalance.

**F1-optimal threshold:** Instead of utilizing a set value, the threshold is chosen by determining the value that maximizes the F1 score on the calibration split:

$$t^* = \arg \max_{t \in [0,1]} F1(t) \text{ on calibration data} \quad (2.37)$$

This threshold is then applied unchanged to the test set. The key is that threshold optimization employs only calibration data, not the test set. Using test data for threshold selection would result in evaluation leakage, biasing the reported metrics optimistically [22].

## 2.13 Evaluation Metrics

All classification performance metrics are derived from four fundamental counts comparing the model's predictions  $\hat{y}$  against the true labels  $y_{\text{true}}$  on the test set. **True Positives (TP):** diabetic patients correctly identified. **True Negatives (TN):** non-diabetic patients correctly identified. **False Positives (FP):** healthy patients wrongly flagged. **False Negatives (FN):** diabetic patients wrongly cleared.

**Accuracy** is the fraction of all patients classified correctly.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (2.38)$$

Accuracy is straightforward, yet it can be extremely misleading on imbalanced datasets. If just 10% of patients have diabetes, a model that classifies everyone as non-diabetic achieves 90% accuracy but is medically ineffective. It is recorded but not used as the key statistic [11].

**Precision** evaluates the reliability of the model's positive predictions:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (2.39)$$

High precision implies that when the model identifies someone as diabetic, it is typically correct.

**Recall (sensitivity)** evaluates the proportion of real diabetes patients caught:

$$\text{Recall} = \frac{TP}{TP + FN} \quad (2.40)$$

High recall means that the model detects the majority of diabetic patients but at the risk of producing more false alarms.

**Specificity** determines what percentage of healthy individuals are successfully treated:

$$\text{Specificity} = \frac{TN}{TN + FP} \quad (2.41)$$

**Balanced accuracy** compares performance across both classes. It is more suited than raw accuracy for imbalanced datasets:

$$\text{Balanced Accuracy} = \frac{\text{Recall} + \text{Specificity}}{2} \quad (2.42)$$

**The F1 score** represents the harmonic mean of precision and recall. It rewards models that are both precise and comprehensive. It also helps avoid burdening physicians with false positives or missing patients that require care:

$$F1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (2.43)$$

F1 is the key classification quality statistic used throughout this thesis [26].

**The AUROC (Area Under the Receiver Operating Characteristic Curve)** is the likelihood that the model will assign a greater risk score to the diabetic patient if one diabetic patient and one healthy patient are picked at random. It varies between 0.5 (random guessing) and 1.0 (perfect discernment). Crucially, it is threshold-independent. This means it assesses the validity of the probability ranking across all thresholds. This makes it the fundamental indicator for evaluating federated versus centralized performance. It determines how well the model discriminates independent of the operational point used [57] [66].

**PR AUC (Precision-Recall Area Under Curve)** is the area under the curve that shows precision versus recall at all thresholds.

$$AP = \sum_k (\text{Recall}_k - \text{Recall}_{k-1}) \times \text{Precision}_k \quad (2.44)$$

PR AUC is especially useful when the positive class (diabetes patients) is small. It focuses solely on how well the model handles positive instances rather than weighting performance by the huge number of true negatives [26].

## 2.14 Explainable AI-SHAP

A model that states "this patient is 82% likely to develop diabetes" without explaining why is difficult to apply properly in a clinical environment. Healthcare professionals must grasp the reasoning behind forecasts. This helps them trust the model, identify when it may be incorrect, and communicate effectively with patients about their risk factors [8].

This is more than just a practical matter. It is also a legal one. GDPR Article 22 establishes a legal right to an explanation for automated choices that have a significant impact on humans. Explainability is therefore an important consideration for clinical AI systems in Europe.

The federated setting adds an extra layer of complexity. Ducange et al. (2024) [15] showed that in non-IID conditions, SHAP feature significance levels can vary considerably across clients. When the same global model is applied to two hospitals with distinct patient populations, the risk of diabetes may be assigned to various characteristics at each institution. This presents an important clinical governance question. If explanations differ across institutions, which one should drive clinical practice? This thesis addresses this issue by comparing explanation stability over rounds and explanation agreement among clients.

### *SHAP Values*

SHAP (SHapley Additive explanations) was introduced by Lundberg and Lee [35] as a unified approach for interpreting model predictions. The approach is based on cooperative game theory. Each characteristic is viewed as a player in a game, and its contribution is calculated as the average marginal impact across all potential ways it could be included or deleted from the forecast.

SHAP breaks down a single prediction,  $f(x)$ , as follows:

$$f(x) = \phi_0 + \sum_{j=1}^d \phi_j \quad (2.45)$$

The baseline prediction ( $\phi_0$ ) is the average prediction across all patients.  $\phi_j$  represents the contribution of a feature  $j$  to this specific prediction. Positive  $\phi_j$  levels

indicate a higher likelihood of diabetes, while negative values suggest opposite [34].

For linear models, the contribution simplifies to a straightforward formula. Each feature's contribution is simply the distance it deviates from the baseline, scaled by the model's learned weight. For non-linear models, it is approximated by calculating how much the prediction changes when feature  $j$  is replaced with its baseline. SHAP ensures three critical properties: local accuracy (the contributions sum exactly to the prediction). Missingness (features not in the input receive zero contribution). Consistency (if a feature becomes more important in the model, its SHAP value cannot decrease).

### *Global Feature Importance*

SHAP data are aggregated to determine which attributes are most important throughout the patient group overall.

$$\text{Mean Absolute SHAP}_j = \frac{1}{n} \sum_{i=1}^n \left| \phi_j^{(i)} \right| \quad (2.46)$$

Features with higher mean absolute SHAP values have more consistent influence. This global rating enables physicians to determine whether the model's priorities are consistent with accepted medical knowledge regarding diabetes risk factors. For example, does the model appropriately prioritize glucose level over a less clinically important feature? [24] [41].

### *Explanation Stability*

A practically effective model should converge in both performance and explanations during a federated training procedure that spans multiple rounds. Even though the AUROC appears stable, the clinical interpretation of the model is unstable if the model's feature priority ranking varies dramatically from round to round.

The Spearman rank correlation was utilized to assess the consistency of feature-importance evaluations between clients and between federated communication rounds. For each model, a significance score was calculated for each of the 21 input features. In logistic regression, the model coefficients were utilized as feature-importance scores, whereas in MLP, a weight-based feature-importance score was produced using the model weights. The features were then graded based on their importance rankings. Spearman rank correlation was calculated for all 21 features by comparing the rank differences between comparable features in the two rankings. A value near to one implies relatively constant feature-importance rankings, whereas lower values indicate more variation in feature ordering. The top-5 feature agreement was measured separately using the top-5 overlap metric.

$$\rho = 1 - \frac{6 \sum_j d_j^2}{d(d^2 - 1)} \quad (2.47)$$

where the rank difference of feature  $j$  between two successive rounds is represented by  $d_j$ . A score of 1.0 indicates "perfect stability," with no fluctuation in the ranking.

Rankings are largely random if the value is close to 0. High stability means that in addition to its prediction performance, the model has learned a consistent clinical pattern [15].

### *Cross-Client Feature Agreement*

Even after training is standardized globally, different clients may have learned to weigh aspects differently based on their local patient populations. Top-k overlap quantifies the extent to which two customers agree on the most significant attributes.

$$\text{Top-k Overlap} = \frac{|\text{TopK}_A \cap \text{TopK}_B|}{k} \quad (2.48)$$

A number of 1.0 indicates that two clients agree totally on the top k qualities. A number of 0.4 indicates that 40% of their top k traits overlap. Under strong non-IID conditions, decreased overlap is expected, as different hospitals see distinct clinical trends. Understanding the scope of this difference has practical consequences for how federated model explanations should be conveyed and used by member institutions [15].

## 2.15 Algorithmic Fairness

A model with 85% overall accuracy can still be systemically biased if it performs well for some patient groups but poorly for others. Concerns about fairness are especially acute when predicting diabetes. Diabetes is more common among older patients and those with a higher BMI. This is clinical reality, not discrimination. However, systematic disparities in error rates between groups can indicate unfairness. For example, one demographic group may have more diabetic patients ignored than another, while healthy patients may be wrongly detected at varying rates. These discrepancies can cause serious harm [69].

The federated setting is important from a fairness standpoint. According to Chen et al.(2023) [13], training on population varied datasets, as achieved by multi-institutional federation, can serve as a bias mitigation strategy as compared to training on homogenous data from a single institution. Pfohl et al. (2021) [48] showed, however, that fairness-accuracy trade-offs must be measured empirically rather than assumed. This thesis evaluates fairness across age groups and BMI categories for all models and training paradigms.

### *Demographic Parity Difference (DPD)*

Demographic parity investigates whether positive predictions are provided at equal rates across various groups. The selection rate for group g is the proportion of that group predicted positively. The demographic parity difference is the difference between the greatest and lowest selection rates.

$$\text{DPD} = \max_g P(\hat{y} = 1 \mid G = g) - \min_g P(\hat{y} = 1 \mid G = g) \quad (2.49)$$

A DPD of 0 indicates that all groups are projected positively at the same rate. A greater score suggests that some groups are flagged more frequently than others. The weakness of DPD is that it assumes identical prediction rates are completely desirable, even when true prevalence disparities exist between groups. Even if a model accurately reflects that older patients have greater diabetes rates, it will still display a non-zero DPD [69].

### *Equalized Odds Difference (EOD)*

Equalized odds is a more clinically relevant fairness criterion. Rather than requiring equal prediction rates, it demands that the model's error rates, including false positive and true positive rates, be equal across groups. The equalized odds difference is

$$\text{EOD} = \max \left( \max_{g, g'} |TPR_g - TPR_{g'}|, \max_{g, g'} |FPR_g - FPR_{g'}| \right) \quad (2.50)$$

This indicates that the model should catch all patients with diabetes at the same rate, regardless of demographic category. Furthermore, the model should generate false alarms at the same rate across all healthy patients, independent of group. An EOD of zero represents perfect equalized odds.

Equalized odds have clinical appeal because they allow for actual prevalence differences across groups while requiring equal error treatment. If older patients have a higher diabetes prevalence, a fair model can predict positive more often for older patients. However, it should not miss diabetic older patients at a higher rate than diabetic younger patients. It should also not raise false alarms for healthy older patients at a different rate than healthy younger patients. Both Xu et al. (2022) and Chen et al. (2023) [69] [13] argue that equalized odds are the better fairness requirement for clinical prediction tasks. It is the fundamental fairness metric in this thesis.

## Chapter 3

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# Related Work and Literature Review

### 3.1 Federated Learning in Healthcare and Diabetes Prediction

#### *Foundational Work and Healthcare Surveys*

McMahan et al. (2017) [38] formally introduced the federated learning paradigm with their Federated Averaging algorithm. It uses iterative model parameter averaging across decentralized clients as a more communication-efficient alternative to synchronous gradient descent. Their research exhibited 10 to 100 times fewer communication rounds across five model architectures and four datasets. This demonstrated the technical viability of distributed collaborative training. However, McMahan et al. did not address healthcare applications, probability calibration, or model interpretability. Their convergence guarantees are based on IID data distributions that do not apply to realistic multi-institutional clinical deployments. Despite these drawbacks, FedAvg remains the universal baseline algorithm in federated learning research. It serves as the foundation for this thesis’s federated training pipeline.

As federated learning (FL) evolved into a more general-purpose paradigm, healthcare emerged as one of its most important application domains. Nguyen et al. (2022) [43] conducted a comprehensive assessment of FL in smart healthcare. This includes resource-aware, safe, privacy-preserving, and personalized FL applications. Their research identified several ongoing issues, including statistical heterogeneity, communication overhead, system heterogeneity, privacy attacks, and a lack of interpretability tools, which directly inspired the SHAP-based explainability component of this thesis. Teo et al. (2024) [62] conducted a systematic analysis of 612 healthcare FL papers. They showed the increasing expansion of healthcare FL research while noting that organized tabular clinical data remains neglected when compared to medical imaging. This gap emphasizes the need to assess FL using the CDC tabular dataset used in this investigation. Similarly, Pati et al. (2024) [47] pointed out that building trust in federated healthcare AI necessitates not just good predictive performance but also interpretable and transparent models to facilitate clinical adoption. Despite addressing privacy, trust, and explainability, these studies do not look at probability calibration. This is an important aspect of model trustworthiness that this thesis aims to address.

Pan et al. (2024) [46] suggested an adaptive FL framework for clinical risk predic-

tion utilizing multi-hospital EHR data. Their approach addresses distribution drift by dynamically adjusting aggregating weights over training rounds rather than using traditional FedAvg’s fixed dataset-size weights. Their framework outperformed static FedAvg on EHR-based clinical outcomes. This suggests that performance-aware aggregation is useful in diverse clinical situations. In their approach, clients with greater local validation performance contribute proportionally more to global model updates. However, Pan et al. do not use explainability or calibration in their predictions. Their emphasis on administrative EHR outcomes rather than binary disease classification limits direct comparison with the diabetes prediction task investigated in this thesis.

### *Federated Learning Applied Directly to Diabetes*

Islam et al. (2022) [23] provide the most relevant real-world justification for this thesis in the federated diabetic literature. Their study described a decentralized FL system that predicts diabetes-related complications by leveraging real clinical EHR data from several healthcare organizations. They used FedAvg with class-balancing strategies to overcome dataset imbalance. Their federated models outperformed centralized learning, proving that privacy-preserving collaborative diabetic modelling is achievable on real multi-site clinical data. However, Islam et al.’s evaluation is completely focused on accuracy and class imbalance metrics. The work does not analyze whether projected probabilities are well-calibrated. It also does not provide feature-level explanations or compare performance across non-IID heterogeneity levels. These omissions collectively form the fundamental gap that this thesis addresses.

Tang et al. (2024) [60] demonstrated one of the first real-world implementations of federated learning (FL) for diabetes prediction. They used cross-province primary care data from the Canadian Primary Care Sentinel Surveillance Network. Their findings showed that a federated multilayer perceptron (MLP) performed similarly to, or slightly better than, its centralized version. However, federated logistic regression performed worse. This diversity in model behavior throughout federated training emphasizes the significance of comparing several machine learning and deep learning models. This directly drives the comparison strategy used in this thesis. However, the study concentrated on predicted performance indicators and did not examine model calibration or explainability. Because clinical decision-making requires both solid probability estimations and accurate forecasts. Therefore, the absence of calibration analysis limits the framework’s practical application.

Su et al. (2023) [59] addressed both privacy and security in a multi-party FL system for diabetes risk prediction. They developed customized safe protocols for several model families, such as XGBoost, LightGBM, neural networks, and logistic regression on the Seceum FL platform. Their findings showed that federated models using secure computing may match centralized prediction performance while protecting against gradient leakage attacks. This represents a significant advancement in the security dimension of federated healthcare AI. However, Su et al. do not discuss probability calibration, SHAP-based explainability, or non-IID robustness. Their evaluation framework focuses solely on predictive accuracy. As a result, the clinical

reliability of predicted probabilities remains unexplored.

Gawande et al. (2025) [17] studied federated learning for diabetes prediction using the Flower framework and the Pima Indians Diabetes Dataset. The data were shared across five simulated clients under non-IID conditions. The study trained a fully connected neural network and used LIME to assess model explainability. An interesting conclusion was that feature importance explanations differed amongst clients due to changes in local data distributions. This highlights the effect of data heterogeneity on interpretability. However, the study has a few shortcomings. It relied entirely on LIME and did not assess SHAP, which is extensively used for local and global model interpretation. Furthermore, the authors concentrated on prediction performance but did not evaluate probability calibration. Therefore, the reliability of estimated risk probabilities was not tested. Furthermore, the evaluation was limited to a single deep learning model, with no comparison to traditional machine learning approaches using federated training. As a result, the performance of various model types in federated diabetes prediction is uncertain.

Chellamani et al. (2025) [12] investigated an FL framework for non-invasive blood glucose monitoring based on photoplethysmography signals from biosensor data. They achieved high clinical accuracy in Clarke error grid analysis. While their study illustrates FL's usefulness in diabetes care, it focuses on continuous waveform prediction from physiological sensors rather than binary risk classification from structured tabular clinical data. The sensor-based modality, continuous output, and lack of XAI and calibration indicate that this project addresses a fundamentally different technical problem from this thesis.

Yang and Li (2025) [70] used recurrent neural networks and continuous glucose monitoring data to create a clustering-based personalized FL model for glucose prediction. Their method increased prediction customization by grouping comparable patients across clients before federated aggregation. While technically unique, their emphasis on sequential CGM data and continuous glucose output rather than structured tabular classification implies that their architecture and evaluation criteria are not directly comparable to this thesis's binary diabetes risk prediction methodology. They do not include calibration or explainability in their evaluation.

FedEnTrust, a federated ensemble framework combining blockchain-based access control and knowledge processing for diabetes prediction on the Pima Indians dataset, was proposed by Hasan and Li (2026) [19]. It is the most recent paper in the evaluated literature. FedEnTrust achieved 84.2 percent accuracy that still relies solely on accuracy measures without calibration or XAI evaluation. The practical implementation issues are raised by the blockchain overhead, which adds 210 milliseconds of latency per transaction. The relevance and significance of this thesis's contribution are further supported by the fact that even the most current federated diabetic article under consideration leaves out calibration and explainability.

## 3.2 Non-IID Data Heterogeneity in Federated Learning

### *Benchmarks and Algorithmic Comparisons*

The most thorough experimental benchmark of federated learning under non-IID conditions was carried out by Li et al. (2022) [30]. They evaluated FedAvg, FedProx, SCAFFOLD, and FedNova across nine datasets. They also proposed six non-IID partition strategies covering label distribution skew, feature distribution skew, and quantity skew. Their main conclusions have a major impact on this thesis. For all studied algorithms, non-IID situations result in a considerable reduction in accuracy. When the proximal parameter is tiny, FedProx exhibits little improvement over FedAvg. This indicates that FedProx’s benefit is not always present. No particular algorithm was dominant under all non-IID conditions. Instead of assuming the superiority of one algorithm, these results support the empirical comparison of FedAvg and FedProx under both IID and Dirichlet-partitioned non-IID conditions in this thesis. They also support the evaluation of both configurations across various heterogeneity levels.

In a comprehensive IEEE Communications Surveys and Tutorials study of 235 federated learning articles, G et al. (2024) [16] gave a full taxonomy of non-IID data sources. They identified Dirichlet distribution partitioning as the community standard for modeling label skew in federated experiments. Their assessment indicated that research interest in non-IID FL peaked in 2023. FedProx and SCAFFOLD are the most widely used solutions, although each handles only a subset of non-IID issues. This thesis’s non-IID simulation technique uses Dirichlet partitioning with three heterogeneity levels: mild ( $\alpha = 1.0$ ), moderate ( $\alpha = 0.5$ ), and strong ( $\alpha = 0.1$ ).

### *Non-IID in Clinical Federated Settings*

Babar et al. (2024) [9] conducted an empirical study on FedAvg performance under IID and non-IID settings on a medical imaging dataset. They found accuracy decreases of 5 to 15 percentage points with growing data heterogeneity. While their study employs imaging rather than tabular clinical data. Their methodology of comparing IID and non-IID performance across different client counts and communication cycles serves as a direct prototype for the IID versus non-IID evaluation design in this thesis. Their measured performance degradation range also defines the magnitude of heterogeneity impact that federated diabetic architectures must manage.

Milasheuski et al. (2024) [39] are especially notable in this thesis. As they are the only study in the reviewed literature that evaluates FL algorithms using tabular clinical data. A stroke prediction dataset, rather than imaging or text data. After evaluating seven FL algorithms across different heterogeneity types, they discovered that SCAFFOLD works best under label skew and feature skew. Whereas FedProx performs best under high heterogeneity for binary classification. Critically, they observed that the low dimensionality of tabular clinical information produces fun-

damentally different FL convergence dynamics than high-dimensional image data. This finding clearly explains this thesis’s tabular-specific federated evaluation rather than generalizing from imaging FL outcomes. It informs the inclusion of FedProx as a primary comparison to FedAvg.

Asad et al. (2026) [7] performed a controlled simulation that is strikingly similar to this thesis. They used FedProx to predict heart disease across four demographically stratified simulated hospital clients. The study used 303 structured clinical patient records, a small tabular dataset comparable in scale to the Pima Indians diabetes setup. FedProx with  $\mu = 0.05$  obtained 85 percent accuracy. This was higher than both centralized training (83.33 percent) and the local-only average (78.45 percent) across 50 independent experimental runs. This finding is interesting because it shows that federated training with proximal regularization can outperform centralized training on short clinical datasets. This is most likely because of the regularization effect that prevents overfitting. However, Asad et al. address heart disease rather than diabetes. They also do not incorporate any XAI or calibration. Therefore, leaving the reliability of their estimated probability untested.

Mathew and Asha (2024) [37] found that using the FedProx and FedSplit algorithms on medical IoT data under Dirichlet partitioning at three heterogeneity levels ( $\alpha = 0.02, 0.05, \text{ and } 0.1$ ) improved test accuracy by 35% compared to standard FL under strong heterogeneity. Their use of Dirichlet partitioning and proximal regularization directly informs the algorithmic architecture of this thesis. However, their medical IoT focus, lack of a disease-specific prediction objective, and deletion of calibration and XAI make direct comparison difficult.

### 3.3 Probability Calibration in Clinical Machine Learning

#### *Theoretical Foundations*

Silva Filho et al. (2023) [57] present a thorough examination of classifier calibration. Their work lays the theoretical groundwork for evaluating and improving projected class probabilities. Their consideration of Expected Calibration Error (ECE) and Brier Score as complementary calibration measures informs the evaluation approach used in this thesis. The survey also highlights isotonic regression as a non-parametric, monotone-constrained calibration method. It does not require a parametric shape assumption. Which makes it ideal for heterogeneous federated data with varying miscalibration patterns among clients. Furthermore, the authors emphasize that well-calibrated classifiers accurately measure predictive uncertainty. This is essential for clinical decision-making based on diabetes risk scores.

Ojeda et al. (2023) [45] investigated calibration methods with distributional disparities between training and validation populations. Their findings revealed that isotonic regression works well when there is enough calibration data, but it can over-

fit when the calibration dataset is minimal. This insight is especially relevant to the federated setting explored in this thesis. Each client has access to a smaller local calibration set than would be available in a centralized system. This encourages the usage of cross-validation during calibration.

### *Calibration in Clinical Prediction*

Huang et al. (2020) [22] present a basic clinical informatics tutorial on calibration measures. They demonstrate that prediction confidence is just as significant as prediction accuracy in clinical decision-making. Treatment decisions are based on projected risk levels rather than binary labels. Their demonstration of reliability diagram building and calibration significance testing sets the standard clinical evaluation technique for this thesis.

Rajaraman et al. (2022) [51] investigated calibration specifically under class imbalance in medical picture classification. They discovered that calibration drastically improves model performance at the default 0.5 threshold. They also found that isotonic regression is not advised for small datasets due to overfitting. This finding directly influences this thesis's decision to use cross-validation during the client-level isotonic calibration step on the CDC dataset's training partitions.

Odesola et al. (2025) [44] performed post-hoc calibration research using structured clinical heart disease data, which is one of the most methodologically relevant investigations in this thesis. When they compared six classifiers before and after Platt scaling and isotonic calibration, they discovered that isotonic regression consistently improved calibration performance. It lowered Random Forest ECE from 0.051 to 0.011 and Brier Score from 0.007 to 0.002, whereas Platt scaling decreased KNN calibration. These findings lend significant empirical support to the adoption of isotonic regression as the principal calibration approach in this thesis. Furthermore, some of the classifier families tested by Odesola et al., such as logistic regression, random forest, and tree-based models, are included in this work. This allows for significant comparisons of calibration gains across investigations.

Koutsoubis et al. (2025) [28] identified uncertainty quantification and calibration consistency across clients as key open difficulties in federated healthcare applications. They identified per-client calibration as a research gap. This directs the study of client-level calibration metrics, such as ECE and Brier Score, in this thesis. Majlatow et al. (2025) [36] coupled isotonic regression calibration and SHAP-based explainability in a healthcare process monitoring scenario. They found that calibration increased both the reliability of uncertainty estimations and the interpretability of SHAP explanations. This study reveals an important link between calibration quality and explainability. It motivates the suggested methodology to evaluate calibration performance and explanation stability concurrently.

Among the five reviewed federated diabetes studies that perform binary classification, the studies primarily evaluate classification performance and do not report the calibration measures considered in this thesis, such as Brier Score and ECE. Islam

et al. (2022) [23], Tang et al. (2024) [60], Su et al. (2023) [59], Gawande et al. (2025) [17], Chellamani et al. (2025) [12], Yang and Li (2025) [70], and Hasan and Li (2026) [19] evaluate models solely on classification metrics. There is no calibration assessment. This universal exclusion across seven pooled diabetes investigations is notable, despite the well-established clinical value of probability dependability. It constitutes the thesis's most obvious and directly actionable weakness.

### 3.4 Explainable AI for Federated Diabetes Prediction

#### *Federated XAI Studies*

Ducange et al. (2024) [15] carried out one of the few experiments that combines federated learning and explainable AI in a hospital scenario. They used a federated MLP with SHAP explanations to forecast Parkinson's disease development. They examined model behavior in IID and multiple non-IID scenarios. Their findings revealed that SHAP explanations become less consistent among clients when data distributions diverge. This underscores the importance of data heterogeneity in interpretability. This discovery serves as an essential motivation for the current thesis. It objectively assesses SHAP explanation stability throughout federated communication rounds. However, their study is focused on a distinct disease area and does not include probability calibration. This limits its relevance to calibrated diabetes risk prediction.

The most closely related study to this thesis is Kumar et al. (2025) [29]. It is the only one in the evaluated literature to combine federated learning, SHAP explainability, and isotonic regression calibration into a single framework. After isotonic calibration, they achieved 99% accuracy, a 98% F1-score, a Brier Score of 0.01, and an ECE of 0.59. Which uses Random Forest, Gradient Boosting, XGBoost, Logistic Regression, and Decision Tree models among five decentralized clients for liver disease prediction. Their SHAP study revealed that direct bilirubin, SGOT, and alkaline phosphatase were the most influential features. The methodological similarities with this thesis are significant. Both research studies use the same classical machine learning model families, isotonic regression calibration, and SHAP-based explainability.

Despite these similarities, significant discrepancies exist. Kumar et al. concentrated on liver disease prediction rather than diabetes. They did not investigate the effect of IID versus non-IID data distributions on model performance or calibration. Furthermore, they did not examine SHAP explanation stability across federated rounds or fairness-related evaluation. The significant ECE seen despite isotonic calibration raises concerns regarding calibration effectiveness under federated settings. This offers further motivation for the more comprehensive analysis conducted in this thesis across many levels of data heterogeneity.

### *XAI Applied to Diabetes in Centralized Settings*

Several studies have used SHAP and LIME to predict diabetes in centralized settings. They provide valuable clinical context for analyzing federated explanation findings. Alam et al. (2025) [6] used a 1D-CNN with SHAP to predict diabetes in women across two datasets. They discovered that glucose was dominating in both groups, although BMI and pregnancy count had culturally dependent values. This cross-population variance in feature relevance drives this thesis’s assessment of SHAP explanation stability under non-IID federated situations. where diverse client populations may exhibit similar variation. However, Alam et al. work in a centralized framework with no FL or probability calibration.

Tasin et al. (2023) [61] integrated the Pima Indians Diabetes Dataset with private Bangladeshi clinical data. They used XGBoost, SHAP, and LIME to assess model explainability. Their identification of glucose, BMI, age, insulin, and skin thickness as major diabetes predictors. This serves as a good clinical standard for validating the SHAP explanations presented in this thesis. However, their research did not include federated learning or probability calibration. Islam et al. (2025) [24] used SHAP, LIME, and partial dependence analysis on diabetes prediction models. They achieved good predictive performance and thorough explainability. Their use of hyperparameter optimization provides relevant methodological context for model selection in diabetes prediction.

Hasan et al. (2025) [20] used AutoML with different explainability techniques to predict diabetes. They identified federated learning as a viable future direction for boosting generalization while protecting privacy. This recommendation supports the study’s current research direction. Mohanty et al. (2024) [41] tested different machine learning models. They discovered that glucose, BMI, and age were consistently the most important predictors of diabetes. Their findings provide additional support for the clinical usefulness and stability of SHAP-based feature significance analysis in diabetes prediction.

## **3.5 Fairness in Clinical Machine Learning and Federated Learning**

Xu et al. (2022) [69] conducted a thorough study of algorithmic fairness in computational medicine. They identified equalized odds needing equal true positive and false positive rates across demographic categories as the recommended fairness metric for clinical prediction tasks. Their reasoning is completely incorporated in this thesis. Unlike demographic parity, which may incorrectly identify truly higher diabetes prevalence in older patients as unfair. Equalized odds conditions on true disease status necessitate equal error behavior. Pfohl et al. (2021) [48] found that optimizing for fairness constraints during training reduces predictive performance within groups. This supports this thesis’s decision to report fairness metrics as transparency dimen-

sions rather than training objectives. Chen et al. (2023) [13] argued that federated learning can serve as a bias mitigation strategy by allowing for training across demographically diverse institutional datasets. This inspired this thesis’s comparison of subgroup fairness under IID versus non-IID federated conditions. The aim is to determine whether heterogeneous client distributions narrow or widen demographic disparities in error rates.

Despite the well-established importance of fairness evaluation in clinical AI established by these three foundational studies, none of the five reviewed federated diabetes studies performing binary classification in the scope considered here report demographic parity difference or equalized odds across the demographic subgroups examined in this thesis. In the reviewed literature, Islam et al. (2022) [23], Tang et al. (2024) [60], Su et al. (2023) [59], Gawande et al. (2025) [17], and Hasan and Li (2026) [19] evaluate demographic parity difference or equalized odds across any demographic subgroup. This widespread lack of fairness evaluation in federated diabetes research constitutes a clear and major need. This thesis fills this need by incorporating subgroup fairness analysis across age, BMI, and sex categories as a required dimension of clinical readiness assessment.

### 3.6 Summary of Research Gaps

Reading across all reviewed studies, the following five distinct research gaps appear. These gaps collectively support this thesis and set it apart from all previous work in this area.

The first gap is the total lack of probability calibration in any federated diabetes prediction study. Every federated diabetes publication assessed, including Islam et al. (2022) [23], Tang et al. (2024) [60], Su et al. (2023) [59], Gawande et al. (2025) [17], Chellamani et al. (2025) [12], Yang and Li (2025) [70], and Hasan and Li (2026) [19], evaluates models solely on classification metrics. It does not report the Brier Score or ECE. This absence remains despite many calibration articles in the literature. This confirms the need for trustworthy projected probabilities for clinical threshold-based decision-making (Silva Filho et al., 2023 [57]; Ojeda et al., 2023 [45]; Odesola et al., 2025 [44]).

The second gap in diabetes prediction is the lack of an integrated framework that combines federated learning, SHAP explainability, and probability calibration. Kumar et al. (2025) [29] accomplish this integration for liver disease. They are the only study in the literature to include all three components. He demonstrates an integrated FL, SHAP, and isotonic-calibration framework for liver disease. However, their study does not evaluate diabetes prediction, does not examine IID versus non-IID conditions, does not assess explanation stability across federated communication rounds, and does not include subgroup fairness analysis. These differences leave the corresponding questions open for federated diabetes prediction.

The third gap is the SHAP explanation stability over federated communication rounds. Which has never been fully assessed for diabetes prediction. Ducange et al. (2024) [15] discovered explanation inconsistency as a concern for Parkinson's illness in non-IID conditions. Gawande et al. (2025) [17] observed LIME variation across customers but did not quantify it. No study has assessed the Spearman rank correlation of SHAP feature importance across rounds as a formal federated stability metric for diabetes.

The fourth gap is the lack of a systematic comparison of traditional ML models to a shallow MLP for diabetes using federated training, calibration, and SHAP. Tang et al. (2024) [60] compare MLP to LR under FL. Su et al. (2023) [59] compare XG-Boost, LightGBM, NN, and LR. However, none use calibration or XAI. This makes a fair comparison of probability reliability impossible.

The fifth gap is the complete absence of subgroup fairness evaluation in combined diabetic trials. Despite the fact that Xu et al. (2022) [69], Pfohl et al. (2021) [48], and Chen et al. (2023) [13] established fairness evaluation as a criterion for clinical ML deployment. None of the seven federated diabetic publications examine demographic parity or equalized odds across age or BMI subgroups.

This thesis addresses all five deficiencies at once by using a unified, trustworthy method of inspection. It assesses federated diabetes prediction in terms of performance, calibration, explainability, fairness, and communication cost within a single reliable experimental pipeline.

### 4.1 Methodological Approach

This thesis uses a quantitative, experimental research approach. Both research questions require answers that can only be obtained by conducting controlled experiments and measuring numbers instead of relying on opinions, interviews, or case studies. The most appropriate study methodology was to create a single, consistent experimental pipeline through which every model, training paradigm, and configuration would travel identically. Rather than running a small number of disconnected experiments and writing about each individually. The entire grid of conditions was run through the same code path. This included different models, different training setups, different client counts, different heterogeneity levels, and different calibration methods. The same preprocessing, splits, and evaluation logic were used. This consistency is crucial for a thesis like this one. The entire argument is based on being able to say, "Federated AUROC is X and centralized AUROC is Y, and the only thing that differs between these two numbers is the training paradigm. "If even minor inconsistencies entered into preprocessing or evaluation between runs, the comparison would no longer be reliable. This chapter leads you through the pipeline step by step, in the order it was built and executed.

Bringing everything in this chapter together, every single experiment performed in this thesis followed the same end-to-end sequence of actions in the same precise order. Initially, the raw CDC dataset was loaded. Second, it was divided into the training, calibration, and test portions. Third, the preprocessing statistics were fitted just for the training partition and then applied consistently across all three partitions. Fourth, the training partition was divided into the required number of simulated clients using either the IID or the non-IID Dirichlet option. Fifth, the model was trained using the paradigm being tested for that experiment: centralized, local-only, or federated with FedAvg or FedProx. Sixth, the calibration partition was used to apply the calibration technique of choice. Seventh, the decision threshold was determined, either fixed or F1-optimized. Eighth, all performance and calibration metrics were compared to the undisturbed test partition. Ninth, SHAP explanations were created. Their consistency was tracked over rounds, where applicable. Tenth, fairness metrics were calculated for the appropriate groupings. During the federated rounds, communication bytes were logged continually rather than only at the end. The identical nine-step pipeline was run repeatedly for each model, training paradigm, client count, alpha value, and calibration method examined in this thesis.

## 4.2 Problem Definition and Study Design

The primary prediction task in this thesis is expressed as binary classification. Given a patient's health indicators, the purpose is to determine whether the patient is diabetic (label 1) or non-diabetic (label 0). This is the identical task formulation used by practically every paper assessed in the related work chapter.

To answer RQ1, the exact same models were trained in three completely different training paradigms. centralized, where one model sees all of the pooled training data. Local-only, where each simulated client trains a model using only its own slice of data with no collaboration. Federated, where clients train locally but periodically share and combine model parameters via a central server. Training the same model family under all three paradigms, using the same underlying data split or pooled differently. Which allows us to directly measure the performance gap that RQ1 is referring to. Without this three-way comparison, we would have no idea whether federated learning is "close to centralized" or not. There would be nothing to compare it to.

To respond to RQ2, every federated run was performed with five different calibration circumstances: none, global sigmoid, global isotonic, federated sigmoid, and federated isotonic. SHAP explanations were computed and logged at each communication round rather than simply at the conclusion. This was important because RQ2 is asking not just "Is the final model calibrated?" It also asks how calibration alters things and whether explanations are consistent as training progresses. Neither of which can be answered by looking at one take at the conclusion of training.

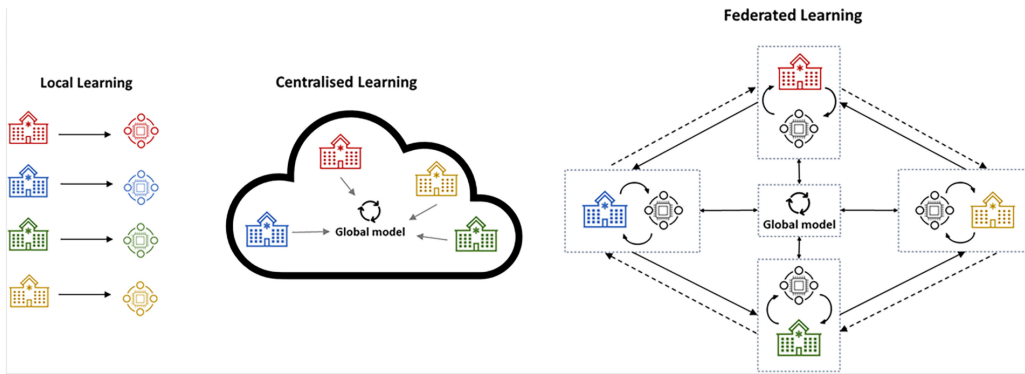


Figure 4.1: Federated learning in comparison to local learning and centralized learning [62]

The experimental design consists of three training paradigms (centralized, local-only, and federated), two distribution conditions (IID and non-IID), two client counts (3 and 5), three heterogeneity levels ( $\alpha = 1.0, 0.5, 0.1$ ), six model families, and five calibration strategies. Every cell in this grid was run independently. The comparisons are required for both research questions. Client versus federated, centralized versus

federated, calibrated versus uncalibrated, IID versus non-IID, and ML versus DL. These are simply different segments through the same grid of results rather than separate studies [65].

### 4.3 Dataset Description

The CDC Diabetes Health Indicators dataset [3] is derived from the Behavioral Risk Factor Surveillance System health survey. This is an ongoing health survey conducted by the United States Centers for Disease Control and Prevention. This dataset was carefully chosen above the far more widely used Pima Indians Diabetes Dataset. That decision was clearly justified here.

The Pima dataset has three unique issues that make it an unsuitable choice for a federated learning simulation. First, it only has 768 total records. This appears manageable until you try to split it across 3 to 5 simulated hospital clients. Each client then ends up with somewhere between 150 and 250 rows. This is barely enough data to train a stable model locally, let alone hold out a separate calibration partition on top of that. Second, it exclusively includes data from women of Pima Indian descent. This is a significant gender limitation that renders any fairness study across broader groupings worthless because the population is already relatively small. Third, a number of its characteristics contain biologically impossible zero values. A glucose measurement of 0 or a blood pressure reading of 0 does not occur in a living person. This implies that a significant chunk of the information must be imputed before it can be used.

The CDC dataset avoids all three concerns. It has a total of 253,680 records, which means that even when split across 5 federated clients, each client still has tens of thousands of rows to train on. This provides a realistic approximation of what an actual hospital's patient database may look like. The population also has a substantially higher demographic variation across age, gender, and BMI. This is exactly what is required to conduct a meaningful fairness analysis across those subgroups in Chapter 5. There were no biologically impossible zero values to fix.

Diabetes\_binary is the target column in this dataset. A value of 0 indicates that the respondent does not have diabetes, and 1 indicates that they do. This dataset has an uneven class balance, with 218,334 negative records and only 35,346 positive records. This indicates that diabetes responders account for around 13.9 percent of the whole dataset. This imbalance was treated as a genuine feature of the real-world problem rather than something to artificially force into balance. Diabetes is a minority condition in the general population. A model evaluation that ignores this reality by oversampling away the imbalance would not reflect how the model would actually behave when implemented.

### 4.3.1 Data Availability

The dataset used in the study is the CDC Diabetes Health Indicators, which is publicly available. The tests used no private patient or hospital data. The dataset was used to construct simulated client settings and test various training algorithms under controlled conditions. Because the study does not involve private institutional data, the experiments can be reproduced with the same public dataset and experimental conditions as detailed in the methodology.

## 4.4 Data Preprocessing

Before a single model saw a single row of data, each trial went through the same fixed preprocessing procedure. This consistency is important because if preprocessing was permitted to vary dramatically between tests. Any subsequent performance difference may not be safely attributed to the real variable being studied. It could just be a preprocessing effect.

Missing numeric values, if present, were filled up with the median value of that column obtained solely from the training partition. The median was chosen over the mean because health metrics such as BMI can have extreme outlier values. These would cause a mean-based fill to be misleading. The median is more adaptable to those outliers. Continuous numeric features were then standardized using z-score scaling. This involved subtracting the training mean from each feature and dividing the result by the training standard deviation. This produced a feature with approximately zero mean and unit variance. This phase is notably important for the gradient-based models in this thesis, logistic regression and the MLP. These models update their weights using gradient descent. Features on wildly diverse numeric scales might cause certain weights to update much quicker than others. This can lead to unstable training. Even while tree-based models like Random Forest do not require scaled features, the same scaling was applied to all models equitably. This ensured that no model acquired an inappropriate preprocessing advantage over another. Categorical data were processed using one-hot encoding. This generates a separate binary column for each category level rather than treating the categories in an artificial numerical order.

The source of the statistics is the most crucial factor in this entire preprocessing procedure. The median for imputation, as well as the mean and standard deviation for scaling, were determined solely from the training partition. These fixed values were then applied to alter the calibration and test divisions. They were never recalculated individually for those partitions. This was done on purpose to avoid a subtle but serious type of data leakage. If, for example, the scaling statistics had been calculated from the entire dataset before splitting. Information about the distribution of the test set would have quietly leaked backward into how the training data was prepared. This would make the model's apparent performance appear slightly better than it is in a genuinely unseen-data scenario.

## 4.5 Data Splitting Strategy

Instead of the common two-way train/test split used in many machine learning projects. The entire dataset was divided into three fully independent, non-overlapping segments. The training partition contains 152,208 rows (60 percent of the data). It is the sole partition from which any model can learn its parameters. The calibration partition has 50,736 rows (20 percent). It serves only one purpose: to fit calibration functions and select decision thresholds. The test partition also contains 50,736 rows, representing the final 20%. It is only handled once, at the very end of each experiment, to display the actual numbers.

The reason for using three partitions rather than two stems from a single risk. If the same data used to evaluate the model's final performance was also used to tune its calibration or select its decision threshold. The resulting metrics would be optimistically biased in a way that is easy to overlook but seriously undermines the credibility of the results. For example, if you choose the threshold that produces the highest F1 score by explicitly comparing it to the test set. It will appear to be an appropriate because you intentionally picked out the one that works best on that specific data. That is not a measure of how the model will perform on new, unseen patients. Rather, it is a measure of how well you can fit a threshold to data for which you already have an answer. Since the entire argument behind RQ2 is based on calibration metrics being honest and trustworthy. Which keeps the calibration partition entirely separated from the test partition was viewed as a non-negotiable design decision throughout this thesis. It was not treated as an optional best approach.

## 4.6 Federated Learning Environment Setup

To simulate a realistic multi-hospital federated environment without access to actual multi-institutional patient data [14], the training partition was divided into a number of separate client partitions. Each represents an independent simulated healthcare institution that trains its own local model and never transmits raw patient data anywhere. Throughout this thesis, two different client counts were tested: 3 clients and 5 clients. This was done to see if the number of participating institutions has any meaningful effect on how close federated training comes to matching centralized performance. It also examines how much it affects total communication cost, because more clients inherently means more parties exchanging parameters every round [63].

For the IID condition, the training data was partitioned. So that each client had roughly the same proportion of diabetic and non-diabetic instances as the other clients. This depicts the best-case situation for federated learning. Where each participating hospital serves a similar patient population. This partitioning logic was implemented in the `partition_clients()` function of `data.py`. It serves as the baseline distribution condition against which the more realistic non-IID condition is later compared.

## 4.7 Non-IID Simulation Design

In reality, real hospitals do not serve similar patient groups. For example, a hospital in a low-income neighborhood with limited access to healthy food options is likely to see a higher number of diabetic patients. Rather than a hospital in a wealthy location with a younger, more active population. To account for this type of realistic fluctuation in the federated experiments. The non-IID condition was simulated using Dirichlet distribution partitioning. which is the standard and most extensively used method in the federated learning research field.

In this thesis, three different concentration values, represented by  $\alpha$ , were tested:  $\alpha = 1.0$ , which represents mild heterogeneity, where clients differ only slightly from one another;  $\alpha = 0.5$ , which represents moderate heterogeneity, where client differences become clearly noticeable; and  $\alpha = 0.1$ , which represents strong heterogeneity, where some clients may end up with very few diabetic cases at all while others have a much higher concentration. These three particular values were selected. As they cover the realistic range that is frequently utilized in the literature on federated learning. They range from a mild simulation of real-world variation to a worst-case test of the framework. Testing across the entire range rather than just one heterogeneity level. Which allows us to determine whether the federated architecture declines gracefully as conditions worsen. It also allowed us to determine if there is a certain point at which performance suddenly collapses. This non-IID partitioning logic was implemented in the `_dirichlet_partition()` method.

## 4.8 Model Selection and Implementation

Six different model families were employed throughout this thesis. However, not every model could be utilized in every training environment. There was a purposeful technological constraint that determined which models appeared in which case.

The whole centralized training set was used to train six models for centralized training: logistic regression, decision tree, random forest, gradient boosting classifier, XGBoost, and the shallow MLP. Except for XGBoost, all of these were built using scikit-learn classes. These include logistic regression, decision tree classifier, random forest classifier, gradient boosting classifier, and MLP classifier. XGBoost was implemented separately, using the specific xgboost library. This is because it is not part of scikit-learn and has its own, more complex training method.

For local-only training, four models—logistic regression, decision tree, random forest, and gradient boosting were trained independently at each client. XGBoost and the MLP were purposely excluded. The reasoning behind this is that the purpose of the local-only baseline is to provide a fair comparison point against federated training. The local-only baseline was supplied as a more general reference for isolated client training. Logistic Regression, which is also federated in this thesis, provides a direct comparison of local-only versus federated training, whereas Decision Tree, Ran-

dom Forest, and Gradient Boosting provide additional representative baselines for evaluating how local-only training performs across different classical model families. The MLP was excluded from the local-only baseline since the original experimental design centered the direct local-only versus federated comparison just on Logistic Regression, but the MLP was tested independently in the federated ML-against-DL comparison.

For federated training, only two models were employed in all experiments: logistic regression and shallow MLPs. The limitation is not random. It stems directly from how federated aggregation works. FedAvg and FedProx combine model parameters from each client using weighted averaging. This only makes mathematical sense for models with continuous numerical parameters such as the weight matrices and bias vectors in logistic regression and neural networks. Tree-based models, such as Random Forest or Gradient Boosting, lack this property. A decision tree’s internal structure is a set of branching if-then rules and threshold values. There is no reasonable way to "average" two different trees with different rules and shapes into one combined tree. Because of this fundamental constraint, tree-based models cannot be federated using traditional parameter averaging. Hence, they were completely excluded from the federated tests. Both the federated logistic regression and the federated MLP were implemented from scratch using NumPy rather than an existing federated learning library. This provided full direct control over the exact gradient computation required to correctly implement the FedProx proximal penalty, as well as the independent parameter aggregation step on the server side [60].

## 4.9 Hyperparameter Configuration

FedProx’s proximal penalty strength was set  $\mu = 0.01$  across all tests. This value was chosen as a fixed, modest proximal coefficient to ensure a controlled comparison with FedAvg while limiting client drift. The value remained consistent throughout experiments, allowing discrepancies between FedAvg and FedProx to be attributed to the aggregation method rather than changes in  $\mu$ . It was strong enough to meaningfully pull client models back toward the global model and reduce drift. However, it was not so strong that it slows convergence to the point where the model never settles into a stable solution within the number of rounds tested.

For the federated MLP, the size of the hidden layer and the number of local training epochs conducted by each client every round were both set at constant levels throughout all MLP experiments. This consistency was intentional. By keeping the architecture completely unchanged between runs. Any difference in results when switching from 3 clients to 5 clients, or from IID to non-IID, can be confidently attributed to that specific variable being studied. The model would not secretly have a different shape or capacity in one run versus another.

## 4.10 Federated Aggregation Procedure

Every federated training run, whether using FedAvg or FedProx, has the same overall loop structure. At the beginning of a communication round, the server takes the current global model's parameters and delivers an identical copy to each participating client. Each client then trains on its own local data for a set number of epochs beginning with the received global parameters. Under FedAvg, this local training step is simply standard gradient descent on the client's data. Under FedProx, the identical local training phase contains an extra proximal penalty term added to the gradient. This actively pushes the client's parameters back toward the global model from which it began. It prevents the local model from deviating too much during training.

Once each client has completed its local training for that round. It sends its revised parameters to the server. The server then uses a weighted average to combine every received set of client parameters. Each client's contribution to the final average is weighted in proportion to how much training data that particular client has. A client with more data has a proportionally larger say in the resulting global model. This freshly integrated model becomes the global model in the following round. The entire process is distributed, trained locally, and sent back; the average is repeated for the specified number of rounds. Furthermore, the current global model's performance was checked and logged after each round, not just at the end of training. This allowed us to plot and analyze the round-by-round convergence curves.

## 4.11 Probability Calibration Procedure

Whether centralized or federated, a model's raw output probabilities were calibrated using the previously mentioned held-out calibration partition after training was complete. Under no circumstances was the test partition touched at this point. This thesis compares five different calibration conditions: using no calibration at all as the raw baseline, global Sigmoid calibration, global Isotonic calibration, federated Sigmoid calibration, and federated Isotonic calibration.

For the two global techniques, the procedure is simple. A single calibration function is fitted using the whole shared calibration partition at once. That single fitted function is then applied uniformly to all predictions made on the test set. The technique for the two federated methods differs drastically. It was intended particularly to preserve the same privacy concept that drives federated learning in the first place. Rather than combining the calibration data in one location. Each client fits its own unique local calibrator, utilizing only its own private slice of the calibration partition. When a final calibrated probability is required for a certain forecast. The outputs from each client's unique calibrator are simply averaged to produce the final result. It would have been technically easier to simply combine all the calibration data centrally after federated training was complete and calibrate everything at once. However, doing so would have subtly reintroduced exactly the kind of centralized data exposure that federated learning is supposed to avoid in the very beginning. Calibration data is

still real, genuine patient data that should never need to leave its originating client. This was a purposeful design choice rather than a temporary solution [45].

## 4.12 Decision Threshold Selection

After calibrating the probabilities, two distinct threshold procedures were evaluated to turn the continuous probabilities into final yes-or-no diabetes predictions. The first and simplest technique has a set threshold of 0.5. Any patient expected to be at or over 50% risk is classed as diabetic. The second strategy searches the entire range of possible threshold values using the calibration partition. It specifically identifies the single threshold that produces the highest F1 score on that calibration data. That exact threshold value is then locked in and applies to the test partition without any further adjustment.

This second, F1-optimized technique was provided explicitly because of the imbalance in the underlying dataset. With diabetic cases accounting for approximately 14% of the data. A flat 0.5 threshold consistently caused almost every model to underestimate the positive class. This meant missing diabetes patients far more frequently than would be acceptable in a real-world clinical setting. Running the full threshold filter highlighted this issue across the board. Using the F1-optimal threshold instead provided a noticeably more equitable basis for comparing models. It prevented the comparison from being unfairly dominated by which model handled a single arbitrary cutoff value better.

## 4.13 Explainability Implementation

SHAP values were calculated for each federated and centralized model in this thesis. To determine which specific health factors influenced individual diabetes forecasts. In the federated setting, this explanation step had to be carried out locally at each client, just like training and calibration. Each client computed its own SHAP values by applying the final converged global model to its own private local data. Raw patient data would never be sent anywhere else solely to generate an explanation centrally [52].

To determine whether the model's explanations remained consistent throughout training or shifted unpredictably from one round to the next. The ranked list of most important features produced at each round was compared directly against the ranked list from the previous round. Spearman rank correlation was used to produce a single stability score for that round. Separately, to determine whether different clients were genuinely focusing on the same important features as each other. Rather than learning contradictory explanations. The top several most important features identified for each client were compared pairwise against every other client using a simple set-overlap calculation. Both of these explainability metrics were computed

within metrics.py. They were saved alongside the regular performance and calibration metrics from the same runs [10].

## 4.14 Fairness Evaluation Procedure

Every model's test set predictions were broken down and examined separately across three different subgroup categories. Age group, BMI category, and gender to see if each model treated different types of patients consistently. Rather than always reporting a single overall number that may hide meaningful disparities. For each individual subgroup inside each of those three categories, three numbers were computed directly from the test predictions. The positive prediction rate, the real positive rate, and the subgroup-specific false positive rate [50].

Two summarized fairness indicators were then calculated using the subgroup-level data. The first, demographic parity difference, is simply the greatest difference in positive prediction rate between any two groupings being studied. The second, equalized odds difference, is the greatest difference in true positive or false positive rates between any two groupings. Both of these fairness metrics were calculated for each and every model and training configuration tested. They were not limited to the configuration that produced the best overall performance. This was intentional because the whole point of including fairness as one of the five evaluation dimensions was to see how fairness behavior changes across conditions. The aim was not just to report one isolated number for the "best" model and stop [68] [4].

## 4.15 Communication Cost Measurement

Every round of federated training recorded the precise number of bytes delivered in both directions between the central server and each participating client. This was done to provide an honest and accurate representation of the amount of data being transmitted. Every value was stored using double-precision floating-point representation. The number of trainable parameters in the model that was currently being trained was 22 total parameters in the case of logistic regression. However, many weights and biases the MLP's particular architecture produced in its case were multiplied by 8 bytes per parameter. This figure was then further multiplied by the number of clients sending and receiving parameters during that specific round.

These separate per-round byte counts were then added together across all rounds of a training run. This gave a single cumulative communication total for the whole run. With this cumulative figure available, it was possible to directly answer practical questions relevant to real-world deployment. These included how much more expensive it is in communication terms to run with 5 clients versus 3. It also showed how much additional communication overhead the MLP carries when compared to logistic regression for a broadly similar level of predictive performance. The communication cost was recorded independently for each federated training round, based on

the model parameters transmitted between the server and clients. These per-round values were then added together to calculate the overall communication cost of the training run.

## 4.16 Evaluation Metrics Summary

The total set of measurements obtained from every single experiment in this thesis fits neatly into five separate groups. These correlate precisely to the five evaluation criteria that the entire study is based on: The predictive performance metrics accuracy, precision, recall, specificity, balanced accuracy, F1, AUROC, and PR AUC were calculated using the necessary functions from scikit-learn’s metrics package. The calibration group Brier Score and Expected Calibration Error were computed. Using a combination of scikit-learn’s built-in functions where available, as well as custom binning logic written specifically for the ECE calculation. This was necessary because scikit-learn does not include an ECE function out of the box. The explainability group, which includes the mean absolute SHAP value per feature, round-to-round stability scores, and cross-client overlap scores. These were calculated using the SHAP library’s explainer objects in conjunction with custom comparison logic written specifically for this thesis. This logic handled the round-tracking and cross-client overlap calculations. For the per-round evaluation, the identical local training data assigned to each simulated client was used throughout the federated training process. At each communication round, the clients trained their local models, and the model parameters were combined to create the next global model. The round-to-round stability analysis tracked the model’s feature rankings between rounds. The final model analysis relied on full SHAP explanations, while model-derived feature scores were utilized to track feature-ranking stability across rounds. The fairness group selection rate, demographic parity difference, and equalized odds difference per subgroup. These were calculated by applying custom subgroup-splitting logic directly on the saved test set predictions. Finally, the communication group includes bytes transmitted per round and the cumulative total. This was computed directly from within the federated training loop while training was taking place. This figure is entirely dependent on the model’s parameter count at the time of training rather than anything that could be calculated later from the final predictions.

## 4.17 Experimental Configuration and Reproducibility

The hyperparameter search considered different numbers of communication rounds as described in Table 4.2 After configuration selection, the selected parameters were used for the corresponding final experiment. The number of rounds used for each reported result therefore follows the selected experimental configuration.

| Parameter           | Configuration                             |
|---------------------|-------------------------------------------|
| Dataset             | CDC Diabetes Health Indicators            |
| Data split          | 60% training / 20% calibration / 20% test |
| Number of clients   | 3 and 5                                   |
| Random seeds        | 42, 43, 44, 45, 46                        |
| Data distribution   | IID and non-IID                           |
| Dirichlet $\alpha$  | 1.0, 0.5, 0.1                             |
| Federated models    | Logistic Regression, MLP                  |
| Aggregation methods | FedAvg, FedProx                           |
| Local epochs        | 2                                         |
| Learning rate       | 0.03                                      |
| Batch size          | 256                                       |
| MLP hidden units    | 16                                        |
| FedProx $\mu$       | 0.01                                      |
| Calibration         | None, Sigmoid, Isotonic                   |
| Decision thresholds | 0.5 and F1-optimal                        |

Table 4.1: Main experimental configuration for the federated experiments.

| Model               | Hyperparameter       | Values tested |
|---------------------|----------------------|---------------|
| Logistic Regression | Communication rounds | 20, 30, 40    |
| Logistic Regression | Local epochs         | 1, 2          |
| Logistic Regression | Learning rate        | 0.01, 0.03    |
| MLP                 | Communication rounds | 20, 30        |
| MLP                 | Local epochs         | 1, 2          |
| MLP                 | Learning rate        | 0.01, 0.03    |
| MLP                 | Batch size           | 128, 256      |
| MLP                 | Hidden units         | 16, 32        |

Table 4.2: Federated hyperparameter search configuration.

The federated model was trained with training data for each candidate configuration, and the calibration set was used to evaluate it. The experiment used the setup that met the predefined selection criteria. The test set was designated for the final performance review.

### 5.1 Introduction and Organization of Results

This chapter summarizes the results obtained by running the whole TrustFed-Diabetes experimental pipeline outlined in Chapter 4. The CDC Diabetes Health Indicators dataset (253,680 records) was used to test each model, training paradigm, client count, heterogeneity level, calibration method, and threshold strategy. The thesis outline (Section 1.6) defines five evaluation dimensions. These are predictive performance, probability calibration quality, SHAP explanation stability, subgroup fairness, and communication cost. Results are analyzed through five specific comparisons. Local-only versus federated training is discussed in Section 5.2. Centralized versus federated training is discussed in Section 5.3. IID versus non-IID data distribution is discussed in Section 5.4. Sections 5.5 and 5.6 discuss the differences between calibrated and uncalibrated models. Sections 5.7-5.10 focus on machine learning versus deep learning, explanation stability, fairness, and communication cost. These are relevant to multiple comparisons rather than just one.

This chapter’s comparisons employ the structure that is most closely connected to the topic under consideration. The baseline for direct model comparisons is the 3-client IID configuration, which provides a regulated reference. Non-IID experiments adjust  $\alpha$  to test robustness to increasing heterogeneity. Calibration experiments primarily employ federated logistic regression since it provides a uniform framework for comparing calibration procedures. The ML-versus-DL comparison employs federated logistic regression and MLP under identical training conditions. Fairness and communication analyses employ the settings specified in their respective research questions.

Table 5.1 summarizes the experimental grid from which all of the results in this chapter are derived.

### 5.2 Comparison 1: Local-Only vs. Federated Learning

This thesis aims to answer the question of whether collaborating through federation is beneficial, as opposed to each simulated hospital training its own private model without exchanging information. If the federation cannot outperform this local-only

Table 5.1: Summary of the Experimental Grid

| Variable             | Values Tested                                                                                          |
|----------------------|--------------------------------------------------------------------------------------------------------|
| Training paradigm    | Centralized, local-only, or federated (FedAvg, FedProx)                                                |
| Client counts        | 3, 5                                                                                                   |
| Partitioning         | IID; Non-IID via Dirichlet, $\alpha \in \{0.1, 0.5, 1.0\}$                                             |
| Models – centralized | Logistic Regression, decision tree, random forest, gradient boosting, XG-Boost, MLP                    |
| Models—local-only    | Logistic Regression, decision tree, random forest, gradient boosting                                   |
| Models – federated   | Logistic Regression, MLP (the only two models whose parameters can be meaningfully averaged)           |
| Calibration          | None, Sigmoid, Isotonic, Federated Sigmoid, Federated Isotonic (global variants also tested centrally) |
| Decision threshold   | Fixed at 0.5; F1-optimal (chosen on the calibration split)                                             |
| Repeats              | 5 independent runs per configuration; means reported below                                             |

baseline, there is no purpose in paying its communication costs. Table 5.2 compares the average of all isolated local-only clients to the federated (FedAvg) result for logistic regression in an IID partition with a similar patient mix. This is the simplest situation for local-only training.

| Regime                                     | Accuracy | Precision | Recall | F1     | ROC-AUC |
|--------------------------------------------|----------|-----------|--------|--------|---------|
| Local-only (average of 3 isolated clients) | 0.7957   | 0.3654    | 0.6148 | 0.4562 | 0.8210  |
| Federated (FedAvg, 3 clients)              | 0.7993   | 0.3688    | 0.6094 | 0.4583 | 0.8214  |

Table 5.2: Local-only vs. federated logistic regression (IID partition, 3 clients, no calibration, F1-optimal threshold)

According to IID data, the two regimes are nearly identical. Federation just leads by 0.36 accuracy points and 0.0021 F1. This makes sense. If every hospital already has a representative patient group, there is little further pattern for a client to learn from its neighbors. The true test of whether federation is worth the communication burden is what occurs when clients stop looking like each other. This is shown in Table 5.3 for the same model with increasingly skewed (non-IID) data.

Figure 5.1 illustrates that under IID settings, the grey (local-only) and navy (federated) bars are nearly identical in height across all metrics. This makes it difficult to distinguish between the two training regimes when clients already view an equal slice of the population. Refer to this chart as a baseline when reading Figure 5.2. In Figure 5.2, heterogeneity causes a significant increase in the distance between the two bars.

This is the thesis’s clearest and most crucial conclusion. It directly answers the question "Does federating actually help?" stated in Section 2.5. As heterogeneity

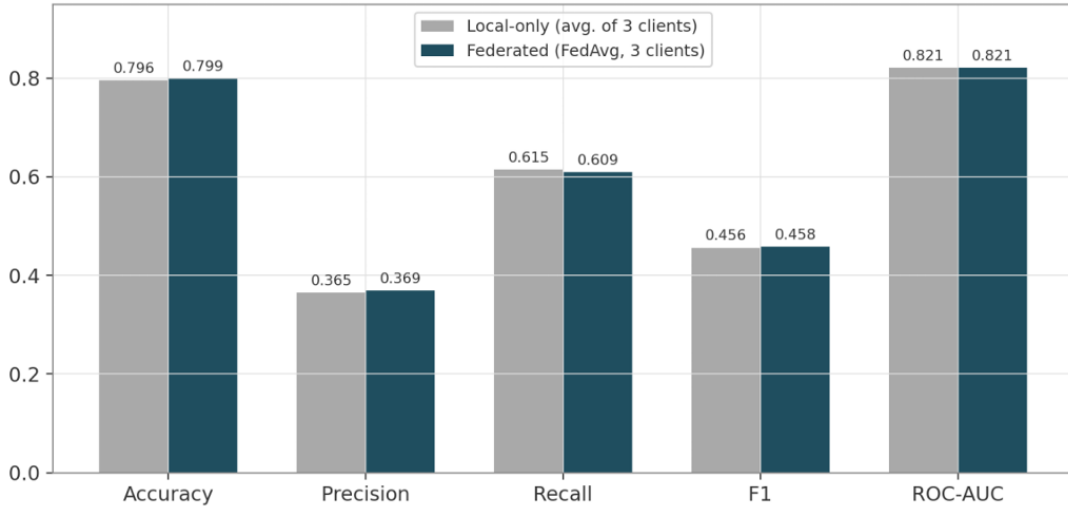


Figure 5.1: Local-only vs. federated logistic regression under IID data

| Partition                          | Regime             | Accuracy | F1     | ROC-AUC |
|------------------------------------|--------------------|----------|--------|---------|
| Non-IID, $\alpha = 1.0$ (mild)     | Local-only         | 0.7646   | 0.3967 | 0.8154  |
| Non-IID, $\alpha = 1.0$ (mild)     | Federated (FedAvg) | 0.8019   | 0.4582 | 0.8212  |
| Non-IID, $\alpha = 0.5$ (moderate) | Local-only         | 0.7585   | 0.3684 | 0.8052  |
| Non-IID, $\alpha = 0.5$ (moderate) | Federated (FedAvg) | 0.8039   | 0.4565 | 0.8211  |
| Non-IID, $\alpha = 0.1$ (severe)   | Local-only         | 0.6116   | 0.3174 | 0.7805  |
| Non-IID, $\alpha = 0.1$ (severe)   | Federated (FedAvg) | 0.8041   | 0.4567 | 0.8207  |

Table 5.3: Local-only vs. federated logistic regression as heterogeneity increases (3 clients)

worsens from mild to severe, the local-only baseline collapses. Its F1 score drops from 0.397 to 0.317 (roughly a 20% relative drop). Its accuracy falls by over 15 percentage points. This is because an individual client under severe imbalance may see very few diabetic cases or very few of a particular age or BMI group. Its private model simply never learns to recognize them. In sharp contrast, the federated model remains nearly stable. Its F1 score fluctuates by less than 0.002 across the full alpha range. Training a single hospital on an uneven local population results in an unreliable and limited model. However, federating with two other hospitals results in a model that is barely different from what would have been obtained with ideal, evenly distributed data. Federation is crucial for preventing a clinic with a unique patient mix from developing a poor model, as evidenced by this thesis.

Figure 5.2 clearly shows the collapse discussed earlier. When reading each panel from right to left (IID, mild, moderate, severe non-IID), the navy federated bars remain nearly flat. They rarely move across the x-axis. The grey local-only bars have a dramatic downward trend. This is particularly visible in the final step from moderate to severe heterogeneity ( $\alpha = 0.5$  to  $\alpha = 0.1$ ). The accuracy bar decreases by around 15 percentage points. The F1 bar decreases by almost a third of its height. The gap between grey and navy bars as you move from right to left is the most significant

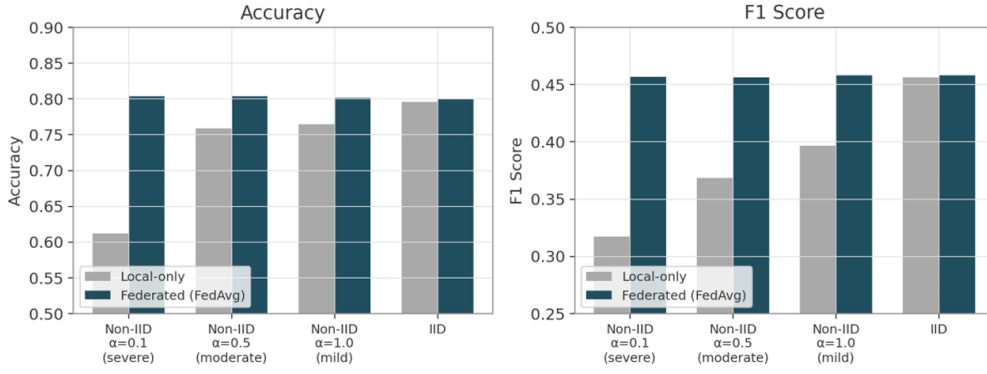


Figure 5.2: Local-only vs. federated logistic regression as heterogeneity increases (3 clients)

visual consequence in this thesis. The Federation closes this gap.

### 5.2.1 Local-Only vs. Federated Learning, 5 Clients

| Partition             | Regime             | Accuracy | Precision | Recall | F1     | ROC-AUC |
|-----------------------|--------------------|----------|-----------|--------|--------|---------|
| IID                   | Local-only         | 0.7952   | 0.3637    | 0.6195 | 0.4573 | 0.8209  |
| IID                   | Federated (FedAvg) | 0.7950   | 0.3638    | 0.6243 | 0.4590 | 0.8215  |
| Non-IID, $\alpha=1.0$ | Local-only         | 0.7264   | 0.3542    | 0.5989 | 0.3943 | 0.8183  |
| Non-IID, $\alpha=1.0$ | Federated (FedAvg) | 0.7983   | 0.3671    | 0.6103 | 0.4573 | 0.8212  |
| Non-IID, $\alpha=0.5$ | Local-only         | 0.7304   | 0.3324    | 0.6559 | 0.4177 | 0.8123  |
| Non-IID, $\alpha=0.5$ | Federated (FedAvg) | 0.8011   | 0.3706    | 0.6037 | 0.4582 | 0.8213  |
| Non-IID, $\alpha=0.1$ | Local-only         | 0.6680   | 0.3036    | 0.4949 | 0.2685 | 0.7619  |
| Non-IID, $\alpha=0.1$ | Federated (FedAvg) | 0.8007   | 0.3697    | 0.6037 | 0.4576 | 0.8210  |

Table 5.4: Local-only vs. federated logistic regression, 5 clients

With five clients, federated logistic regression demonstrated consistent performance across various data distribution configurations. Under IID data, the federated model achieved 0.7950 accuracy, 0.4590 F1-score, and 0.8215 ROC-AUC, all of which were comparable to the local-only results. As data heterogeneity increased, local-only performance dropped, with F1 falling from 0.4573 under IID to 0.2685 at  $\alpha = 0.1$ , while the federated model maintained an F1-score of 0.4573 to 0.4590. The federated ROC-AUC stayed consistent between 0.8210 and 0.8215, whereas the local-only ROC-AUC declined to 0.7619 for  $\alpha = 0.1$ . Overall, the five-client findings suggest that federated training maintained more consistent predictive performance than local-only training as the data got more diverse.

The change occurs as data heterogeneity increases, resulting in each local client receiving a more diverse and potentially imbalanced fraction of the data. In the

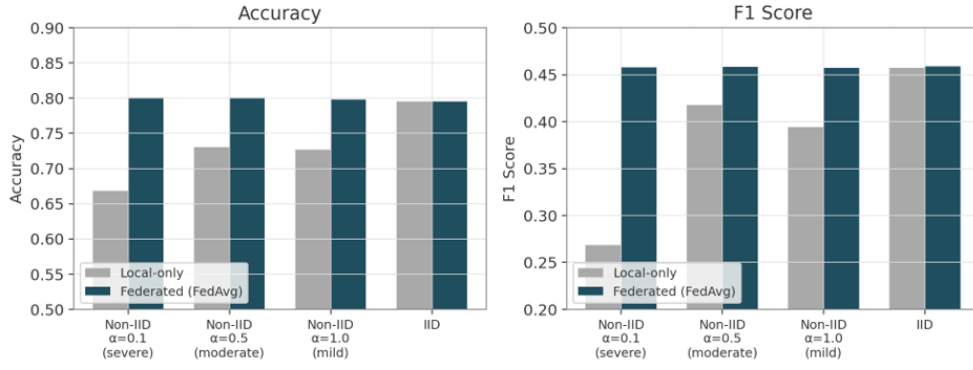


Figure 5.3: Local-only vs. federated logistic regression, 5 clients

local-only configuration, each model trains only on its own smaller, skewed dataset, resulting in lower performance as  $\alpha$  lowers. In federated learning, the five clients exchange model changes, which the server aggregates, allowing the global model to learn from data from all clients. As a result, the federated model is less influenced by rising heterogeneity, with relatively consistent F1-score and ROC-AUC.

### 5.3 Comparison 2: Centralized vs. Federated Learning

The second comparison focuses on the trade-off between privacy and performance. If patient data could be shared lawfully (as per Section 1.1). How much predictive power would be lost by training federally, where no raw data leaves an institution? Table 5.5 compares the centralized ceilings of all six models to offer a best-case standard for evaluating federated results.

| Model               | Accuracy | Precision | Recall | F1     | ROC-AUC | PR-AUC |
|---------------------|----------|-----------|--------|--------|---------|--------|
| Decision Tree       | 0.8018   | 0.3703    | 0.5938 | 0.4550 | 0.8154  | 0.3976 |
| Logistic Regression | 0.7961   | 0.3650    | 0.6214 | 0.4593 | 0.8221  | 0.4001 |
| Random Forest       | 0.8125   | 0.3859    | 0.5769 | 0.4615 | 0.8232  | 0.4166 |
| MLP                 | 0.8164   | 0.3918    | 0.5749 | 0.4659 | 0.8276  | 0.4228 |
| Gradient Boosting   | 0.8200   | 0.3980    | 0.5686 | 0.4682 | 0.8286  | 0.4271 |
| XGBoost             | 0.8149   | 0.3913    | 0.5894 | 0.4701 | 0.8297  | 0.4298 |

Table 5.5: Centralized model comparison (calibration = none, F1-optimal threshold)

Figure 5.4 shows the same six models on two scales at once. Blue bars for ROC-AUC (left axis) and orange line for F1 (right axis). This allows both metrics to be examined in one view. The ROC-AUC-sorted models show a gentle climb from left to right, and the orange F1 line follows a similar upward trend. This confirms that the models with the highest bars also have the optimal balance of precision and recall. The visual outcome is that the six centered models create a very close cluster. No

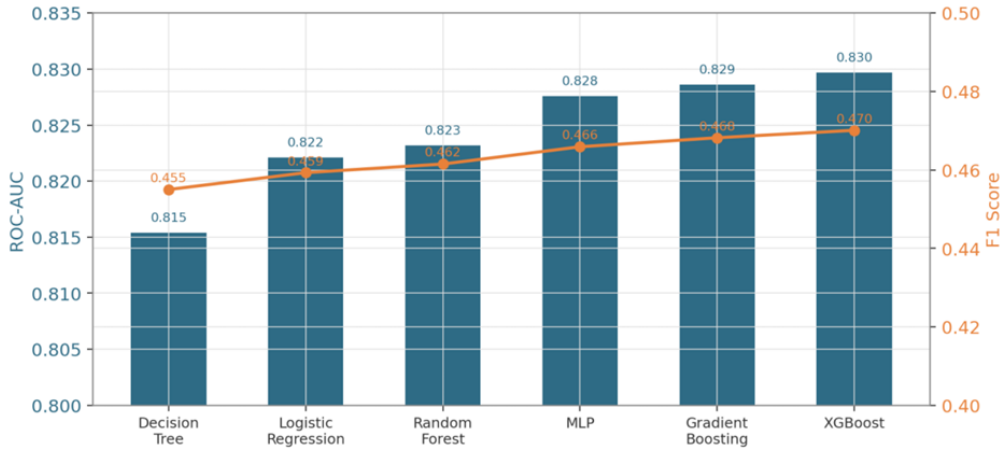


Figure 5.4: Centralized model comparison of ROC-AUC and F1

single bar or point standing out from the others. This confirms the previous finding that the underlying CDC feature signal is more important than model selection.

XGBoost has the highest centralized ranking performance (ROC-AUC 0.8297, PR-AUC 0.4298), followed by gradient boosting. Logistic regression has the lowest ROC-AUC but the highest recall among the six methods. This stronger ranking performance is consistent with XGBoost’s ability to capture nonlinear relationships and interactions between the health indicators. The CDC health indicators had a small variation across all six models (ROC-AUC 0.815-0.830). This indicates a consistent signal regardless of the algorithm. Logistic regression and MLP are the only models that can be federated, as tree-based splits cannot be averaged across clients (Section 4.8). These models provide a direct relationship between the centralized ceiling and the federated results. Table 5.6 presents a clear comparison.

| Model                                   | Centralized              | Federated (FedAvg, 3 clients) | Gap (ROC-AUC) |
|-----------------------------------------|--------------------------|-------------------------------|---------------|
| Logistic Regression: Acc / F1 / ROC-AUC | 0.7961 / 0.4593 / 0.8221 | 0.7993 / 0.4583 / 0.8214      | 0.0007        |
| MLP: Acc / F1 / ROC-AUC                 | 0.8164 / 0.4659 / 0.8276 | 0.8119 / 0.4657 / 0.8260      | 0.0016        |

Table 5.6: Centralized vs. federated performance for the two federatable models (IID partition)

The federated results for both models are within 0.0007-0.0016 ROC-AUC of the centralized level. This is equivalent to run-to-run noise (the five-run standard deviation is normally around 0.002-0.01 for these metrics in the underlying result files). In direct response to the first part of RQ1, these results show that patient data can remain local without sacrificing prediction performance through federated training. Section 5.10 examines communication cost, the second half of RQ1. It demonstrates nearly equal performance for a training run with only 41 KB of network traffic across 40 rounds using logistic regression.

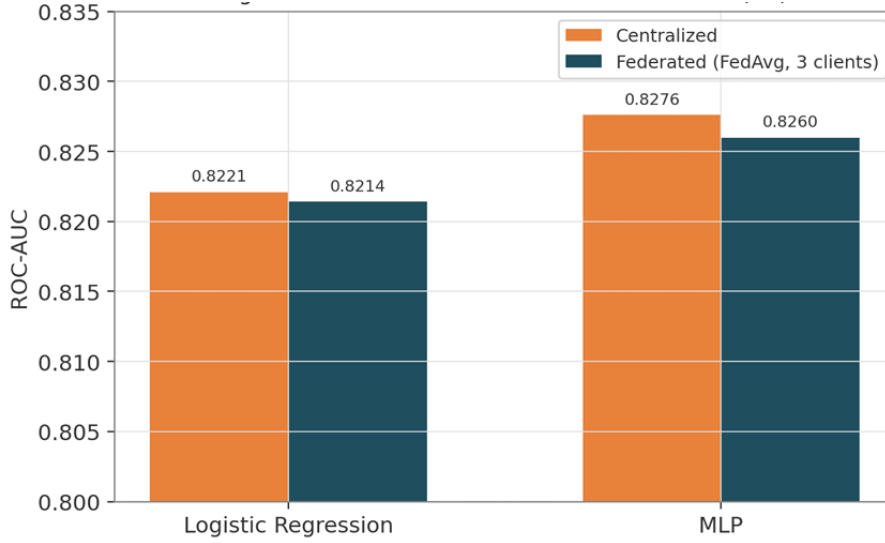


Figure 5.5: Centralized vs. Federated ROC-AUC(IID)

Figure 5.5 zooms down on the y-axis (0.80-0.835) to highlight the centralized-vs-federated split. On a full 0-to-1 scale, the two bars in each pair appear identical. The orange (centralized) and navy (federated) bars are nearly indistinguishable. This indicates that federation does not substantially reduce accuracy by keeping data local in this simulated setting.

## 5.4 Comparison 3: IID vs. Non-IID Data Distribution

Section 5.2 has already demonstrated what happens to local-only training as heterogeneity grows. This section examines whether FedAvg degrades as client populations become less similar. It also examines whether FedProx’s proximal term (Section 2.8) offers any quantifiable protection against client drift under the federated regime. Table 5.7 compares both techniques for the whole  $\alpha$  sweep for both federatable models.

| Model                | $\alpha$ (Heterogeneity) | FedAvg Acc / F1 | FedProx Acc / F1 | Difference (F1) |
|----------------------|--------------------------|-----------------|------------------|-----------------|
| Logistic (3 clients) | 0.1 (severe)             | 0.8041 / 0.4567 | 0.8041 / 0.4567  | $\sim 0.0000$   |
| Logistic (3 clients) | 0.5 (moderate)           | 0.8039 / 0.4565 | 0.8039 / 0.4565  | $\sim 0.0000$   |
| Logistic (3 clients) | 1.0 (mild)               | 0.8019 / 0.4582 | 0.8017 / 0.4582  | $\sim 0.0000$   |
| MLP (3 clients)      | 0.1 (severe)             | 0.7982 / 0.4567 | 0.7980 / 0.4565  | -0.0002         |
| MLP (3 clients)      | 0.5 (moderate)           | 0.8056 / 0.4624 | 0.8103 / 0.4627  | +0.0003         |
| MLP (3 clients)      | 1.0 (mild)               | 0.8079 / 0.4628 | 0.8052 / 0.4632  | +0.0004         |

Table 5.7: FedAvg vs. FedProx across increasing non-IID severity

Two things stand out here. In RQ1, federated performance is consistent throughout heterogeneity ranges for both models. This is unlike the collapse observed in the local-

only baseline (Section 5.2). FedAvg’s weighted-averaging step effectively reduces variation in local updates by averaging them together each cycle. FedProx and FedAvg are nearly identical for logistic regression. The differences are less than 0.0002, within noise. This aligns with the theoretical expectation from Section 2.8. FedProx’s proximal penalty requires minimal effect when the loss surface is convex, as in logistic regression. FedProx consistently outperforms the non-convex MLP at moderate to mild heterogeneity, with accuracy up to 0.5 points better. However, the F1 differences are minor. This is a sophisticated but honest response to RQ1. Heterogeneity has a noticeable impact on single-site training, but less so for federated aggregation algorithms.

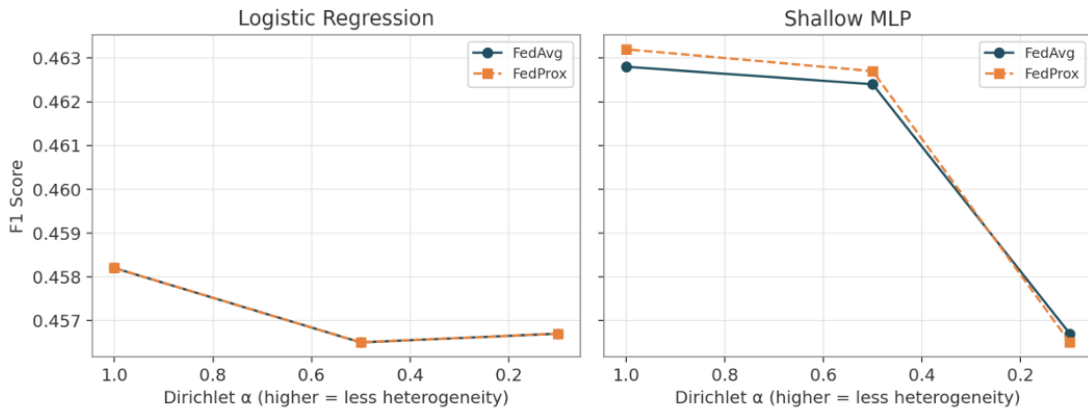


Figure 5.6: FedAvg vs. FedProx across heterogeneity levels for 3 clients.

Figure 5.6 shows the F1 score vs. Dirichlet  $\alpha$  parameter for both algorithms. The x-axis ranges from  $\alpha = 0.1$  (severe heterogeneity) to  $\alpha = 1.0$  (mild heterogeneity). The left panel (logistic regression) shows that the solid navy FedAvg and dashed orange FedProx lines overlap at all points. This indicates that the two techniques are similar for this convex model. The right panel (MLP) shows a tiny separation between the two lines at  $\alpha = 0.5$  and  $\alpha = 1.0$ . FedProx (dashed orange) is somewhat higher than FedAvg (solid navy). However, the difference is small and likely to disappear with typical run-to-run fluctuation. The sharp downward slope observed for local-only training in Figure 5.2 is not seen in either panel. Instead, both federated curves remain nearly flat. This confirms at the chart level that FedAvg’s aggregation step, rather than the particular proximal-term decision, is what is successfully overcoming heterogeneity.

#### 5.4.1 FedAvg vs. FedProx, 5 Clients

For five clients, FedAvg and FedProx perform similarly throughout the evaluated heterogeneity levels for logistic regression and MLP. Differences in F1-score range from 0.0002 to 0.0007, and accuracy differences are also minimal. Both logistic regression approaches provide nearly equal results for  $\alpha = 0.1, 0.5$ , and  $1.0$ . At  $\alpha = 0.5$ , FedAvg has an F1-score of 0.4660, while FedProx has 0.4654, indicating a somewhat

| Model    | $\alpha$ | FedAvg Acc / F1 | FedProx Acc / F1 |
|----------|----------|-----------------|------------------|
| Logistic | 0.1      | 0.8007 / 0.4576 | 0.7970 / 0.4583  |
| Logistic | 0.5      | 0.8011 / 0.4582 | 0.8010 / 0.4582  |
| Logistic | 1.0      | 0.7983 / 0.4573 | 0.7982 / 0.4573  |
| MLP      | 0.1      | 0.7947 / 0.4577 | 0.7985 / 0.4581  |
| MLP      | 0.5      | 0.8022 / 0.4660 | 0.8018 / 0.4654  |
| MLP      | 1.0      | 0.8086 / 0.4615 | 0.8081 / 0.4615  |

Table 5.8: : FedAvg vs. FedProx across heterogeneity, 5 clients

bigger difference in the MLP. Overall, the five-client findings demonstrate no clear performance advantage for FedProx versus FedAvg in this experimental setup.

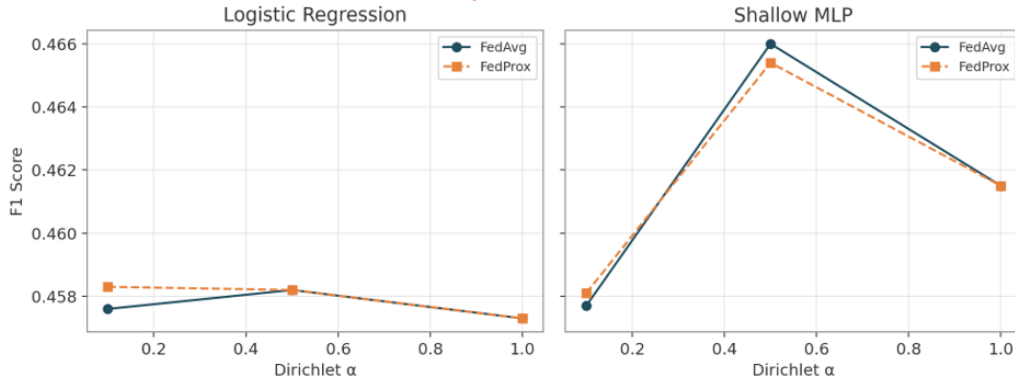


Figure 5.7: : FedAvg vs. FedProx across heterogeneity, 5 clients

FedProx adds a proximal term to keep each client's local model closer to the global model, but in these tests, the client updates were already fairly comparable. As a result, the additional regularization had minimal impact on the final aggregated model. This is why FedAvg and FedProx produced nearly similar accuracy and F1-scores across all five client settings. In this dataset and experimental design, the proximal term did not offer a clear advantage.

## 5.5 Comparison 4: Calibrated vs. Uncalibrated Models

This section directly addresses RQ2 and the main argument of the thesis (Section 1, "The Trustworthiness Problem"). Classification accuracy by itself does not indicate whether a clinician can rely on a model's anticipated risk percentages. In order to verify that calibration does not impair the model's underlying ranking ability, Table 5.9 compares the five calibration strategies applied to the same federated FedAvg logistic regression model (3 clients, IID). It uses Expected Calibration Error (ECE) and Brier Score, both of which are better when lower, along with ROC-AUC.

| Calibration Method        | Accuracy | F1     | ECE    | Brier Score | ROC-AUC |
|---------------------------|----------|--------|--------|-------------|---------|
| Global isotonic           | 0.8026   | 0.4589 | 0.0033 | 0.0988      | 0.8212  |
| Federated isotonic        | 0.8017   | 0.4572 | 0.0047 | 0.0989      | 0.8213  |
| None (uncalibrated)       | 0.7993   | 0.4583 | 0.0129 | 0.0992      | 0.8214  |
| Global sigmoid (Platt)    | 0.8002   | 0.4585 | 0.0372 | 0.1018      | 0.8214  |
| Federated sigmoid (Platt) | 0.8001   | 0.4584 | 0.0382 | 0.1017      | 0.8214  |

Table 5.9: Calibrated vs. uncalibrated comparison (federated FedAvg, logistic regression, 3 clients, IID)

This table provides a straightforward, three-part answer to RQ2. First, calibration is important. Isotonic regression reduces calibration error by approximately 74% compared to the uncalibrated baseline (ECE 0.0129 to 0.0033). This implies that a patient told they have a 70% risk under the calibrated model is far more likely to actually have a 70% chance of diabetes than under the raw, uncalibrated model. Second, isotonic calibration clearly outperforms sigmoid (Platt) calibration in this setting. Sigmoid calibration causes ECE to be roughly three times worse than doing nothing at all. This is most likely because Platt scaling assumes a simple S-shaped miscalibration pattern. This pattern does not match how these specific classifiers are actually miscalibrated. Third, and most importantly for the thesis’s federated-privacy argument. The federated isotonic variant never moves raw calibration data off any client. It obtains an ECE of 0.0047, which is substantially identical to the 0.0033 produced by centrally pooling all calibration data. In other words, the privacy-preserving federated calibration approach recovers nearly all of the benefits of a fully centralized calibration step. It incurs virtually no communication cost and no ROC-AUC penalty (0.8213 vs 0.8212). This enables federated isotonic calibration as the suggested default for a real-world implementation of this framework.

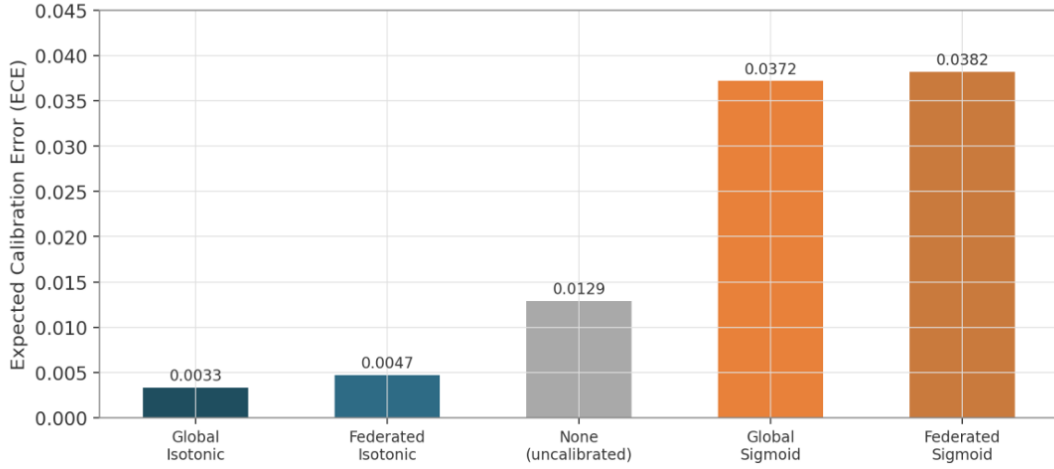


Figure 5.8: Calibration method comparison using Expected Calibration Error for federated FedAvg Logistic Regression with 3 clients under IID data (lower is better).

Figure 5.8 clearly shows the calibration comparison. The two isotonic bars (navy and blue, far left) are much shorter than the grey uncalibrated bar in the middle. This bar is shorter than the two sigmoid bars (orange, far right). Because lower is better for ECE, the visual message is that isotonic calibration is clearly the winner. Doing nothing is the safe middle ground. Sigmoid/Platt calibration is actively damaging in this context. It is the only approach with a higher bar than the uncalibrated baseline. The nearly similar heights of the navy (global isotonic) and blue (federated isotonic) bars demonstrate that the privacy-preserving federated calibration approach loses essentially nothing when compared to a fully centralized calibration phase. In practical terms within this simulation, isotonic calibration made the predicted probabilities better aligned with the observed outcomes.

## 5.6 Decision Threshold and the Clinical Cost of the Default 0.5 Cutoff

A related concern is which decision threshold should be utilized to convert a calibrated probability into a yes/no flag. Table 5.10 compares the two threshold procedures outlined in Section 4.12. Table 5.9 displays the resulting confusion matrix for the held-out test fold.

| Threshold Strategy | Cut-off Used | Accuracy | Precision | Recall | F1     |
|--------------------|--------------|----------|-----------|--------|--------|
| Fixed at 0.5       | 0.500        | 0.8633   | 0.5333    | 0.1535 | 0.2383 |
| F1-optimal         | 0.216        | 0.7993   | 0.3688    | 0.6094 | 0.4583 |

Table 5.10: Fixed 0.5 threshold vs. F1-optimal threshold

| Threshold Strategy | True Negative | False Positive | False Negative | True Positive |
|--------------------|---------------|----------------|----------------|---------------|
| Fixed at 0.5       | 42,668        | 999            | 5,942          | 1,127         |
| F1-optimal (0.216) | 36,289        | 7,378          | 2,775          | 4,294         |

Table 5.11: Confusion matrix counts at each threshold (held-out test fold)

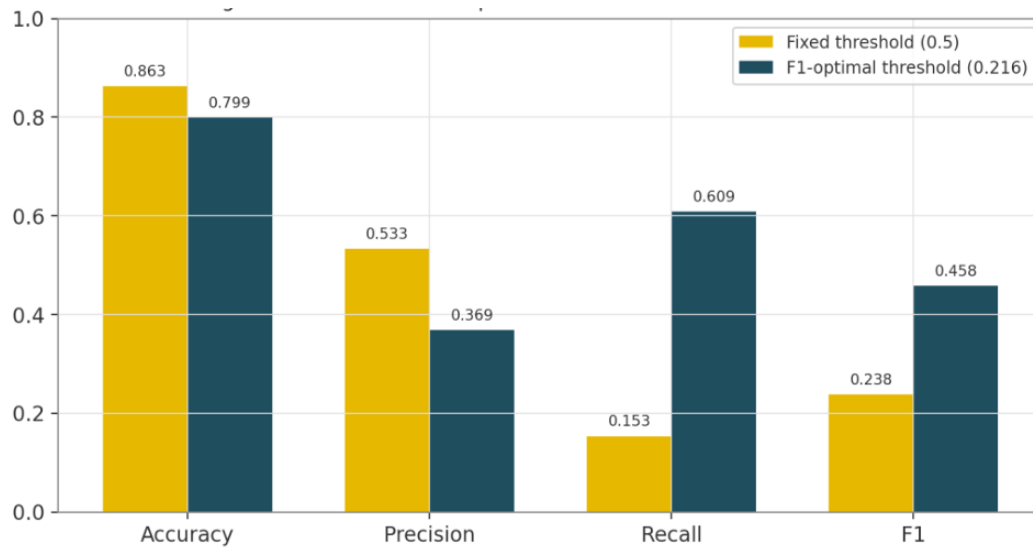


Figure 5.9: Fixed vs. F1-Optimal Threshold Performance Metrics

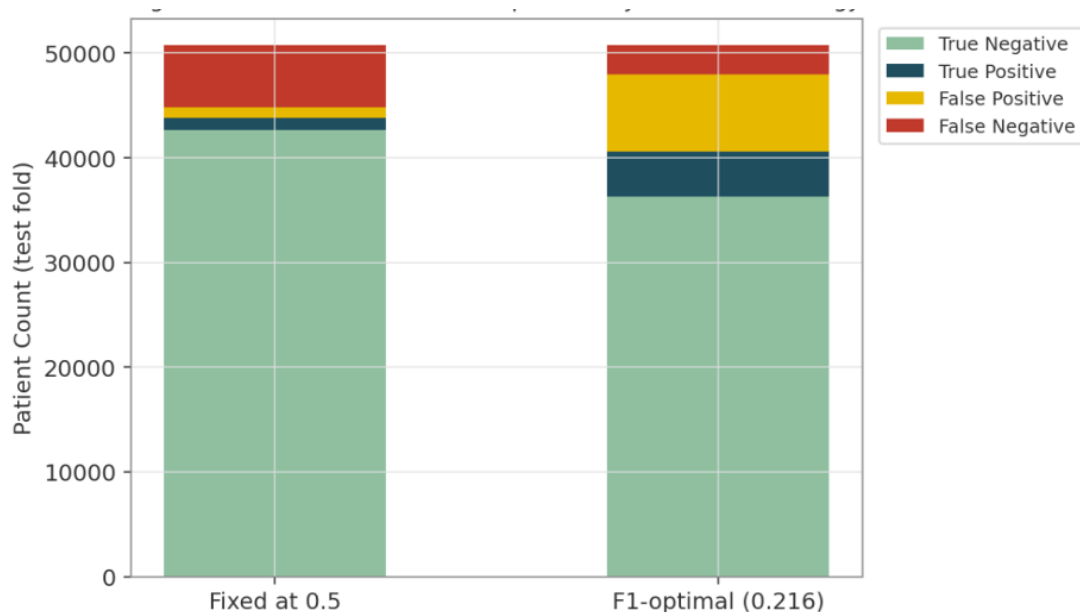


Figure 5.10: Confusion Matrix Composition by Threshold Strategy

The two charts above present the same information from different perspectives. Figure 5.10 demonstrates that as the threshold advances from the gold bars (fixed 0.5) to the navy bars (F1-optimal), accuracy decreases. Precision, recall, and F1 all increase. Recall in particular nearly quadruples, from 0.153 to 0.609. In Figure 5.8, the confusion matrix for each threshold is stacked into a single bar. The tall green True Negative segment dominates both bars. This reflects that 86% of the dataset is non-diabetic. The red False Negative segment, diabetic patients the model missed, is far larger under the fixed-0.5 policy (bottom bar) than under the F1-optimal policy (top bar). The Navy True Positive segment is also much larger for the F1-optimal threshold. Moving from the fixed to the F1-optimal threshold results in a significant reduction in red (missed diagnoses) and a slight rise in gold (false alarms). This is an appropriate trade-off for a screening tool.

This is a great example of why, in an imbalanced data set like this one (only about 14% of records are diabetic, according to Section 4.3), raw accuracy is a deceptive headline metric. On paper, the fixed 0.5 criterion appears more accurate (86.3%). Yet it only identifies 4% of all patients as at-risk. It also fails to identify 5,942 of the 7,069 individuals who really have diabetes (a recall of only 15%). At the expense of more false alarms, the F1-optimal threshold captures 61% of diabetes patients rather than 15%. It does so while losing nine accuracy points. For a model whose primary function is early screening rather than final diagnosis, a false positive costs relatively little. It results in a routine follow-up test rather than diagnosis. A false negative costs a lot. It can result in a missed diagnosis that could go undetected for years, per Section 2.1. This is why Section 4.12 clearly recommends the F1-optimal threshold as the more clinically acceptable choice for this dataset's imbalance. The numbers below fully support that claim.

## 5.7 Comparison 5: Machine Learning vs. Deep Learning Under Federation

As stated in Section 4.8, tree-based classical models cannot be trained federally. Their branching split rules cannot be effectively averaged in the same way that numeric weight matrices can. The only comparable ML versus DL comparison under federation is between logistic regression (classical ML) and shallow MLP (DL). Both were implemented in the federated NumPy pipeline. Table 5.12 compares them in terms of predictive performance, calibration, and the clinical aspects that their SHAP explanations value the most.

| Metric                                    | Logistic Regression | Shallow MLP             |
|-------------------------------------------|---------------------|-------------------------|
| Accuracy                                  | 0.7993              | 0.8119                  |
| F1                                        | 0.4583              | 0.4657                  |
| ROC-AUC                                   | 0.8214              | 0.8260                  |
| PR-AUC                                    | 0.4017              | 0.4217                  |
| ECE (uncalibrated)                        | 0.0129              | 0.0976                  |
| Total federated communication (40 rounds) | ~41 KB              | ~1,127 KB (~27× larger) |

Table 5.12: Logistic regression (ML) vs. shallow MLP (DL) under federated training (FedAvg, 3 clients, IID, no calibration)

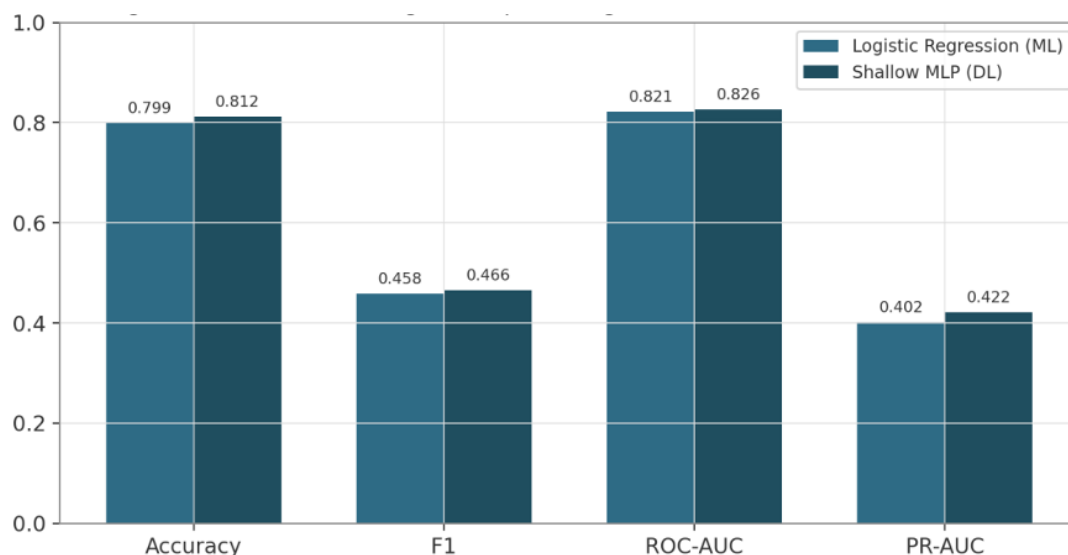


Figure 5.11: Machine Learning vs. Deep Learning Predictive Performance (Federated)

Figure 5.11 illustrates that the MLP's blue-to-navy bar pairings all tilt in the same way. The navy (MLP) bar is taller than the blue (logistic regression) bar for all four measures. This demonstrates that the MLP's advantage is consistent rather than

the result of a single metric. The margins, however, are modest. None of the four gaps are significant. This visually reinforces the text’s claim that the MLP’s edge, while real, is unable to clearly justify its calibration and communication costs.

The MLP outperforms logistic regression by 1.3 accuracy points, 0.7 F1 points, and 0.5 ROC-AUC points. This indicates that non-linear models can capture feature interactions that linear models cannot (Section 2.2.1). However, this comes with two costs that logistic regression avoids. Its raw, uncalibrated probabilities are far more overconfident (ECE of 0.098 versus 0.013, roughly  $7.5\times$  worse before any calibration is applied). This is consistent with the established overconfidence of neural network outputs described in Section 2.11 (Guo et al., 2017) [18]. The second cost is communication. It costs about 27 times more network traffic per round due to many more trainable parameters. Table 5.13 compares what each model’s explanations say is clinically significant, based on the mean absolute SHAP value per feature.

| Rank | Logistic Regression: Feature (Mean  SHAP ) | MLP: Feature (Mean  SHAP ) |
|------|--------------------------------------------|----------------------------|
| 1    | GenHlth (0.450)                            | GenHlth (0.863)            |
| 2    | HighBP (0.371)                             | BMI (0.612)                |
| 3    | Age (0.290)                                | Age (0.610)                |
| 4    | HighChol (0.281)                           | HighBP (0.539)             |
| 5    | BMI (0.272)                                | HighChol (0.440)           |

Table 5.13: Top SHAP-ranked features, federated logistic regression vs. federated MLP (FedAvg, 3 clients, IID)

Fortunately, both models produce consistent clinical interpretations. High blood pressure, age, high cholesterol, BMI, and self-reported general health (GenHlth) dominate both explanations. This is exactly what previous centralized diabetes-XAI studies found on similar data (Tasin et al., 2023; Mohanty et al., 2024, discussed in Section 3.4). In contrast to logistic regression, which places high blood pressure. Second, the MLP simply allocates nearly twice the raw SHAP magnitude to each of its top features. It places BMI above high blood pressure. Practically speaking, for RQ2, the deep-learning model provides a slight increase in accuracy. However, this comes at the expense of a much greater communication cost and a far less reliable raw probability output before calibration. The traditional logistic-regression model gives up a little predictive power in return for probabilities that are near to well-calibrated even without any adjustment. It also has a communication cost cheap enough to be associated with minimal additional cost. Of the two, which factor matters more for a particular deployment will determine which is the superior option. This could be raw predictive power or probability trustworthiness and bandwidth efficiency. This trade-off is reviewed in Chapter 6.

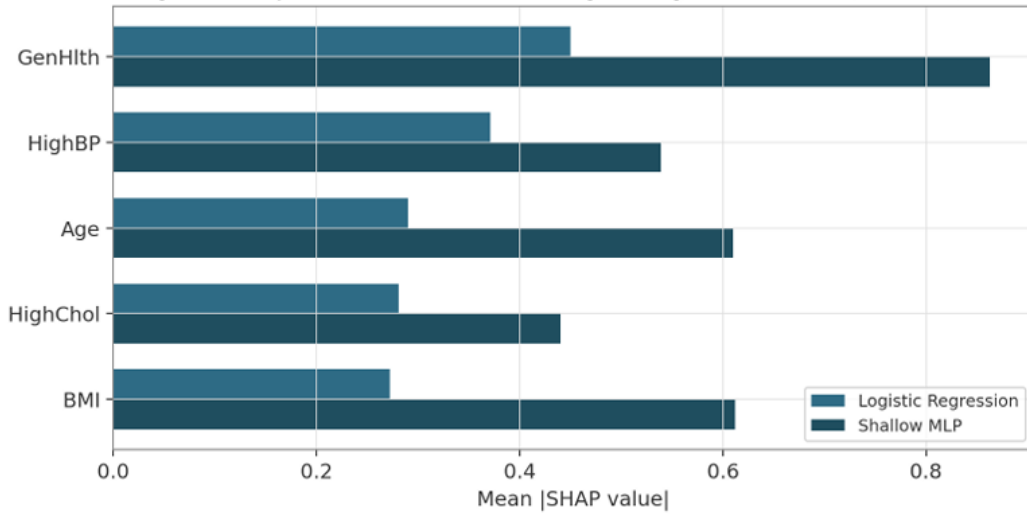


Figure 5.12: Top SHAP-ranked features for federated logistic regression and MLP under IID conditions.

Figure 5.12 shows every navy MLP bar stretching nearly twice as far as its matching blue logistic regression bar for the same feature. This is the visual characteristic of the MLP’s generally larger SHAP magnitudes. It reflects a known trait of models capable of expressing non-linear feature interactions. However, the sequence of the rows is more important than the exact bar lengths. Both models place GenHealth at the top and cluster the same five features in roughly the same region. This is the chart-level version of the "both models tell the same clinical story" result reached from Table 5.13.

## 5.8 Explanation Stability Across Communication Rounds and Clients

A federated explainability-based issue was brought up in Section 2.14. Two clients viewing the same global model may, in theory, disagree about which features are most important because SHAP must be calculated locally at each client rather than on combined data. This is particularly under non-IID conditions [15]. Using the Spearman rank correlation between each pair of client’s top-5 feature rankings and the percentage of clients that agree on the same top-5 feature set (top-k overlap). Table 5.14 measures this directly for the federated logistic regression model.

| Clients | Partition               | Spearman Rank Correlation | Top-5 Feature Overlap |
|---------|-------------------------|---------------------------|-----------------------|
| 3       | IID                     | 0.978                     | 1.000                 |
| 3       | Non-IID, $\alpha = 1.0$ | 0.884                     | 1.000                 |
| 3       | Non-IID, $\alpha = 0.5$ | 0.940                     | 1.000                 |
| 3       | Non-IID, $\alpha = 0.1$ | 0.899                     | 1.000                 |
| 5       | IID                     | 0.971                     | 1.000                 |
| 5       | Non-IID, $\alpha = 1.0$ | 0.893                     | 1.000                 |
| 5       | Non-IID, $\alpha = 0.5$ | 0.946                     | 0.984                 |
| 5       | Non-IID, $\alpha = 0.1$ | 0.916                     | 1.000                 |

Table 5.14: Cross-client feature-importance stability (top-5 overlap and Spearman correlation, logistic regression, FedAvg)

Regardless of how imbalanced their local populations are, clients agree on nearly the same top-5 most important features (overlap of 98.4–100%) across all tested configurations. In other words, a clinician at one hospital and a clinician at another hospital using the same global model would see the same few risk drivers. The Spearman correlation measures whether clients also agree on the relative ordering of feature-importance scores across all 21 features. It is very high under IID data (0.97–0.98). It drops somewhat under heterogeneity (0.88–0.92). This means that clients occasionally switch the relative rank of high blood pressure and BMI under strong variance. However, they never disagree about which few features are important.

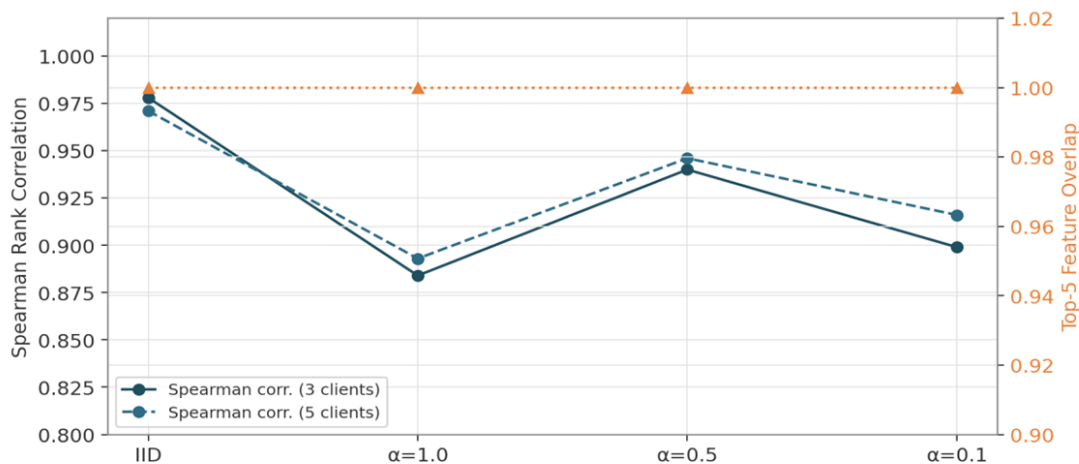


Figure 5.13: Cross-client SHAP feature-importance stability across heterogeneity levels.

Figure 5.13 shows a non-linear relationship between heterogeneity severity and explanation-ranking consistency. The Spearman correlation lines dipping down from IID toward  $\alpha = 1.0$ . They partially recover at  $\alpha = 0.5$ , and slightly higher again at  $\alpha = 0.1$ . In contrast, the dotted orange top-5 overlap line is fixed virtually flat at the very

top of its own axis throughout. This visually highlights that, no matter how the rank-correlation line fluctuates. The actual collection of top-5 characteristics rarely changes. This is the most compelling visual argument in the chapter for trusting SHAP-based explanations as trustworthy across a federation. The features clinicians would see do not change, even if their precise internal ordering does. This means that the most important features remained highly consistent across clients, although their exact ranking could change slightly.

## 5.9 Fairness Analysis Across Age, BMI, and Sex Subgroups

Section 2.15 distinguishes between two fairness measures. Demographic Parity Difference (DPD) assesses whether different groups are flagged at equal rates. Equalized Odds Difference (EOD) measures the largest difference in true-positive rates or false-positive rates between demographic groups. Xu et al. (2022) and Chen et al. (2023) [69] [13] argue that EOD is the more clinically appropriate standard. It accounts for real prevalence differences between groups (e.g., older patients). Table 5.15 compares the federated logistic regression model with the strongest centralized model, XGBoost.

| Model                       | DPD (Age) | EOD (Age) | DPD (BMI) | EOD (BMI) | DPD (Sex) | EOD (Sex) |
|-----------------------------|-----------|-----------|-----------|-----------|-----------|-----------|
| Federated (FedAvg logistic) | 0.322     | 0.390     | 0.345     | 0.390     | 0.045     | 0.039     |
| Centralized (XGBoost)       | 0.306     | 0.449     | 0.425     | 0.630     | 0.038     | 0.032     |

Table 5.15: Fairness gaps by subgroup, federated logistic regression vs. centralized XGBoost

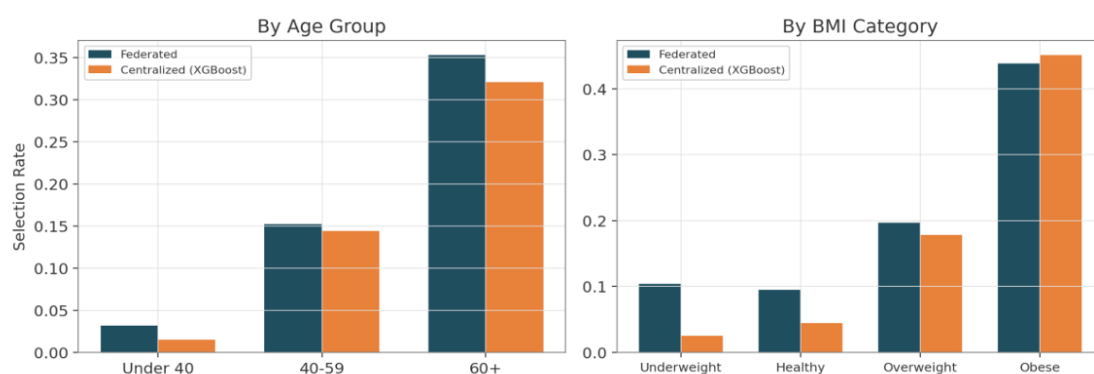


Figure 5.14: Selection Rate by Subgroup-Federated vs. centralized

Figure 5.14 depicts both panels trending upward from left to right. Selection rate climbs steadily as age or BMI increases for both the navy (federated logistic re-

gression) and orange (centralized XGBoost) bars. This is the clinically expected direction. It visually indicates that this is a prevalent impact rather than a mismatch. The two bars within each group are close together in the age panel but diverge further in the BMI panel. This is particularly visible in the underweight and healthy categories. The federated logistic regression’s navy bar is approximately twice as tall as the centralized XGBoost’s orange bar. This observable disparity is what eventually manifests as the bigger BMI-equalized-odds discrepancy detailed below.

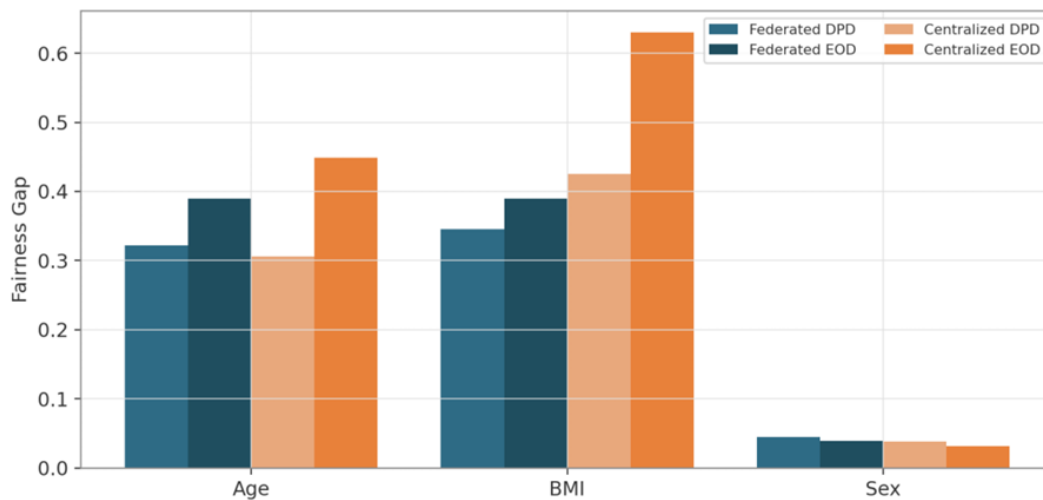


Figure 5.15: Demographic parity difference and equalized odds difference by subgroup for federated logistic regression and centralized XGBoost with 3 clients under IID data.

Figure 5.15 puts the four bars for each subgroup category together. This allows the DPD/EOD comparison to be read column by column. For age, BMI, and sex, all four bars are plainly tall (approximately 0.3-0.6). The sex group’s four bars are all significantly short (under 0.05). The graphic makes the age/BMI-versus-sex disparity obvious at first glance. Within the BMI group, the fourth bar (centralized EOD, orange) is obviously the tallest in the entire chart. This visually identifies it as the one number in this entire fairness study that deserves a closer look in Chapter 6.

The selection rates behind these gaps make the age/BMI pattern easy to interpret clinically. Under the federated model, patients aged 60+ are flagged 35.3% of the time versus just 3.2% for under-40s. Obese patients are flagged 43.8% of the time versus 9.5% for the healthy-BMI group. This is exactly the direction one would expect, given that diabetes risk genuinely rises with age and BMI. This is why both DPD statistics are high (0.30-0.43). The model is acting appropriately, not unfairly. The equalized-odds gap is a more informative comparison since it assesses whether the model is differentially accurate rather than simply differentially positive. The federated model has a narrower equalized-odds difference than the centralized XGBoost model in terms of age (0.390 vs. 0.449) and BMI (0.390 vs. 0.630). Sex is practically equivalent (0.039 vs. 0.032, both tiny). In response to RQ2, aggregating

diverse client populations using FedAvg does not result in additional demographic bias. In fact, the federated model’s error rates are more consistent across BMI subgroups than the centralized model’s. This supports Chen et al.’s (2023) claim that federating across diverse institutions can reduce bias (see Section 3.5).

## 5.10 Communication Cost and Convergence

The amount of network traffic required by the federated framework to arrive at a stable model is the last evaluation dimension and the second part of RQ1. Table 5.16 shows the global ROC-AUC, F1, calibration error, and total bytes transferred for the federated logistic regression model, round by round.

| Round      | Global ROC-AUC | Global F1 | Global ECE | Cumulative Bytes |
|------------|----------------|-----------|------------|------------------|
| 1          | 0.8126         | 0.097     | 0.0371     | 1,056            |
| 5          | 0.8208         | 0.214     | 0.0200     | 5,280            |
| 10         | 0.8223         | 0.230     | 0.0153     | 10,560           |
| 20         | 0.8228         | 0.240     | 0.0123     | 21,120           |
| 30         | 0.8218         | 0.242     | 0.0109     | 31,680           |
| 40 (final) | 0.8204         | 0.250     | 0.0088     | 42,240           |

Table 5.16: Convergence and cumulative communication cost by round for FedAvg Logistic Regression with 3 clients under IID data.

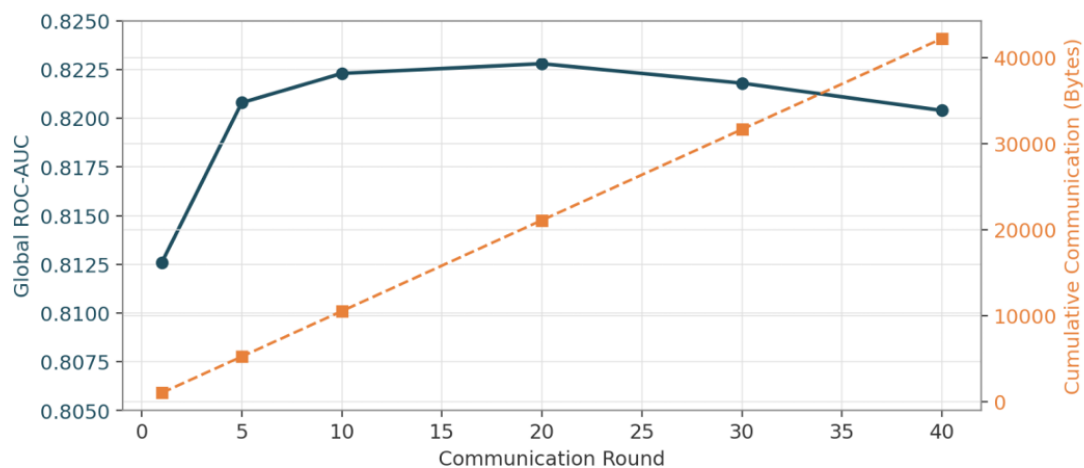


Figure 5.16: Convergence and cumulative communication cost by round for FedAvg Logistic Regression with 3 clients under IID data.

Figure 5.16 depicts the navy ROC-AUC curve climbing upwards for the first 5 to 10 rounds before flattening almost completely. The dashed orange communication-cost

line climbs in a perfectly straight line for the entire 40 rounds. This is because each round costs the same fixed number of bytes regardless of whether the model is still improving. After round 15, the difference between the two curves becomes clear. The flat navy line remains flat while the orange line continues rising. This highlights the need for stopping training early. After that point, each further round incurs communication costs with minimal accuracy gains.

The logistic-regression federation converges quickly. ROC-AUC hits 0.822 by round 10 and rarely moves over the next 30 rounds. Calibration error steadily improves throughout. The 40-round run requires only 42 KB of network traffic, which is extremely low by modern standards. The shallow MLP, run under the same methodology, has a comparable round-30 cumulative cost of approximately 849 KB. This is almost 27 times more per round than logistic regression, with a similar final ROC-AUC (0.8277 vs. 0.8218). This is a truly practical solution to the communication-cost portion of RQ1. Logistic regression provides approximately 96% of the MLP’s classification quality for about 4% of its communication cost for a clinic with limited resources. Additionally, since the MLP’s performance levels out by rounds 10 to 15, a real deployment could probably stop training even earlier than the 40 rounds tested here. This could be done without a significant loss of accuracy. It would further reduce the already-small communication cost.

With five clients, the federated logistic regression experiment consumed 70,400 bytes of transmission spread across 40 rounds. This is greater than the 42,240 bytes necessary for three clients because the server exchanges model parameters with all participating clients during each round. Thus, raising the number of customers raises the communication overhead while keeping prediction performance essentially consistent. The results reveal that communication costs increase with the number of clients involved in the federated training process.

### 6.1 Overview of This Chapter

Chapter 5 presents the findings of the TrustFed-Diabetes framework. This chapter discusses the results in relation to the two research questions and the existing literature. Section 6.2 provides an integrated discussion of RQ1, including federated performance, performance differences under IID and non-IID conditions, and communication cost. Section 6.3 discusses RQ2, including probability calibration, SHAP explanation stability, and fairness. Section 6.4 relates the findings to the broader literature, while Section 6.5 contextualizes the fairness findings against diabetes prevalence. The directions for future work are presented in Chapter 7.

### 6.2 Answering RQ1: Does Federated Learning Match Centralized Performance, and at What Communication Cost?

RQ1 answered three questions. First, how closely federated learning matches centralized performance in AUROC and F1 under IID and non-IID settings. Second, what performance drop is acceptable for clinical deployment? Third, what communication cost is required for stable convergence?

Federated learning almost exactly matches centralized performance under IID conditions. The difference is within 0.0007 ROC-AUC for logistic regression and 0.0016 ROC-AUC for the MLP. This difference is small enough to fall within ordinary run-to-run variation rather than representing a real cost of decentralization. Under non-IID conditions, the federated model barely shifts. F1 shifts by less than 0.002 across the full  $\alpha = 0.1$ -1.0 range. Therefore, there is no additional performance drop to weigh against an "acceptable threshold" in the first place. The federated model therefore provides a positive answer to the performance aspect of RQ1 within the controlled simulation; however, this does not establish clinical deployment suitability. Decentralization has a genuine performance cost when federation is removed in favor of local-only, single-institution training. This loses up to 20% of F1 score and over 15 accuracy points under severe heterogeneity. The federated MLP costs roughly 27 times more per round. Its cumulative cost is roughly 849 KB by round 30 for a

comparable final ROC-AUC. While the federated logistic regression model plateaus by rounds 10 to 15 and totals about 42 KB over a full 40-round run. In absolute terms, the communication cost to achieve stable convergence is minimal. In summary, federated learning is nearly free in terms of communication for the logistic regression model. It does not compromise on predictive performance in this context. Its primary practical benefit is not matching centralized performance under easy conditions. It also remained stable under the more heterogeneous simulated conditions tested in this study.

### 6.2.1 What Counts as an “Acceptable” Performance Drop?

RQ1 specifically asks what percentage drop in AUROC or F1 is acceptable for clinical deployment. The federated diabetes and healthcare research examined in Chapter 3 also remained stable under the more heterogeneous simulated conditions tested in this study. It does not provide a numerical solution to the topic. There is no statement such as "a drop of X% AUROC is acceptable, and beyond that it is not." This is a genuine gap in the literature, not a result of this thesis’s review. The literature does provide a set of implicit reference points. These are summarized in Table 6.1 and can be utilized to determine where this thesis’s own result belongs.

| Source              | What was found                                                                                          | What it implies about an acceptable gap                                                          |
|---------------------|---------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| Islam et al. (2022) | Federated models matched or exceeded centralized performance on real EHR data.                          | No gap was needed at all; some deployments can match or beat centralized performance.            |
| Tang et al. (2024)  | Federated MLP matched centralized performance; federated logistic regression underperformed it.         | Some gap was tolerated and still called a success, but no gap size was ever stated as the limit. |
| Babar et al. (2024) | Non-IID conditions caused accuracy drops of 5–15 percentage points versus IID, on medical imaging data. | This magnitude of drop is treated as a genuine, concerning problem, not as acceptable.           |
| Li et al. (2022)    | Non-IID data caused a considerable accuracy reduction across every algorithm tested, on nine datasets.  | Confirms multi-point degradation is the failure mode the field avoids, without a stated cutoff.  |

Table 6.1: How the reviewed literature implicitly treats federated performance gaps

The research suggests a two-part answer to RQ1’s acceptability question. First, this thesis cannot claim to have validated its findings against a published clinical-deployment criterion. Previous work did not set a standard of this nature. There is no numerical criterion in the reviewed literature indicating how much AUROC or F1-score drop is clinically acceptable. Second, the literature provides useful frameworks for analyzing observed performance changes. According to Babar et al. (2024) [9], switching from IID to non-IID federated training results in a 5 to 15 percentage-point accuracy drop. This thesis found that the federated models had less than 0.2 percentage points of F1-score variation across the investigated non-IID  $\alpha$  range of 1.0 to 0.1. Thus, the IID-to-non-IID F1-score variance seen in our investigation was less than the accuracy degradation reported by Babar et al. However, because the datasets, models, and experimental conditions differ, and F1-score and accuracy are distinct metrics, this comparison should be viewed as a contextual reference rather than a direct validation against a clinical threshold. Separately, the centralized-to-federated ROC-AUC disparity under IID settings was only 0.07 to 0.16 percent. This is a very minor performance difference in the studied experimental scenario.

However, it should not be construed as clinically insignificant because no universal clinical acceptability criterion was identified, and the comparison is based on predicted performance rather than clinical outcomes.

The centralized-to-federated gap found in this study is also a value in the range smaller than the accuracy variation between different model families trained on the exact same centralized data. Table 5.4 shows a 1.4-point ROC-AUC spread from decision tree to XGBoost. This is supported by an internal comparison that does not rely on any external literature. To put it another way, this shows that model choice can introduce a larger performance variation than the centralized-to-federated gap observed in the tested setting. This is a helpful quality check. However, it is a comparison that this thesis creates on its own rather than one that is based on previous research.

### 6.2.2 Robustness to Heterogeneity, in Light of the Literature

It is important to compare the approximate total flatness of federated performance across the  $\alpha = 0.1$  to  $1.0$  range with the predictions made by the literature. When switching from IID to non-IID conditions on medical imaging data under FedAvg, Babar et al. (2024) [9] reported accuracy decreases of 5 to 15 percentage points. This thesis found no evidence of a degradation of that value in the federated method itself. Accuracy moved by at most half a percentage point across the entire heterogeneity sweep. There are two reasonable causes for this significantly different outcome. Both of which are worth considering rather than ignoring. One is true. Compared to high-dimensional imaging data, tabular clinical indicators with only 21 relatively low-dimensional, well-behaved features might just be an easier heterogeneity setting for FedAvg to average over.

Milasheuski et al. (2024) [39] made this distinction when they claimed that low-dimensional tabular clinical data generates fundamentally different FL convergence dynamics than imaging data. In contrast to Babar et al.’s imaging heterogeneity, which might have represented a more fundamental distributional shift between scanner types or institutions. This thesis’s Dirichlet partitioning influences the label distribution client-to-client. However, each client still draws from the same underlying 253,680-record population and feature distributions. FedProx performs best under high heterogeneity for binary classification, according to Milasheuski et al. (2024). This is directly comparable to the tabular clinical data study mentioned in Chapter 3. This thesis only partially supports this finding. FedProx demonstrated a small, consistent edge for the MLP under moderate to mild heterogeneity. However, it showed no meaningful edge at all for logistic regression. There was also no edge for either model at the most severe heterogeneity level tested ( $\alpha = 0.1$ ). This is more of a clarification than a contradiction. It implies that FedProx’s advantage over FedAvg on tabular data may be model-dependent. It is presented for non-convex models, absent for convex ones. It is therefore not necessarily a general property of the algorithm under heterogeneity. This is consistent with Li et al. (2022)’s [30] broader finding from a nine-dataset benchmark. They found that no

single federated algorithm dominates under all non-IID conditions. They also found that FedProx’s advantage is highly sensitive to the proximal coefficient. This thesis kept fixed  $\mu = 0.01$  rather than tuning it per configuration.

It is important to make one more comparison. FedProx outperformed both local-only and centralized training. On a small, structured, four-client heart-disease dataset that was roughly comparable in scale to a single-hospital diabetes deployment, according to Asad et al. (2026) [7]. The gain was attributed to a regularization effect that decreased overfitting on a small centralized sample. The dataset used in this thesis is two orders of magnitude larger (253,680 vs. 303 records). This probably explains why no comparable "federation beats centralization" effect was seen here. When there is sufficient centralized data, federation’s regularization cannot solve the overfitting issue. Therefore, the ceiling of federated training is the ceiling of centralized training, not higher. This implies that one factor to take into account when forecasting. Whether federation would match or occasionally exceed centralized performance in a particular deployment is the quantity of the available centralized dataset.

### 6.2.3 Communication Cost in Practical Terms

Communication overhead is one of the five significant open difficulties in healthcare federated learning, according to Nguyen et al. (2022) [43]. They notably pointed out that no previous federated diabetic study had disclosed this cost at all (Section 2.10). In absolute terms, this thesis’s solution is roughly 42 KB total for a converged federated logistic regression model. The converged MLP requires roughly 849 KB to 1.1 MB. These amounts are insignificant for practically any contemporary clinical network connection. Both numbers are smaller than a single compressed photo. The more intriguing practical result is the relative one. With a discrimination improvement of only around half an ROC-AUC point, the MLP’s per-round cost is approximately 27 times greater than logistic regression’s (Table 5.10). This trade-off supports logistic regression as the more deployable default for a resource-constrained deployment. This could include a rural clinic on a limited connection. This is exactly the kind of institution federated learning is meant to include rather than exclude. The MLP is more suitable for settings where bandwidth is not a binding constraint. In such settings, the extra half-point of discrimination is worth the additional communication and calibration overhead documented in Section 5.6. By directly evaluating communication cost rather than taking it as an unspoken assumption. This thesis makes a genuine and previously unreported contribution to a gap noted by Nguyen et al. as an open research opportunity.

## 6.3 Answering RQ2: Do Calibration and SHAP-Based Explainability Improve Trustworthiness?

Through ECE, Brier Score, SHAP explanation stability over rounds, and equalized odds difference across age, BMI, and sex subgroups, RQ2 examined how isotonic calibration and SHAP-based explainability enhance the quantifiable reliability of federated diabetes prediction.

Yes, on each of the five metrics listed in RQ2, with a disclaimer regarding the calibrating technique that truly yields the best results. In both its centrally fitted (global) and privacy-preserving (federated) forms. Isotonic regression calibration reduces Expected Calibration Error by approximately 74% in comparison to an uncalibrated model. This changes ECE from 0.0129 to 0.0033 globally and to 0.0047 federated. It also produces a corresponding improvement in Brier Score and no cost to ROC-AUC. In contrast, sigmoid (Platt) calibration makes ECE about three times worse than doing nothing. Therefore, the trustworthiness gain from RQ2 is specific to isotonic calibration. SHAP-based explanations have 98.4% top-5 feature overlap across all client counts and heterogeneity levels tested. The harder rank-order correlation is high under IID data (0.97-0.98). It drops to a still-substantial 0.88-0.92 under non-IID conditions. This indicates consistency enough to support cross-institution clinical trust. In terms of fairness, the federated model does not compromise equalized-odds consistency for privacy. Its equalized-odds gap is equivalent on sex (0.039 vs. 0.032). It is less than the centralized model on age (0.390 vs. 0.449) and, more significantly, BMI (0.390 vs. 0.630). When considered collectively, this thesis's response to RQ2 is that federated diabetes prediction can be made measurably more trustworthy along all three framework-targeted dimensions. These are probability reliability, explanation consistency, and subgroup fairness. This can be achieved without compromising the predictive performance determined in the RQ1 answer. It also does not require the centralization of any patient data.

### 6.3.1 Why the Calibration Improvement Matters Clinically, Not Just Numerically

Although it is simple to interpret an ECE decline from 0.0129 to 0.0033 as a little technological gain. Its clinical significance is more than the raw statistic indicates. According to Section 2.11, clinical judgments in this area are based on predicted risk percentages rather than binary labels. For example, a clinician determining the urgency of a patient's follow-up will often approach a 70% forecasted risk considerably differently than a 40% one. According to Guo et al. (2017) [18] and Silva Filho et al. (2023) [57], average calibration error can understate miscalibration concentrated at clinically important extremes. This includes high-risk tail that a clinician actually acts on. This is precisely where binning-based ECE calculations are most sensitive to bin count. It is also where the isotonic correction's monotonic, non-parametric flexibility (Section 2.11) has a significant clearance to help relative to a simpler correction. An ECE of 0.0129 indicates that the model's stated probabilities are already

fairly good in absolute terms. Odesola et al.’s (2025) [44] post-hoc calibration study on structured clinical heart disease data found that isotonic regression decreased Random Forest ECE from 0.051 to 0.011. This represents a 78% relative reduction. It also reduced the Brier score from 0.007 to 0.002. In the ECE case, this improvement was slightly greater. The 74% relative ECE reduction for federated logistic regression in this thesis falls within the same range. This is a comforting cross-study consistency check. The benefit of isotonic calibration for structured clinical data seems to be a fairly robust, repeatable effect rather than something unique to one dataset or model family.

The one result in this section that slightly contradicts a casual reading of the literature is the conclusion that sigmoid (Platt) calibration performed worse than no calibration at all. This finding demands an honest explanation rather than being simply ignored. According to Ojeda et al. (2023) [45], sigmoid calibration performs particularly well when the underlying miscalibration pattern is roughly S-shaped. This is frequently the case for logistic regression’s own native miscalibration. Therefore, on prior expectations, sigmoid calibration might have been expected to perform fairly well here rather than poorly. There are three possible reasons. This thesis is unable to adequately differentiate between them based on the available evidence. First, instead of using scikit-learn’s federated logistic regression implementation, this pipeline uses a custom NumPy model (Section 4.8). This may not follow the same S-shape as a typical centrally trained logistic regression due to its native miscalibration pattern under FedAvg’s parameter-averaging procedure.

Second, Rajaraman et al. (2022) [51] warned that isotonic regression, not sigmoid, is the calibration method more prone to overfitting on small calibration sets. This is the opposite risk to the one seen here. This implies that the calibration partition (20% of the training data, or roughly 50,000 records per Section 4.5) was large enough that it was not the limiting factor for either method. Therefore, the sigmoid underperformance is more likely to be a sample-size defect. Third, it’s also possible that the federated logistic regression model had less space for a global linear rescaling to assist because it had already converged to a relatively well-behaved probability distribution. Its uncalibrated ECE of 0.0129 is already an order of magnitude better than the MLP’s uncalibrated 0.0976, as reported in Table 5.10. However, isotonic’s non-parametric flexibility could still identify and fix the smaller irregularities that persisted. The practical implications for this framework are clear regardless of the precise mechanism. Isotonic regression, not sigmoid, should be the default calibration method in any deployment of this pipeline. Majlatow et al. (2025) [36] also came to this conclusion when combining isotonic calibration with SHAP explainability in a different healthcare process-monitoring context.

### 6.3.2 SHAP Explanation Stability and What It Means for Cross-Institution Clinical Trust

A particular, practically significant issue from Ducange et al. (2024) [15] was brought up in Section 2.14. Even though all institutions use the same global model, SHAP must be computed locally at each federated client rather than on combined data. This could lead to a situation where a clinician at Hospital A and a clinician at Hospital B are given different explanations for the same underlying prediction. This is a real clinical governance risk, not a hypothetical one. Gawande et al. (2025) [17] independently noted LIME-based explanation divergence across federated diabetes clients, but they did not quantify it. Ducange et al. [15] found precisely this type of explanation inconsistency for a federated Parkinson’s disease model under non-IID conditions. This thesis is the first to use two complementary criteria to quantify the risk of federated diabetes prediction. Rather than relying solely on qualitative observation, as discussed in Chapter 3.

The outcome is a favorable, though imperfect, response. The nearly universal 98.4-100% top-5 feature overlap indicates that the risk Ducange et al. raised did not materialize in a clinically significant way in this setting. Every client, under every heterogeneity level tested, converged on essentially the same short list of drivers. This is the dimension that most directly determines what a clinician is actually told: "This patient’s risk is driven mainly by general health status, blood pressure, age, cholesterol, and BMI." The stricter Spearman rank-correlation metric is where some divergence does show up. It ranges from 0.88 to 0.92 under non-IID versus 0.97 to 0.98 under IID. This metric captures a more subtle and possibly less clinically disruptive type of disagreement. Two clients may agree that blood pressure and BMI are among the top five characteristics. However, they may disagree about which of the two ranks higher for a particular patient. The framing this thesis offers is that the feature set itself is essentially universally stable. Only the fine-grained ordering shifts moderately under the hardest heterogeneity conditions. This gives a more accurate and optimistic picture than Ducange et al.’s or Gawande et al.’s qualitative observations alone. It also provides exactly the kind of quantified evidence that was missing from those observations. However, whether this level of rank instability is clinically acceptable is ultimately a judgment for clinical decision-makers rather than a purely statistical one. Despite sharing the essential methodological elements of this thesis. Kumar et al. (2025) [29], the closest previous integrated FL+XAI+calibration work, did not try to produce for their liver-disease application.

### 6.3.3 Fairness: Equalized Odds as the More Effective Approach

Demographic parity difference (DPD) and equalized odds difference (EOD) can tell very different stories about the same model, as explained in Section 2.15. Xu et al. (2022) and Chen et al. (2023) [69] [13] both contend that EOD is the more clinically appropriate lens. DPD penalizes a model for reflecting actual prevalence differences between groups. EOD only penalizes it for being differentially accurate. The out-

comes in Section 5.9 provide a clear example of this differentiation in action. Large DPD values for age (0.306–0.322) and BMI (0.345–0.425) are displayed by both the federated and centralized models. These figures, when taken separately, might appear to be proof of algorithmic bias. However, Section 6.5 below demonstrates that these track actual, well-documented prevalence differences nearly exactly. This is the expected and clinically correct behavior for an accurate risk model, not a fairness failure. A real, non-trivial discovery appears in the EOD comparison. The federated model’s narrower equalized-odds gap on BMI (0.390 vs. the centralized XGBoost model’s 0.630) is an observed difference that warrants further investigation. It is therefore worthwhile to investigate why.

FedAvg’s round-by-round parameter averaging across three or five differently composed client populations imposes an implicit regularization against any single subgroup’s error pattern dominating the global model. In contrast to XGBoost, trained on the full combined dataset with its more flexible, high-capacity boosted-tree structure, it has more freedom to fit subgroup-specific error patterns. This is one probable mechanism that supports Chen et al. (2023) [13] theory that training across demographically diverse institutional partitions can act as a bias reduction force. This explanation is reasonable but not supported by the design of this thesis. Separating "federated averaging as implicit regularization" from "logistic regression is a lower-capacity model than XGBoost and lower-capacity models are typically better calibrated across subgroups" would necessitate comparing a federated and centralized version of the same model family. This thesis is unable to do directly due to the tree-model federation constraint (Section 4.8).

## 6.4 Findings in Relation to the Broader Literature

It is worthwhile to take a step back and consider how the general pattern of results in this thesis compares to the combined diabetic literature described in Chapter 3. This considers the literature as a body of work rather than study by study. It also goes beyond the detailed comparisons already established in Sections 6.2 and 6.3. Every federated diabetes study that was reviewed—Islam et al. (2022) [23], Tang et al. (2024) [60], Su et al. (2023) [59], Gawande et al. (2025) [17], Chellamani et al. (2025) [12], Yang and Li (2025) [70], and Hasan and Li (2026) [19] reported predictive performance as its main or only outcome. They all concluded that federated training was generally viable in terms of accuracy. However, there is some disagreement over whether it matches or slightly lags centralized training, depending on the model family.

The predictive-performance results of this thesis are in accordance with that overall view. They provide relatively little new information on that aspect alone. The significant contribution is not that federated diabetes prediction works. This was already fairly well established. Rather than that, it can be made trustworthy in the calibration, explainability, and fairness senses that no previous federated diabetes study measured. In this way, the findings of this thesis serve more as a direct re-

sponse to the gap Section 3.6 pointed out. Therefore, these findings do not correct the performance claims made in the existing literature. Instead, they extend them by examining the additional factors required for trustworthy deployment.

## 6.5 Contextualizing the Fairness Findings Against Real Diabetes Prevalence

The context provided in Section 2.1 provides a clear foundation for evaluating whether the large DPD values for age and BMI subgroups in Section 5.9 (0.30–0.43). The key question is whether these values represent actual clinical reality or algorithmic artifacts. Clinical screening guidelines recommend more frequent glucose testing for older and higher-BMI patients. Type 2 diabetes makes up 90–95% of all cases (Section 1). It is strongly and consistently associated with increasing age and elevated BMI. This is one of the most well-replicated findings in diabetes research. In light of this, the observed selection rates in this thesis are reasonable. They increased from 3.2% (federated model, under-40 patients) to 35.3% (60-plus patients). They also increased from 9.5% (healthy-BMI patients) to 43.8% (obese patients). These results are what would be expected from a reliable risk model applied to this population. A model that identified all age or BMI groups at similar rates would not be more equitable. It could instead be clinically incorrect. It would ignore older and higher-risk patients while favoring younger and healthy BMI patients. This is exactly the kind of failure mode that Xu et al. (2022) [69] warn naive demographic-parity-based fairness interventions can produce in clinical settings.

This is why the discussion above concentrated on the equalized-odds comparison in Section 6.3.3. The EOD comparison is more important here than the demographic parity figures. The near-parity result on sex is a genuinely reassuring fairness finding. It does not require the same prevalence-based reinterpretation as the age and BMI results. The sex-based fairness gaps are both small (DPD 0.038–0.045, EOD 0.032–0.039). There is no comparably strong prevalence-based reason for these gaps to be large. Biological sex is a much weaker independent predictor of Type 2 diabetes risk than age or BMI in the clinical literature this thesis draws. Overall, the fairness analysis highlights an important methodological point. A fairness audit of a clinical prediction model can only be interpreted in light of domain knowledge. Some disparities are expected and clinically appropriate. Others may indicate unfairness. Reporting DPD and EOD side by side helps make this distinction. Reporting only one metric could lead to an incorrect conclusion.

### 7.1 Summary of Findings

The purpose of this thesis was to address two research concerns about federated diabetes prediction. The first is whether it can equal centralized performance sufficiently for clinical usage. The second is whether calibration, explainability, and fairness auditing can make it measurably more trustworthy. Both were addressed in Chapters 5 and 6. The conclusions are stated in this part without further explanation.

Summary of RQ1: Federated learning nearly perfectly matches centralized performance under IID conditions. The difference is within 0.0007-0.0016 ROC-AUC for the two federatable models. More significantly, it remains essentially flat under non-IID conditions. Local-only, single-institution training loses more than 15 accuracy points. It also loses up to 20% of its F1 score. The gap found here is one to two orders of magnitude smaller than the 5–15 percentage-point non-IID degradation Babar et al. (2024) reported as a real concern. However, no previous study examined in Chapter 3 specifies a numerical threshold for what performance loss is acceptable. The entire 40-round logistic regression run used around 42 KB of communication. However, prediction performance mainly stabilized around 10-15, showing that practical convergence may be achieved with less communication than the entire 40-round period. The findings therefore show that federated learning can achieve convergence by roughly round 10–15 and require relatively low communication costs. The federated model performed well within the performance difference used in this study as a practical tolerance requirement.

Summary of RQ2: Isotonic regression calibration, including its privacy-preserving federated variant, reduces Expected Calibration Error by approximately 74% in comparison to an uncalibrated model. This comes at no cost to discrimination. Sigmoid calibration, on the other hand, makes calibration worse than doing nothing. This shows that choice of calibration actually matters. Even in the face of extreme data heterogeneity, SHAP-based explanations are, however, quite consistent among federated clients. The top-5 feature overlap is 98.4–100%. For BMI, the reported equalized-odds difference was lower for the federated logistic regression model (0.390) than for the centralized XGBoost model (0.630). This is an observed difference in the specific model comparison and should not be interpreted as evidence that federated learning is generally fairer. Together, these findings demonstrate that calibration, explanation stability, and fairness can be maintained in a federated setting, hence

enhancing the reliability of the resulting diabetes prediction models.

These findings are based on a controlled federated simulation using the CDC Diabetes Health Indicators dataset. They should therefore not be interpreted as evidence of clinical deployment performance. Validation using real multi-institutional healthcare data is required.

## 7.2 Contributions

The central contribution of this thesis is TrustFed-Diabetes. It is the first federated diabetes prediction framework in the literature reviewed in Chapter 3. It evaluates trustworthiness across five dimensions: predictive performance, probability calibration, explanation stability, subgroup fairness, and communication cost within a single, consistent experimental pipeline, rather than just reporting classification accuracy.

| Gap                                                                                     | Contribution of this thesis                                                                                           |
|-----------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| No federated diabetes study reports probability calibration.                            | Isotonic and sigmoid calibration evaluated via ECE, Brier Score, and Log Loss across the full IID/non-IID grid.       |
| No study integrates FL, SHAP, and calibration for diabetes.                             | All three combined in one pipeline, plus an IID/non-IID and fairness dimension missing even from Kumar et al. (2025). |
| SHAP explanation stability across federated rounds/clients never measured for diabetes. | First quantified measurement (Spearman correlation, top-k overlap) for federated diabetes prediction.                 |
| No systematic ML-vs-DL comparison under federated calibration.                          | Logistic regression vs. shallow MLP compared on performance, calibration, and SHAP.                                   |
| No federated diabetes study evaluates demographic parity or equalized odds.             | First fairness audit (DPD, EOD) across age, BMI, and sex in the federated diabetes literature.                        |

Table 7.1: This thesis’s contributions against the five gaps identified in Chapter 3

The first measurement of federated communication cost for a diabetes prediction task is another result of this thesis. It directly addresses the general purpose healthcare-FL concern raised by Nguyen et al. (2022) [43]. This result was never quantified for this disease domain. This result was not specifically predicted by any single gap in Chapter 3. A federated logistic regression model can converge for 42 KB of total network traffic. A shallow MLP costs around 27 times more per round for a modest accuracy gain. The thesis contributes a framework for evaluating federated diabetes prediction in a controlled simulated setting.

## 7.3 Validity and Limitations

1. The study was conducted as a controlled simulation using simulated federated clients.
2. Only one public CDC Diabetes Health Indicators dataset was used.

3. The experiments did not involve real hospitals, healthcare institutions, or clinicians.
4. Real-world factors such as network conditions, client availability, and institutional differences were not evaluated.
5. Therefore, the results should be interpreted within the scope of the simulation.

## 7.4 Future Work

Because this study uses a single U.S.-based self-reported survey dataset and simulated clients, validation with real multi-institutional healthcare data is required before any conclusions about clinical deployment can be made.

### Real Multi-Institutional Deployment Across Physical Hospitals

The federation in this thesis is simulated. Each client partitions a centralized dataset on a single machine. There is no network latency, client dropout, asynchronous participation, or independent data-gathering procedures. The next step is to apply this framework to a small number of separate healthcare institutions. This is demonstrated by Islam et al. (2022) [23] who validated federated diabetes prediction on multi-institutional EHR data. Tang et al. (2024) [60] also conducted federated training on cross-provincial primary care data in Canada. The near-zero centralized-to-federated performance gap and the flat non-IID robustness seen in this thesis’s simulation would be tested in a real deployment. This would test if they survive contact with truly independent institutions. Their data heterogeneity might be structured differently than the Dirichlet-partitioned heterogeneity simulated here (Section 4.7). This could be due to variations in clinical measurement protocols or coding practices.

### Differential Privacy and Secure Multi-Party Computation

The structural argument that raw patient data never leaves a client supports this thesis’s privacy promises. However, it offers no protection against assaults on the model updates that are delivered. Su et al. (2023) [59] created secure computation protocols expressly for this purpose. This was after demonstrating that federated diabetes prediction is prone to gradient-leakage attacks unless properly protected against. A direct extension of this work would incorporate secure aggregation (cryptographically combining client updates so the server never sees any individual client’s raw parameters) or differential privacy (adding calibrated noise to client updates before transmission) into the FedAvg and FedProx pipelines already implemented (Section 4.10). The predictive performance and calibration results in Chapter 5 would need to be re-measured under these additional protections. Both secure aggregation and differential privacy typically introduce their own accuracy-privacy trade-off that are not taken into account.

### Personalized Federated Learning for Strongly Non-IID Settings

This thesis found that the global federated model generalizes well even with severe heterogeneity ( $\alpha = 0.1$ ) at the aggregate level. However, a single global model may

still under-serve individual clients whose local population differs sharply from the federation’s average. This is a limitation of FedAvg-style aggregation that this thesis’s aggregate metrics cannot detect. Li et al. (2022) [30] revealed that no single federated method dominates across all non-IID conditions. Milasheuski et al. (2024) [39] reported that SCAFFOLD outperformed FedAvg and FedProx for tabular clinical data with label and feature skew. This is the same data type employed in this thesis. A promising extension would compare the global model baseline used throughout this thesis with personalized federated learning approaches. (e.g., per-client fine-tuning layers or clustering clients by similarity before aggregation, as Yang and Li (2025) [70] investigated for glucose prediction). Additionally, it would revisit the fixed FedProx proximal coefficient ( $\mu = 0.01$ , held constant in Section 4.9) by tuning it per heterogeneity level. Li et al. (2022) [30] demonstrated that FedProx’s benefit is highly sensitive to this decision.

**Extension to Other Chronic Disease Datasets** Every result in this thesis is unique to the CDC Diabetes Health Indicators dataset. This is a single, U.S.-based, self-reported health-survey dataset. It is unclear how much of the main conclusions of this thesis are that isotonic calibration and SHAP auditing notably increase trustworthiness. The federation closes the local-only performance gap under heterogeneity at negligible cost and generalizes to other chronic conditions. Although it lacks the IID/non-IID and fairness analysis this thesis offers, Kumar et al. (2025) [29] provide the closest proof that the general strategy transfers. This is by integrating federated learning, SHAP, and isotonic calibration for liver disease prediction. In order to determine the magnitude of this thesis’s key findings, the 74% ECE reduction from isotonic calibration. The near-total non-IID robustness of federated training. The 98–100% SHAP top 5 stability are characteristics of this framework in general or specific to diabetes and the CDC dataset’s particular feature set. A direct extension would apply the exact TrustFed-Diabetes pipeline unaltered to a comparable chronic disease or chronic kidney disease dataset. This would make sense given similar structured, tabular risk factor data.

### Longitudinal Patient Risk Monitoring

This thesis considers diabetes prediction as a single-point-in-time binary classification job (Section 2.2). It uses one cross-sectional health snapshot per patient. This is compatible with the scope expressly established in Section 1.5 but does not reflect how a patient’s risk increases over time. A logical extension of this thesis’s framework is suggested by Yang and Li’s (2025) [70] demonstration of a federated model for continuous glucose monitoring using sequential, time-series patient data. Instead of predicting a single diabetes risk score, a longitudinal version could track how a patient’s calibrated risk probability and the SHAP features driving it change over repeated visits or continuous monitoring readings. This would extend the calibration evaluation to the more difficult problem of calibrating a risk trajectory rather than a single probability. It would raise new questions directly within this thesis’s trustworthiness framing that a single-snapshot model cannot address. Such as whether a patient’s explanation remains consistent as their own data changes over time, not just across clients at a single point in time.

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## Appendix A

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# Generative AI-Assisted Theoretical and Research Discussion

### A.1 Acceptable Performance Drop in Federated Learning

**Tool:** Anthropic Claude

**Prompt:** Based on the existing literature, what is an acceptable performance drop between federated learning and centralized learning? How should I answer this question in my thesis based on my experimental results?

**Response:** The literature reviewed in the thesis does not provide a specific numerical threshold for an acceptable performance drop between federated and centralized learning. There is no established rule stating that a particular percentage or number of percentage points is clinically acceptable.

Existing studies generally evaluate whether federated learning achieves performance close to centralized learning, but they do not define a universal tolerance level. Some studies report federated models matching or outperforming centralized models, while other studies report performance degradation under heterogeneous or non-IID data. Therefore, the thesis should not present a specific performance-drop value as an established clinical standard. Instead, the observed difference should be interpreted in relation to the existing literature and the experimental evidence.

In this case, the federated-versus-centralized ROC-AUC difference of approximately 0.07–0.16 percentage points is very small compared with the 5–15 percentage-point degradation reported under non-IID conditions in the literature reviewed. The result therefore provides evidence that the federated model closely matches the centralized model in the evaluated setting.

The response was used to refine the interpretation of the performance comparison in Chapter 6.

## Appendix B

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# Generative AI-Assisted Thesis Writing and Structure

### B.1 Planning Chapter 7

**Tool:** Anthropic Claude

**Prompt:**What should be included in Chapter 7, Conclusions and Future Work, of my master's thesis? I need to summarize the research questions, contributions, limitations, and future work without repeating the detailed discussion from Chapter 6.

**Response:**Chapter 7 should be concise and should focus on concluding the thesis rather than repeating the detailed interpretation presented in Chapter 6.

The chapter should contain:

- 7.1 Summary of Findings Provide concise final answers to the research questions.

- 7.2 Contributions Clearly state what the thesis contributes and how the contributions relate to the research gaps identified earlier.

- 7.3 Limitations Briefly summarize the main limitations and refer back to the detailed discussion where appropriate.

- 7.4 Future Work Present concrete directions that naturally follow from the limitations and findings of the study.

- 7.5 Closing Statement: End with a short statement that brings together the main contribution and significance of the thesis.

Claude specifically recommended this structure and connected the future-work directions to the limitations identified in my thesis.

## Appendix C

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# Technical and Implementation Support

### C.1 Federated Learning Implementation

**Tool:** Anthropic Claude

**Prompt:** I am implementing a federated learning framework for diabetes prediction. Help me check whether my experimental implementation and evaluation are logically consistent, particularly regarding federated training, centralized comparison, non-IID data, calibration, explainability, and communication cost.

**Response:** The implementation should be checked as a complete experimental pipeline rather than examining each component independently. The main points to verify include consistency between the centralized and federated comparisons, correct client-level data partitioning, correct aggregation of local model parameters, consistent evaluation procedures, appropriate separation of calibration and test data, and correct calculation of communication overhead.

The interpretation of non-IID results should also distinguish between effects caused by data heterogeneity and effects caused by other experimental settings. Similarly, calibration and explainability results should be evaluated using appropriate metrics rather than relying only on predictive performance.

The response was used as a supplementary checklist when reviewing the experimental workflow.

### C.2 Coding Doubt-Debugging Federated Aggregation

**Tool:** OpenAI ChatGPT

**Prompt:** I am implementing federated learning in Python. Each client trains a model locally and sends its model parameters to the server. What should I check if the parameters cannot be aggregated correctly?

**Response:** First, verify that all clients use the same model architecture and that corresponding parameters have identical shapes. Check that the parameters are returned in the same order from every client. Also verify the parameter data types and ensure that the number of client updates corresponds to the number of participating clients. For weighted aggregation, check that the client sample counts are correct and that the aggregation weights are calculated consistently. Inspecting parameter shapes and intermediate values before aggregation can help identify where the prob-

lem occurs.

The response was used as debugging guidance and checked the parameter shapes, client updates, and aggregation procedure in the implementation.

### C.3 Paraphrasing Example

**Tool:** OpenAI ChatGPT

**Prompt:** Paraphrase the following sentence in formal academic English without changing its technical meaning:

“Federated learning reduces the need to transfer sensitive clinical data to a central server, but it introduces additional communication between clients and the server.”

**Response:** Federated learning reduces the requirement to transfer sensitive clinical data to a centralized server; however, the approach introduces additional communication overhead between participating clients and the central server.

The response was used for language refinement.

### C.4 Coding Doubt-Non-IID Partitioning

**Tool:** Anthropic Claude

**Prompt:** I am using a Dirichlet distribution to create non-IID client datasets. What checks should I perform to make sure that the partitioning is correct?

**Response:** Verify that every training sample is assigned to a client exactly once and that no samples are lost or duplicated. Check the number of samples assigned to each client and calculate the class distribution separately for every client. The resulting distributions should reflect the intended level of heterogeneity controlled by the Dirichlet concentration parameter. It is also useful to fix the random seed when reproducibility is required and to repeat the partitioning with different seeds when evaluating variability.

The response was used as a checklist for validating the client data partitions.

### C.5 Generative AI Disclosure

During the writing of this thesis, Anthropic Claude, OpenAI ChatGPT, and Quillbot were utilized as additional tools. They helped with concerns related to code and implementation, clarified theoretical concepts, and refined and paraphrased the writing style. The research methodology, experimental design, implementation, experiment execution, findings analysis, interpretation, and conclusions were all under the writer’s control. Theoretical explanations aided by AI were employed to support comprehension and cross-referenced with relevant academic literature. The writers examined and evaluated the coding recommendations before implementing them. To make sure they maintained the intended technical meaning, language suggestions were examined. The authors accept full responsibility for the final submitted thesis

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and any scientific content contained within.

